

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/309624859>

Passive and active droplet generation with microfluidics: a review

Article in *Lab on a Chip* · November 2016

DOI: 10.1039/C6LC01018K

CITATIONS

1,048

READS

10,783

2 authors:



Pingan Zhu

City University of Hong Kong

92 PUBLICATIONS 3,118 CITATIONS

[SEE PROFILE](#)



Liqui Wang

Xi'an Jiaotong University

282 PUBLICATIONS 7,666 CITATIONS

[SEE PROFILE](#)



Passive and active droplet generation with microfluidics: a review

Pingan Zhu^{a,b} and Liqiu Wang^{*a,b}

Received 00th January 20xx,
Accepted 00th January 20xx

DOI: 10.1039/x0xx00000x
www.rsc.org/

Precise and effective control of droplet generation is critical for applications of droplet microfluidics ranging from materials synthesis to lab-on-a-chip systems. Methods for droplet generation can be either passive or active, where the former generates droplets without external actuation, and the latter makes use of additional energy input in promoting interfacial instabilities for droplet generation. A unified physical understanding of both passive and active droplet generation is beneficial for effectively developing new techniques meeting various demands arising from applications. Our review of passive approaches focuses on the characteristics and mechanisms of breakup modes of droplet generation occurring in microfluidic cross-flow, co-flow, flow-focusing, and step emulsification configurations. The review of active approaches covers the state-of-the-art techniques employing either external forces from electrical, magnetic and centrifugal fields or methods of modifying intrinsic properties of flows or fluids such as velocity, viscosity, interfacial tension, channel wettability, and fluid density, with a focus on their implementations and actuation mechanisms. Also included in this review is the contrast among different approaches of either passive or active nature.

1. Introduction

Droplet-based microfluidics has emerged as a versatile tool for widespread applications, attributed to the advantages: a small volume of reagents consumed, massive production of monodisperse droplets, high surface-area-to-volume ratio that facilitates fast reaction, and independent control of each droplet¹. In general, applications of microfluidic droplets arise from two distinct but complementary aspects². One exploits droplets with well-defined components and structures as templates in material science, for example, synthesis of microcapsules³⁻⁵, microparticles⁶⁻⁹, and microfibers¹⁰ applicable to pharmaceuticals, cosmetics, and foods; another involves lab-on-a-chip applications where droplets are used as microreactors to perform chemical and biochemical reactions. In most of these applications, highly-uniform droplets are desired to ensure constant, controlled and predictable outcomes. In addition, a wide range of tunable droplet-volumes, typically from femtolitre to nanolitre, is preferable. Some applications require even smaller droplets, such as several hundred nanometers in diameter, for the production of nanodroplets and nanoparticles^{11, 12}. Moreover, instead of utilizing uniform droplets, some applications favor well-controlled sequences of droplets with different volumes, for example, multi-volume droplet digital polymerase-chain-reaction (PCR) for precise and quantitative detection of genetic targets¹³. Aside from commonly used passive methods, more sophisticated techniques have been adopted to actively control droplet generation. Reliable generation of droplets with accurate control over their size and size distribution is therefore of vital importance to meet the ever-severed

demands in various applications. To this end, it is critical to have a deep and systematic understanding of microfluidic droplet formation, including both passive and active techniques.

Droplet generation originates from fluid instabilities. In passive microfluidic devices, introduction of one immiscible fluid (dispersed fluid) into another (continuous fluid) typically leads to the formation of droplets in one of the five modes (Fig. 1): squeezing¹⁴, dripping¹⁵, jetting¹⁵, tip-streaming¹⁶ and tip-multi-breaking¹⁷. The squeezing arises from the quite different mechanism from the capillary (Rayleigh-Plateau) instability that is responsible for the other four modes. Channel confinement plays a dominant role in squeezing regime and inhibits capillary instability so that breakup exhibits quasi-static mechanisms until the last stage of thread pinch-off. The other four modes of breakup come from the capillary instability as interfacial tension forces seek to minimize the interfacial area according to the thermodynamic principle of minimum interfacial energy. In these cases, viscous and inertial forces that act to deform the liquid interface counteract interfacial tension forces that resist the deformation. It is the competition of these forces that determines the specific breakup mode of droplet generation for a given set of parameters.

In comparison with passive methods, active techniques add another level of controllability in modulating droplet formation with the aid of additional energy input by external elements in addition to the pressure applied typically through a syringe pump in passive techniques. Apart from the advantage of fast response time, active controls are indispensable in some extreme situations and applications, such as breaking fluid threads in aqueous two-phase systems (ATPS) with ultra-low interfacial tension¹⁸, producing droplets from highly viscous liquids¹⁹, and on-demand droplet generation, for example, to encapsulate cells with deterministic cell numbers²⁰. According to the nature of external energy input, active techniques are categorized into electrical, magnetic, centrifugal, optical, thermal, and mechanical controls. The additional energy

^a Department of Mechanical Engineering, the University of Hong Kong, Hong Kong. E-mail: lqwang@hku.hk

^b HKU-Zhejiang Institute of Research and Innovation (HKU-ZIRI), 311300, Hangzhou, Zhejiang, China

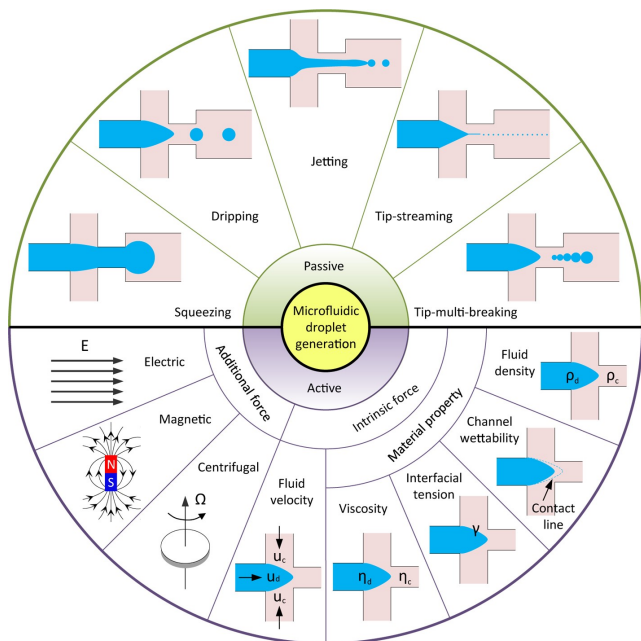


Fig. 1 Schematic of droplet generation in passive and active methods. In passive method, droplets can be produced in squeezing, dripping, jetting, tip-streaming and tip-multi-breaking modes, depending on the competition of capillary, viscous, and inertial forces. In active control, droplet generation can be manipulated by either applying additional forces from electrical, magnetic, and centrifugal controls, or modifying intrinsic forces via tuning fluid velocity and material properties including viscosity, interfacial tension, channel wettability, and fluid density.

input modifies the force balance on the interface and thus manipulates interfacial instabilities. In principle, the interfacial force balance can be modified by two basic strategies in active controls: (i) introducing additional forces like electrical, magnetic, and centrifugal force; (ii) modifying viscous, inertial, and capillary force by varying intrinsic parameters like flow velocity and material properties (Fig. 1).

Previous reviews of droplet formation have concentrated mainly on passive approaches^{1,2,21–24} with an exception of a very recent one exclusively on active droplet generation²⁵. Rather than focusing on either passive or active approach in those reviews, we aim to present a unified review of both in a systematic manner from the point of view of physical mechanisms. Our focus is on droplet generation in liquid-liquid biphasic flows, though some examples of gas-liquid¹⁴ and liquid-gas²⁶ flows are cited for the sake of elucidating breakup mechanisms. We also limit our discussion to cases where continuous phase fluids preferentially wet the channel walls, so that dispersed droplets are completely surrounded by continuous fluids. In what follows, we outline the governing equations and fundamental dimensionless parameters for droplet generation in Section 2. We summarize the commonly-used devices in producing microfluidic droplets in Section 3. In Section 4, we discuss breakup modes of passive droplet generation, specially, the generation process, mechanism, condition of occurrence, characteristics, and nature of fluid instabilities for each mode. Finally, we highlight, in Section 5, the state-of-the-art techniques for active droplet generation, organized according to their mechanisms of manipulation: techniques utilizing additional forces followed by those via modulating intrinsic parameters.

2. Dimensionless numbers

Table 1 Dimensionless parameters in microfluidic droplet generation.

Symbol	Name	Formula	Physical meaning
Re	Reynolds number	$Re = \frac{\rho u L}{\eta}$	inertial force / viscous force
Ca	capillary number	$Ca = \frac{\eta u}{\gamma}$	viscous force / interfacial tension
We	Weber number	$We = \frac{\rho u^2 L}{\gamma}$	inertial force / interfacial tension
Bo	Bond number	$Bo = \frac{\Delta \rho g L^2}{\gamma}$	buoyancy / interfacial tension
λ	viscosity ratio	$\lambda = \frac{\eta_d}{\eta_c}$	dispersed viscosity / continuous viscosity
φ	flow rate ratio	$\varphi = \frac{Q_d}{Q_c}$	dispersed flow rate / continuous flow rate

Fluid motions with various characteristics could occur in microflows, which are normally determined by competing physical effects, such as force balance. Dimensionless numbers characterize the relative predominance of different effects and are capable of contrasting flows in parameter space, thereby unifying flowing features between different systems. To explore the key dimensionless numbers characterizing the droplet generation (Table 1), we start from the governing equations that determine microfluidic two-phase flows.

Liquid-liquid two-phase microflows are well described with continuum hypothesis²⁷. The incompressible continuity equation reads, for both dispersed and continuous phase fluids:

$$\nabla \cdot \mathbf{u}_s = 0, \quad (1)$$

where ∇ is the del operator, $\mathbf{u}$ is the velocity vector of the fluid, and subscript “s” denotes either “d” or “c”, corresponding to dispersed or continuous phase fluid, respectively.

The momentum equation is the well-known Navier-Stokes equation of incompressible Newtonian fluids:

$$\rho_s \frac{\partial \mathbf{u}_s}{\partial t} + \rho_s \mathbf{u}_s \cdot \nabla \mathbf{u}_s = -\nabla p_s + \eta_s \nabla^2 \mathbf{u}_s + \mathbf{f}_s, \quad (2)$$

where t is the time, ρ the density of the fluid, p the pressure, η the dynamic shear viscosity, and $\mathbf{f}$ the body force vector per unit volume. The left side of Eq. (2) describes inertial acceleration composed of time-dependent acceleration $\rho_s \partial \mathbf{u}_s / \partial t$ and convective acceleration $\rho_s \mathbf{u}_s \cdot \nabla \mathbf{u}_s$ (spatial effect), while the right side represents force densities (forces per unit volume)²⁸. The diffusion term $\eta_s \nabla^2 \mathbf{u}_s$ is the divergence of viscous stresses $\boldsymbol{\sigma}_s = \eta_s [\nabla \mathbf{u}_s + (\nabla \mathbf{u}_s)^T]$. The body force densities $\mathbf{f}_s$ can cover effects from those like gravity, electrical, magnetic, and centrifugal effects. If no other external fields are applied except gravity, $\mathbf{f}_s = \rho_s \mathbf{g}$ with $\mathbf{g}$ being the gravitational acceleration.

Droplet formation in microfluidic devices involves the deformation and breaking of liquid-liquid interface. As such, several interfacial boundary conditions should be specified. Among them, the first comes from the continuity of normal velocity at the immiscible interface²⁹:

$$\mathbf{u}_d \cdot \mathbf{n} = \mathbf{u}_c \cdot \mathbf{n}, \quad (3)$$

with $\mathbf{n}$ being the unit normal vector outward to the interface. Secondly, the tangential viscous stress should be continuous (Eq. (4)), and the normal stress difference between the dispersed and continuous phases is balanced by capillary pressure (Eq. (5))²⁹:

$$\boldsymbol{\sigma}_d \cdot \mathbf{t} = \boldsymbol{\sigma}_c \cdot \mathbf{t}, \quad (4)$$

$$\mathbf{T}_d \cdot \mathbf{n} - \mathbf{T}_c \cdot \mathbf{n} = -\gamma \kappa \mathbf{n}, \quad (5)$$

where $\mathbf{t}$ is the unit tangential vector at the interface, $\mathbf{T}_s = -p_s \mathbf{I} + \boldsymbol{\sigma}_s$ the stress tensor that contains pressure p_s and viscous stress $\boldsymbol{\sigma}_s$, $\mathbf{I}$ the

Table 2 Comparisons of different droplet generators.

Droplet generator	Fabrication technique	Droplet production rate	Minimum droplet size	Monodispersity
Cross-flow	For cross-flow, co-flow, and flow-focusing geometries, 2D devices can be fabricated with silicon and glass by etching and with polymers, such as Poly(dimethylsiloxane) (PDMS), Poly(methyl methacrylate) (PMMA), and Poly(urethane), by soft lithography, hot embossing, injection molding, and laser ablation ⁸⁷ ; 3D devices can be fabricated by assembly of glass capillaries, and with polymers by soft lithography and 3D printing. However, 3D devices are more complicated in fabrication: assembly of glass capillaries requires careful alignment; soft lithography of PDMS usually demands multi-layer fabrication ³²³ ; 3D printing is limited in the resolution ³²⁴ .	Up to 7.4 kHz in air-bubble-triggered droplet formation ³⁸	Limited by channel dimension, usually larger than 10 μm ³³	CV < 2% ²²
Co-flow		Several hundred Hz to tens of kHz ⁵⁴	Several hundred nanometers ¹⁶⁸	CV < 3% ⁵⁴
Flow-focusing		Up to tens of kHz ⁶²	Several hundred nanometers ¹⁶⁶	CV < 5% ⁶²
Step emulsification	Step emulsification devices are made with PMMA by micromachining ⁶⁷ , and with SU-8 ⁷⁵ and PDMS ⁷⁰ by photolithography.	Up to 15 kHz ⁷⁰	Several hundred nanometers ⁷⁰	CV < 1% ⁷⁰
Microchannel emulsification	Microchannel array devices are fabricated with silicon by photolithography and wet etching ⁸⁶ .	Up to 43 Hz per microchannel ⁸⁸	1 μm ⁸⁶	CV < 5% ⁸⁶
Membrane emulsification	Shirasu porous glass (SPG) membrane is fabricated by phase separation; Microsieve membranes are fabricated with nickel by LIGA process, aluminium and stainless steel by laser drilling, and silicon nitride by reactive ion etching ⁸⁶ .	Several tons of dispersed phase per hour ⁸⁶	$\sim 0.1 \mu\text{m}$ ⁸⁶	CV $\sim 20\%$ ¹¹⁴

identity matrix, γ the interfacial tension, and $\kappa = R_1^{-1} + R_2^{-1}$ twice of the mean curvature of the interface, with R_1 and R_2 being the principle radii of the curvature.

The above-mentioned governing equations indicate four types of forces that govern the droplet generation: inertial force, viscous force, gravity (for the case without other external force fields) and capillary force. To compare the relative importance of various forces, we unify them in the form of stress (forces per unit area) based on which simple scaling arguments are derived. Considering a volume of fluid flowing at velocity u_s in a microfluidic device of characteristic length L , inertial stress scales as $f_i \sim \rho_s u_s^2$, viscous stress $f_v \sim \eta_s u_s / L$, gravity $f_g \sim \rho_s g L$, and capillary pressure $f_\gamma \sim \gamma / L$. Based on this scaling, we can develop several dimensionless numbers useful in studying droplet generation.

In principle, the ratio of any two stresses of the four defines a dimensionless number. The Reynolds number Re represents the relative importance of inertia to viscous force, and thus $Re = f_i / f_v$:

$$Re_s = \frac{\rho_s u_s L}{\eta_s}. \quad (6)$$

Re usually ranges between 10^{-6} and 10 in microfluidic flows. Therefore, the viscous stress dominates over fluid inertia, yielding a laminar flow of microflow typically.

For interfacial flows that are relevant to droplet generation, Reynolds number is rarely used. During the process of the droplet generation, three major steps are normally involved²³: initially, the dispersed and continuous phase fluids meeting at the junction to form an immiscible interface, followed by the large deformation of the interface to an unstable state, and finally the unstable interface fragmenting spontaneously and decaying into disconnected droplets. Interfacial tension thereby plays a key role in droplet generation. Consequently, we focus on examining the following three dimensionless numbers that characterize the relative importance of interfacial tension to the other three forces. Among them, the most-commonly-used one is the capillary number Ca , which is the ratio of viscous stress to capillary pressure f_v / f_γ :

$$Ca_s = \frac{\eta_s u_s}{\gamma}. \quad (7)$$

The inertial stress f_i does not depend on the device length-scale L directly. Shrinking of device length-scale L would, however, enhance both viscous stress f_v and capillary pressure f_γ and weaken gravitational effect via reducing f_g . Therefore, viscous and capillary

forces would become dominant over the other two if L is sufficiently small. This attributes Ca to be the most-frequently-used dimensionless number in characterizing microfluidic droplet generation. In microflows, Ca is usually in the range of 10^{-3} to 10.

Although fluid inertia is negligible for most microfluidic flows, it can be important for the cases such as jet¹⁵ and nonlinear bubble³⁰ formation at high flow velocity, or in the vicinity of droplet and bubble pinch-off^{31,32}. The competition between inertia and capillary pressure, f_i / f_γ , yields the Weber number We in the form of,

$$We_s = \frac{\rho_s u_s^2 L}{\gamma}. \quad (8)$$

For most microfluidic flows, $We < 1$.

The relative importance of gravity to capillary pressure, f_g / f_γ , yields the Bond number Bo ,

$$Bo = \frac{\Delta \rho g L^2}{\gamma}, \quad (9)$$

where $\Delta \rho$ is the density difference between dispersed and continuous phases. Typically, $\Delta \rho$ and L are small for liquid-liquid microflows, thereby $Bo \ll 1$ in microfluidic droplet generation.

Another two dimensionless numbers are also relevant: the viscosity ratio λ and the flow rate ratio φ of dispersed to continuous phase fluids defined by,

$$\lambda = \eta_d / \eta_c, \quad (10)$$

and

$$\varphi = Q_d / Q_c. \quad (11)$$

3. Device geometry

Microfluidic channel provides the boundary of microflow and thus its geometry would impact the droplet generation as well. Typical microfluidic geometries used in generating droplets are shown in Fig. 2. Among them, some use viscous shear forces to break off droplets with three types of most-frequently-used structures being cross-flow, co-flow and flow-focusing geometries. The others employ variations of channel confinement to facilitate or drive droplet generation, such as step emulsification, microchannel emulsification, and membrane emulsification. Herein, we discuss these droplet generators in terms of their features, developments, and variations, and compare the fabrication technique, typical droplet generation frequency, droplet size and monodispersity for each droplet generator in Table 2.

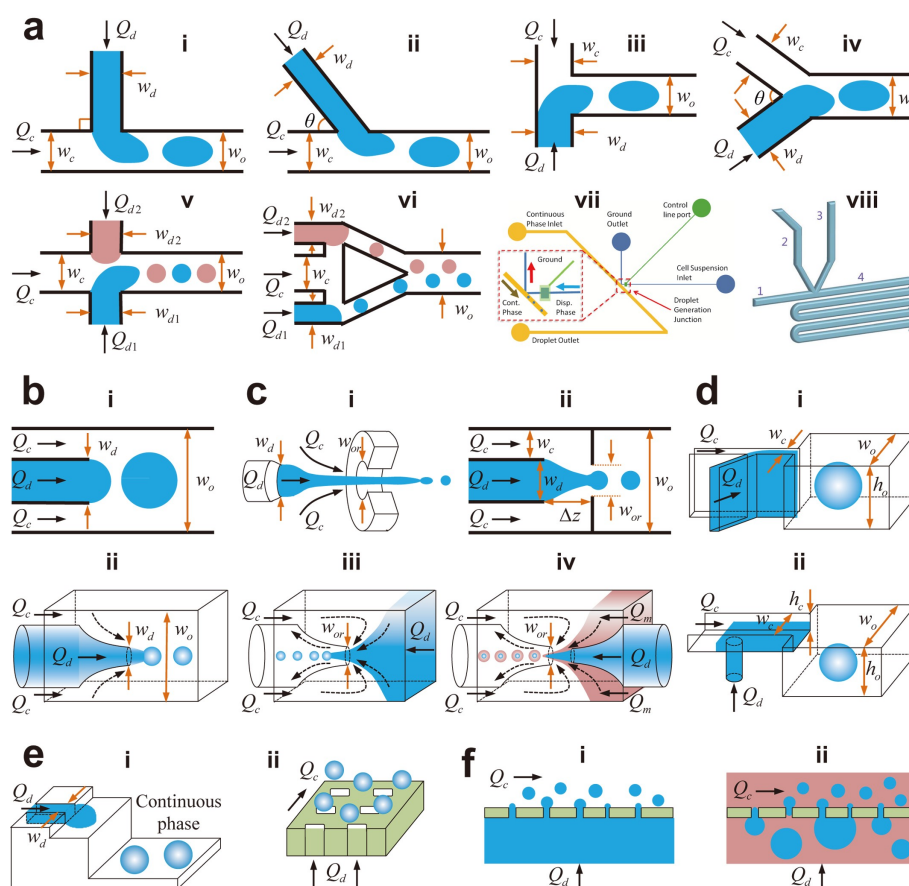


Fig. 2 Schematic of various microfluidic device geometries (not to scale). (a) Cross-flow. (i) “T-junction” where the continuous and dispersed phase fluids meet perpendicularly ($\theta = 90^\circ$). (ii) “T-junction” in which the two fluids intersect at an angle θ ($0^\circ < \theta < 90^\circ$). (iii) “Head-on” geometry ($\theta = 180^\circ$). (iv) Y-shaped junction with intersection angle of θ ($0^\circ < \theta < 180^\circ$). (v) Double T-junction with droplet pairs generated at the same location. (vi) Double T-junction that produces droplet pairs at separated parallel T-junctions. (vii) “K-junction”. (viii) “V-junction”. (b) Co-flow. (i) Quasi-2D planar co-flow. (ii) 3D co-flow. (c) Flow-focusing. (i) Axisymmetric flow-focusing geometry. (ii) Planner flow-focusing geometry. (iii) Microcapillary flow-focusing device. (iv) Microcapillary device combining co-flow and flow-focusing geometries. (d) Step emulsification. (i) Horizontal step. (ii) Vertical step. The inlet channel has a high aspect ratio, and the reservoir is wider and deeper, with an abrupt topographic step in between the inlet channel and the reservoir. (e) Microchannel emulsification. (i) Grooved-type microchannel. (ii) Straight-through microchannel. (f) Membrane emulsification. (i) Direct membrane emulsification. (ii) Premix membrane emulsification. Q , w , h , and Δz denote the volumetric flow rate, channel width, channel height, and horizontal distance from the end of the dispersed microchannel to the orifice entrance, respectively. For planar devices, the channel height h is uniform. In the case of the geometry with a circular cross-section, w represents the channel diameter. The subscripts “c”, “d”, “o”, “or” stand for the continuous phase, dispersed phase, outlet channel, and orifice, respectively. (a,vii) is reprinted with permission from ref. 48. Copyright 2012, American Institute of Physics. (a,viii) is reprinted with permission from ref. 50. Copyright 2015, Royal Society of Chemistry.

3.1 Cross-flow

The cross-flow geometry refers to the one where dispersed and continuous phase fluids meet at an angle θ ($0^\circ < \theta \leq 180^\circ$; Fig. 2a)²⁴. In microfluidics, the cross-flow structure is frequently implemented as a T-junction, where the dispersed and continuous fluids are fed orthogonally (Fig. 2a,i). The T-junction device was firstly reported by Thorsen *et al.*³³ to produce monodisperse water droplets in oil surroundings for generating ordered dynamic patterns in pressure controlled laminar flows. This geometry is widely used because of its simplicity and ability to produce monodisperse droplets. The coefficient of variation (CV, defined as the ratio of standard deviation to the mean of the droplet radius) of droplets is usually less than 2% in T-junctions³⁴. Contrary to the typical operation conditions shown in Fig. 2a,i where continuous phase is introduced from the main channel and dispersed phase is fed from the side channel, the opposite was also proposed in T-junctions by controlling wetting properties of channel walls^{22, 35}. Besides functioning as a droplet generator, T-junctions have recently been applied in designing microvalve³⁶ and microactuator³⁷ as well.

Variations in T-junction have been studied. A slightly different T-junction with arbitrary angle θ (Fig. 2a,ii)³⁸ was found to influence the process of droplet breakup³⁹ and resultant droplets⁴⁰. Another variation from the typical T-junction is the “head-on device”⁴¹, where the dispersed and continuous fluids are fed into the two straight channels from opposite directions, as shown in Fig. 2a,iii. Droplet formation in head-on devices, rich as the flow behaviour was found to be⁴², has similar features to that in typical T-junctions. Furthermore, if the two fluids are not injected oppositely but at a crossing angle θ , the head-on device reduces into a Y-shaped junction⁴³ (Fig. 2a,iv).

A single T-junction with above-mentioned simple structure cannot meet special requirements in some pragmatic applications. In performing chemical reactions via merging two droplet microreactors that contain different reagents⁴⁴, or indexing the targeted droplet by the addition of a droplet marker⁴⁵, for example, it is desired to generate droplet pairs in microchannels. A double T-junction (Fig. 2a,v) was designed to create stable alternating droplet pairs by Zheng *et al.*⁴⁵, where two dispersed streams pinch off at the same junction alternately. Droplets can also be generated separately at different locations in the main channel, and then flushed into a

wider channel to initiate fusion⁴⁶. In addition, a parallel dual T-junction that produces droplet pairs in two parallelized T-junctions (Fig. 2a,vi) was proposed to tune the size ratio of droplet pairs⁴⁷. In biological and chemical assays, on-demand droplet generation with prescribed size, composition and frequency is usually required. To achieve this goal, modified T-junctions such as K-junction^{48,49} (Fig. 2a,vii) and V-junction^{50,51} (Fig. 2a,viii) were introduced. For mass production of droplets, the proposed strategies include the integration of parallel T-junction droplet generators⁵², and splitting primary droplets subsequently into daughter droplets^{46,53}.

3.2 Co-flow

In co-flow geometry, the dispersed and continuous phase fluids meet in parallel streams²⁴ (Fig. 2b). In liquid-liquid systems, the co-flow microfluidic device was firstly introduced by Umbanhowar *et al.*⁵⁴. Co-flow configurations can either be quasi-two-dimensional (2D) planar²⁴ (Fig. 2b,i) or three-dimensional (3D) coaxial⁵⁵ (Fig. 2b,ii). The former can be fabricated by standard soft lithographic methods⁵⁶, while the latter is often made by inserting a tapered cylindrical glass capillary^{54,55,57} into a rectangular microchannel or a square glass capillary. The polydispersity of the resulting droplets in co-flow geometries depends on the flow conditions: droplets generated in dripping mode are highly monodisperse with $CV < 3\%$ ⁵⁴, while those generated in jetting mode are polydisperse¹⁵. Usually, droplet sizes are larger than the dimension of dispersed microchannel w_d (Fig. 2b) when droplets are produced in dripping or widening jetting mode¹⁵. Exceptions include the continuous phase flows at very high speeds so that a narrowing jet¹⁵ and even tip-streaming regime^{58,59} occur.

3.3 Flow-focusing

To produce smaller droplets, it is recommended to use flow-focusing devices, where the two immiscible fluids are hydrodynamically focused and thereafter in elongating flows when passing through a contraction (Fig. 2c). Similar to co-flow geometries, flow-focusing microfluidic devices can also be categorized into two types: 3D axisymmetric^{26,60-62} (Fig. 2c,i) and quasi-2D planar^{24,63} (Fig. 2c,ii). Gañán-Calvo was the first to produce monodisperse micron-sized droplets in a surrounding gas stream²⁶ and mono-sized microbubbles in a focused liquid stream⁶⁰ using the axisymmetric flow-focusing configuration as shown in Fig. 2c,i. Later in 2003, Anna *et al.*⁶³, for the first time, translated the planar flow-focusing geometry (Fig. 2c,ii) into a liquid-liquid microfluidic system. Compared to planar flow-focusing devices, 3D axisymmetric flow-focusing devices avoid issues like wetting of channel walls by the dispersed phase⁶¹, therefore producing monodisperse droplets ($CV < 5\%$) with higher throughputs⁶². A branch of 3D flow-focusing microfluidic devices is the microcapillary device⁵⁵ (Fig. 2c,iii), where the continuous and dispersed fluids are supplied into the device in opposite directions and meet at the entrance of the narrow orifice, then focusing downstream through the orifice and generating uniform droplets. In combination with co-flow geometry, the microcapillary device (Fig. 2c,iv) is able to generate monodisperse multiple emulsions in a single step^{55,64}.

3.4 Step emulsification

In contrast to the aforementioned microfluidic devices that use shear forces to break off fluids in generating droplets, other geometries generate uniform droplets by variations of channel confinement through which a sharp change in capillary pressure occurs and droplet pinch-off is driven by interfacial tension. Among them, a generic structure is the step emulsification geometry where the stabilized co-flow in the high aspect ratio inlet channel gives its way into droplets upon reaching an abrupt geometric step towards the

released channel confinement in the reservoir²³ (Fig. 2d). Two groups of step emulsification are identified based on the geometric feature of the step⁶⁵: horizontal step (Fig. 2d,i) and vertical step (Fig. 2d,ii). To the best of our knowledge, Chan *et al.*⁶⁶ were the first in using a vertical step change in microfluidic device to generate nanoliter droplets for the synthesis of CdSe nanocrystals. Then the geometry of step emulsification was carefully characterized by Priest *et al.*⁶⁷, who produced highly monodisperse droplets ($CV < 1.5\%$) and gel emulsions with high volume fraction of the dispersed phase. More recently, femtolitre droplets produced by step emulsification were used to perform PCR⁶⁸ or solidified to design colloidal building blocks⁶⁹. In vertical step emulsification (Fig. 2d,ii), the resultant smallest droplet has its diameter about two to three times the height of the inlet channel h_c , such that submicron droplets could be generated via scaling h_c down to several hundred nanometers^{70,71}.

Step emulsification takes several advantages. Firstly, droplet generation in step emulsification is less sensitive to flow rate or pressure fluctuations, resulting in probably the highest monodispersity of droplets ($CV < 1.5\%$ reported by Priest *et al.*⁶⁷, and $CV \sim 1\%$ reported by Malloggi *et al.*⁷⁰). Secondly, droplets with unaffected size can be generated in a wide range of flow rates by keeping flow rate ratio ϕ constant, while the production rate can be tuned over two orders of magnitude⁶⁷. Thirdly, taking advantage of producing foam-like emulsions in situ, step emulsification has been employed to study the assembly of microdroplets into 3D clusters^{72,73}. Finally, step emulsification is of the easiness in parallelizing multiple droplet production channels for high throughput, because only the flow of dispersed phase needs to be precisely controlled⁷⁴. For example, pairs of droplets with different contents can be produced by duplicating the dispersed inlet channels⁷⁵, and numbering the dispersed channels up to 16⁷⁶ and 252⁷⁷ were also achieved.

Variations in step emulsification have been proposed. One is using a gradual transition of channel height⁷⁸ instead of a sudden change from the inlet channel to the reservoir. The generated monodisperse droplet ($CV \sim 0.1\%$) is forced away by the gradient of surface energy. The size of droplets mainly depends on the channel geometry and is weakly influenced by flow rates and fluid properties. On-demand chemical reactions can be achieved by guiding droplets with local variations in channel height. In addition, high-throughput generation of droplets was demonstrated by parallelizing 256 parallel nozzles. Another variation is the emulsification method of edge-based droplet generation (EDGE) proposed by van Dijke *et al.*^{74,79,80}. In this device, the dispersed phase is introduced into a shallow but rather wide quasi-2D plateau, and then breaks into droplets at the edge of the plateau adjacent to which a deeper channel feeds the continuous cross flow to flush away the produced droplets. In a regular and organized manner, many sites of droplet generation can occur at the edge of the same plateau⁷⁴. Both experiments and simulations showed that the distance between droplet generation sites depends on the height of the plateau and applied pressure⁸⁰. A parallelization of up to 196 EDGE units on a single chip was presented to generate quasi-monodisperse droplets ($CV < 10\%$) at large production frequency (around 300 kHz)⁷⁹. In addition, Dutka *et al.*⁸¹ recently incorporated a constriction and bypasses into the dispersed inlet channel of step emulsification geometry. In this geometry, monodisperse droplets were produced by injecting only the dispersed phase fluid into the device. The size of droplets depended weakly on the injection rate, therefore increasing the stability of droplet generation against flow rate fluctuations. Furthermore, pump-free step emulsification configurations that incorporate centrifugal⁸²⁻⁸⁴ and magnetic⁸⁵ controls have been exploited, which will be detailed in section 5 of active controls.

3.5 Microchannel emulsification

In microchannel emulsification, the dispersed phase is forced through the inlet channel and then fragments into droplets at the end of the channel. Microchannel emulsification can be divided into two groups^{86, 87}: grooved-type microchannel (Fig. 2e,i) and straight-through microchannel (Fig. 2e,ii). To achieve high throughput, microchannel arrays are usually implemented⁸⁸. The grooved-type microchannel arrays were first fabricated by Kikuchi *et al.*⁸⁹ in 1989, in which the continuous fluid can either be static (dead-end)⁹⁰ or cross-flow^{91, 92}. For the sake of higher rate of droplet production, cross-flow modules are better than dead-end ones, because the former enables more microchannels incorporated on one chip and the continuous flow sweeps away the generated droplets, leaving more space for latter droplets. Compared with grooved-type microchannel arrays, straight-through modules possess higher throughput because of the better utilization of chip surface⁸⁶. In straight-through microchannel arrays, the rectangular microchannel (Fig. 2e,ii) is better than the circular shape, and can be either symmetric⁹³ or asymmetric⁹⁴⁻⁹⁶. However, the monodispersity of droplets is seriously constrained by channel dimensions. To produce uniform droplets ($CV < 2\%$), an aspect ratio of channel height to width must be larger than three⁹⁷.

3.6 Membrane emulsification

To produce fine emulsions at industrial scales, membrane emulsification was proposed in the early 1990s⁹⁸. With this technique, the dispersed phase is forced through a porous membrane after which droplets detach into the continuous phase (Fig. 2f). Apart from directly injecting a pure dispersed phase into the membrane, the so-called direct membrane emulsification⁹⁹ (Fig. 2f,i), pre-emulsions containing coarse droplets are homogenized by pressing the premix through a membrane¹⁰⁰ or a packed bed of uniform particles¹⁰¹, named as premix membrane emulsification (Fig. 2f,ii). There are many operation modes of applying shear stress on the membrane surface to facilitate uniform droplet formation with high throughputs, such as cross-flow⁹⁸, stirring^{102, 103}, rotating^{104, 105} and vibrating^{106, 107}. The most commonly-used membranes are Shirasu porous glass membrane⁹⁸ and micro-engineered membranes¹⁰⁸. Others include ceramic¹⁰⁹, metallic^{110, 111} and polymeric^{112, 113} membranes. Membrane emulsification enables quasi-monodisperse droplets formation ($CV \sim 20\%$)¹¹⁴. The droplet size is affected by characteristics of the membrane (pore morphology, pore size distribution, porosity, spatial arrangement of the pores, surface wettability and charge, to name a few), material properties (viscosity of the two fluids, surfactant type and concentration), and dynamic parameters (transmembrane flux and pressure, shear stress at the membrane surface)¹¹⁵. Both analytical^{116, 117} and computational¹¹⁸⁻¹²⁰ models have been developed to predict droplet size in membrane emulsification.

In addition to the above geometries, some other structures have been introduced. For example, Amstad *et al.*¹²¹ proposed a microfluidic post-array device to process the crude emulsion into a fine one, which is potentially beneficial for industrial applications due to its high throughput. Other devices enable on-site formation of droplets with predetermined volume by using micro-patterned channels^{122, 123}, where a long microchannel connects large numbers of pre-defined microchambers, either in series¹²² or in parallel¹²³, as sites for droplet generation.

4. Passive droplet generation

Droplet generation is of prime importance for various microfluidic applications. In passive method, microfluidic two-phase flow is controlled by either syringe pumps that supply constant flow rates Q_d and Q_c or pressure regulators^{124, 125} and gravity-based pressure units⁸¹ that set stable source pressures P_d and P_c without additional

energy input. During droplet formation, energy introduced from the syringe pumps or pressure controllers is partially converted into interfacial energy and thus facilitates the destabilization of liquid-liquid interface, whereby discrete droplet shedding from the dispersed phase²³. Despite the complexity of channel geometry, droplet generation can be categorized into several groups, due to the similar physical processes and underlying mechanisms involved. We, in the following subsection 4.1, identify and characterize five fundamental modes and one transitional mode of droplet generation in most commonly-used microfluidic cross-flow, co-flow, and flow-focusing configurations, as well as droplet generation in step emulsification structures. Influence of surfactants on droplet generation is also included. Then, in subsection 4.2, we discuss the fluid instabilities of two-phase microflows and relate each of the breakup modes to these instabilities.

4.1 Breakup modes

Five fundamental modes of droplet generation have been observed in shear-based droplet generators: squeezing¹²⁶, dripping⁵⁴, jetting⁵⁷, tip-streaming¹²⁷ and tip-multi-breaking¹⁷. The first three have been observed in cross-flow, co-flow, and flow-focusing geometries; the last two have not been reported in cross-flow yet (Figs. 3a-e). In principle, droplets generated with squeezing mode is larger than the channel dimension and highly monodisperse (Fig. 3a), with dripping is somehow smaller than channel dimension and monodisperse (Fig. 3b), with jetting is polydisperse (Fig. 3c), with tip-streaming can be as small as submicrometer-scale and monodisperse (Fig. 3d), and with tip-multi-breaking are in droplet sequences with polydisperse but geometric-progression size distribution (Fig. 3e). Transitions between different modes can be achieved by changing dispersed and/or continuous phase capillary numbers (Fig. 3f)¹⁰. For example, the squeezing-dripping transitional regime occurs at an intermediate continuous-phase capillary number between squeezing and dripping. Similarly, different modes of droplet generation are identified in step emulsification: shear-induced generation, step-regime and jet-regime. In this sub-section, our focus is mainly on the process of droplet generation, mechanism, the condition of occurrence and characteristics of each mode (Table 3), as well as effects of surfactants.

4.1.1 Squeezing. Squeezing mode occurs at low capillary number, such as $Ca_c < O(10^{-2})$ in T-junctions¹²⁸⁻¹³⁰, where viscous stress gives its way to the confinement of channel walls. In this case, the dispersed-fluid protrusion obstructs the junction region in cross-flow^{129, 131, 132}, co-flow^{16, 133}, and flow-focusing^{14, 126} structures as it grows, thereby restricting the continuous flow around the developing protrusion (Fig. 3a). Consequently, a pressure gradient in continuous fluid across the enlarging droplet is built up as the gap between the liquid interface and the opposing channel wall decreases. Once the pressure gradient is sufficiently large to overcome the pressure inside the dispersed droplet, the interface is “squeezed” to deform and necks into a droplet. The generated droplet is confined by channel walls, adopting a plug-like rather than a spherical shape. Constrained by the channel geometry, squeezing is also referred to as “geometry-controlled” mode in the literature^{17, 41, 127, 134}.

The pressure rise in the continuous fluid is responsible for the dispersed fluid breakup in squeezing mode. During the breakup, the pressure profile in the dispersed phase fluctuates rather than remains constant. This mechanism was firstly unveiled by Sivasamy *et al.*¹³⁵ in a numerical study, and later confirmed experimentally by using Laplace sensors to measure pressures in T-junction¹³⁶ and flow-focusing¹³⁷ microfluidic devices, respectively. As the emerging droplet blocks the junction region and then necks, the continuous-phase pressure fluctuates in the way of first increasing and then decreasing, in anti-phase with the dispersed-fluid pressure variation^{136, 137}. The amplitude of pressure fluctuations diminishes with the

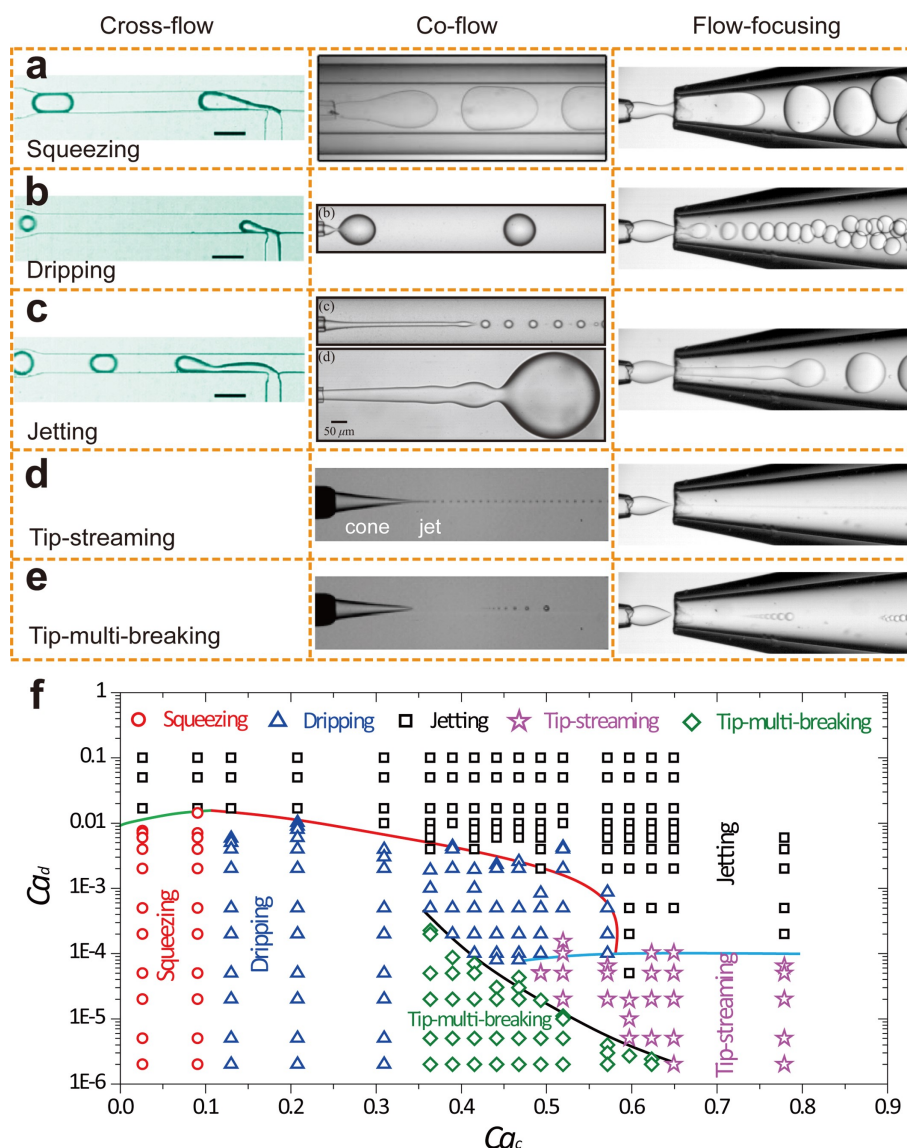


Fig. 3 Images of droplet generation with different modes in cross-flow, co-flow and microcapillary flow-focusing geometries. (a) Squeezing mode. (b) Dripping mode. (c) Jetting mode. The upper image in co-flow is a narrowing jet while the lower one is a widening jet. (d) Tip-streaming mode. (e) Tip-multi-breaking mode. Neither tip-streaming nor tip-multi-breaking modes have been reported in cross-flow geometry yet. (f) Phase diagram in (Ca_c, Ca_d) plane for various modes observed in microcapillary flow-focusing devices¹⁷. Cross-flow of (a), (b) and (c) are reprinted with permission from ref. 131. Copyright 2010, American Chemical Society. Co-flow of (a) is reprinted with permission from ref. 133. Copyright 2007, American Physical Society. Co-flows of (b) and (c) are reprinted with permission from ref. 15. Copyright 2007, American Physical Society. Co-flows of (d) and (e) are reprinted with permission from ref. 58. Copyright 2013, Cambridge University Press.

increasing of capillary number Ca_c .

In a plug-like shape, droplet generated with squeezing mode has its size characterized by the ratio of the plug length L_p to continuous phase channel width w_c (such as in T-junctions, Fig. 2a,i), L_p / w_c . To preliminarily determine the dimensionless length L_p / w_c , the process of droplet generation is usually divided into two stages in T-junctions^{129,138,139}. First, the dispersed-liquid tip fills and blocks the junction region with its length estimated as $L_1 = \varepsilon w_c$; then the dispersed droplet is squeezed by the continuous fluid and has the neck thinning rate proportional to the continuous fluid velocity $u_c = Q_c / (hw_c)$, with h being the channel height. During the second “squeezing” stage, the dispersed plug elongates due to the inflation of dispersed fluid at rate: $u_d = Q_d / (hw_d)$ (w_d is the dispersed channel width, Fig. 2a,i). Accordingly, the length contribution from the second stage is about $L_2 = u_d(d / u_c) = dw_c Q_d / (w_d Q_c) = d\phi w_c / w_d$, where d is the initial neck diameter. Finally, the droplet length $L_p = L_1 + L_2 = \varepsilon w_c + d\phi w_c / w_d$. Therefore, L_p / w_c is approximated as,

$$L_p / w_c = \varepsilon + \delta\phi, \quad (12)$$

where ε and δ ($\delta = d / w_d$) are fitting parameters of order unity. Depending on channel geometry, various groups of ε and δ have been reported from both experimental^{129,138,140} and numerical¹⁴¹⁻¹⁴³ works. Eq. (12) suggests that, in squeezing mode, the droplet length is determined by device geometry and flow rate ratio, independent of physical properties of fluids.

The continuous phase is not utterly blocked during droplet formation. This has been confirmed by the microscopic particle image velocimetry (μ -PIV) measurements of 3D flow fields around the emerging interface^{144,145}, especially in rectangular microchannels, where the leakage of continuous fluid from the gutters between liquid interface and channel corners is significant. By considering the leakage, van Steijn *et al.*¹⁴⁶ proposed a closed-form expression to predict droplet and bubble volume V without fitting parameters in T-junctions,

Table 3 Characteristics of droplet breakup modes.

Breakup mode	Condition for occurrence	Characteristics	Specification	Ref.
Squeezing	$Ca_c < O(10^{-2})$	$L_p / w_c = \varepsilon + \delta\varphi$	Eq. (12). Plug length is only determined by flow rate ratio and channel dimension. ε and δ are of order unity	129
		$V / (hw_c^2) = V_{fill} / (hw_c^2) + \zeta\varphi$	Eq. (13). Expression for predicting droplet volume in T-junctions by taking leakage into account.	146
		$V / (hw_c^2) = \alpha_f + \zeta_f\varphi$	Eq. (14). Prediction of droplet volume in flow-focusing geometries.	147
Dripping-squeezing transition	$Ca_c \approx O(10^{-2})$	$V / (hw_c^2) = \bar{b}^2 + \Gamma\varphi$ with $(1 - \bar{b})^3 = \bar{b}Ca_c$	Eqs. (20) and (21). Prediction of droplet volume in transitional regime.	152
		$V / (hw_c^2) = \alpha_{lag} + \alpha_{fill} + \zeta_{neck}\varphi$	Eq. (23). Prediction of droplet volume based on lagging, filling, and necking stages of droplet formation.	159
		$fre_{drop}^* = \varphi / (\alpha_{lag} + \alpha_{fill} + \zeta_{neck}\varphi)$	Eq. (24). Prediction of droplet generation frequency based on lagging, filling, and necking stages of droplet formation.	159
Dripping	$O(10^{-2}) < Ca_c < O(1)$, and $We_d < O(1)$	$\bar{D}^3 - (1 + \frac{1}{3Ca_c})\bar{D}^2 - \bar{D}\Lambda + \Lambda = 0$	Eq. (15). In co-flow geometry, droplet size is the solution of the third-order polynomial. Only shear forces and interfacial tension are considered.	54
		$\bar{D} = a + \frac{b}{3Ca_c}$	Solution of Eq. (15) when $\varphi \ll 1$. a and b are fitting parameters. Eq. (16) is recovered when $a = b = 1$.	54
		$Ca_{cri}\bar{D}^4 + (\bar{Q}_c + \bar{Q}_d)\bar{D}^3 -$ $(Ca_{cri}\bar{w}_{or}^2 + \bar{Q}_c + \bar{Q}_d)\bar{D}^2 -$ $\bar{Q}_d\bar{w}_{or}^2\bar{D} + \bar{Q}_d\bar{w}_{or}^2 = 0$	Eq. (17). In flow-focusing geometry, droplet size is the solution of the fourth-order polynomial.	153
		$\bar{D} = \frac{-\bar{Q}_c + \sqrt{\bar{Q}_c^2 + 4Ca_{cri}(Ca_{cri}\bar{w}_{or}^2 + \bar{Q}_c)}}{2Ca_{cri}}$	Eq. (18). Solution of Eq. (17) when $\varphi \ll 1$.	153
		$\bar{D}^4 - 2\bar{D}^3 + \frac{\beta}{2}\bar{D}^2 + \frac{\bar{D}_d^2}{2C_\lambda Ca_c} = 0$	Eq. (19). In cross-flow geometry, droplet size is the solution of the fourth-order polynomial.	155
Jetting	$Ca_c + We_d \geq O(1)$	$d_{jet} / w_o = \left(\frac{(1 + \varphi\lambda)^{1/2} - 1 - \varphi}{\lambda - 2 - \varphi}\right)^{1/2}$	Eq. (25), jet diameter in narrowing jetting regime.	161
		$d_{jet} / w_o = (\varphi / 2)^{1/2}$	Eq. (26), jet diameter in narrowing jetting regime when $\varphi \ll 1$.	161
		$L_{jet} = C_{jet} \frac{\eta_d}{\gamma} \frac{8}{\pi w_o} \left(\frac{Q_d Q_c}{2}\right)^{1/2}$	Eq. (27), jet length in narrowing jetting regime.	162
		$D / d_{jet} = [3\pi / (2k^*)]^{1/3}$	Ratio of droplet diameter and jet diameter in narrowing jetting regime.	59
		$D / w_o = \sqrt[6]{\frac{9\pi^2}{32}} (k^*)^{-1/3} \varphi^{1/2}$	Eq. (28), expression for dimensionless droplet diameter in narrowing jetting regime.	59
		$d_{jet} = \sqrt[3]{\frac{4Q_d}{\pi\omega_{t,max}}}$	Eq. (29), jet diameter in widening jetting regime.	59
		$D \sim (Q_d / u_c)^{1/2}$	Eq. (30), scaling of droplet diameter in widening jetting regime.	15
		$\frac{D}{w_d} = \frac{1}{w_d} \left(\frac{6Q_d d_{jet}}{k^* u_{jet}}\right)^{1/3}$	Eq. (31), unified expression for dimensionless droplet diameter in both narrowing and widening jetting regimes.	59
Tip-streaming	$Re_c \ll 1$, $Re_d \ll 1$, $Ca_c > Ca_{cri}$, and $\varphi \ll 1$	$D / w_o = \sqrt[6]{\frac{9\pi^2}{32}} (k^*)^{-1/3} \varphi^{1/2}$	Eq. (28), expression for dimensionless droplet size. Tip-streaming can be regarded as narrowing jetting with $\varphi \ll 1$.	59
Tip-multi-breaking	$\varphi < \varphi_{min} \ll 1$, and $Ca_c < Ca_{cri}$	$R_i = R_i \chi^{i-1}$, $i = 1, 2, \dots, n$	Eq. (32), geometrical progression of droplet size distribution in a decreasing manner.	17
Step-regime	$Ca_d w_c / h_c \lesssim 0.38$	$D / h_c = 3C \left[1 + \frac{1 + (\lambda\varphi)^{-1}}{w_c / h_c}\right]^{-1}$	Eq. (33). Prediction of droplet size in step emulsification geometry.	197

$$V / (hw_c^2) = V_{fill} / (hw_c^2) + \zeta_T\varphi, \quad (13)$$

where V_{fill} is the droplet/bubble volume at the end of the first “filling” stage depending only on channel geometry, and ζ_T depends on both geometry and the leakage. The analytical expressions for V_{fill} and ζ_T can be found in ref. 146. Eq. (13) is consistent with Eq. (12) in that the droplet volume V can be roughly estimated to be $V = L_p w_c h$.

Similar to Eq. (13), Chen *et al.*¹⁴⁷ theoretically predicted the volume of droplet in flow-focusing geometries by accurately

reconstructing the 3D shape of the droplet and determining the pressure drop along the droplet via force balance analysis:

$$V / (hw_c^2) = \alpha_f + \zeta_f\varphi, \quad (14)$$

where α_f denotes the dispersed flow into the droplet during the filling stage, and ζ_f accounts for the continuous flow into the junction during the necking stage. Model of Eq. (14) includes the influence of device geometry, viscosity ratio, flow rate ratio, and continuous-phase capillary number. Based on the prediction of droplet volume, the droplet generation frequency and droplet spacing were also

predicted, with a relative error of $\pm 10\%$.

4.1.2 Dripping. As capillary number Ca_c increases, the mode transforms from squeezing to dripping¹³⁰, where viscous forces that drag the interface to rupture dominate over interfacial tension effects that stabilize the emerging droplet against breakup. In dripping mode, breakup occurs right at the dispersed nozzle in cross-flow and co-flow geometries, or at the focusing orifice in flow-focusing structure (Fig. 3b). As a consequence of large viscous shear force, the dispersed fluid breaks up before the growing droplet obstructs microchannels, so that the droplet retains a spherical shape with its size smaller than the channel dimension (Fig. 3b). Given a constant viscous stress, highly monodisperse droplets can be generated.

The droplet diameter D can be scaled, by simply balancing the viscous stress of magnitude $\eta_c u_c / L$ and capillary pressure γ / D ^{15,33}, to be $D / L \sim Ca_c^{-1}$, where L is the hydraulic length of the nozzle with $L = 2hw_d / (h + w_d)$ in 2D geometries and $L = w_d$ in 3D co-flow and flow-focusing structures. In the squeezing-dripping transitional regime, both shear stress and squeezing pressure play the role in determining the droplet size^{130, 143, 148–150}; the scaling power of Ca_c deviates from -1 to other values such as -0.25 numerically observed by van der Graaf *et al.*¹³⁹, ranging from -0.3 to -0.2 by Liu and Zhang^{143, 151}, -1/3 by Christopher *et al.*¹⁵², -0.34 by Lan *et al.*¹⁵⁰, and -0.4 by De Menech *et al.*¹³⁰. Here, we focus our attention only on the purely shear-dominant dripping regime.

To predict the absolute droplet size, both the viscous shear and interfacial tension forces should be analytically determined. In a 3D co-flow geometry, viscous force can be approximated by a modified Stokes formula after considering the shielding effect of the dispersed nozzle: $F_v = 3\pi\eta_c(D - w_d)(u_c - u_d)$ where $u_d = Q_d / (\pi D^2)$. Interfacial tension force can be expressed by $F_\gamma = \pi w_d \gamma$. By equating F_v and F_γ , Umbanhowar *et al.*⁵⁴ found the dimensionless droplet diameter $\bar{D} = D / w_d$ as a solution to the third-order polynomial,

$$\bar{D}^3 - (1 + \frac{1}{3Ca_c})\bar{D}^2 - \bar{D}\Lambda + \Lambda = 0, \quad (15)$$

with $\Lambda = Q_d / (\pi w_d^2 u_c)$. Commonly, Q_d is negligible compared with Q_c in dripping regime. Therefore, dropping the last two terms with Λ at the left-hand side of Eq. (15) arrives at

$$\bar{D} = 1 + \frac{1}{3Ca_c}. \quad (16)$$

Additional fitting parameters are introduced to collapse the experimental data onto the analytical prediction of Eq. (16). Depending on different values of nozzle diameter w_d , $\bar{D} = a + b / (3Ca_c)$ takes different fitting values of a and b .

In microcapillary flow-focusing devices (Fig. 3b), the similar deduction was carried out to predict the droplet size. Erb *et al.*¹⁵³ defined the same forms of viscous and interfacial tension forces as used by Umbanhowar *et al.*⁵⁴, and approximated $u_d = 4Q_d / (\pi D^2)$ and $u_c = 4Q_c / [\pi(w_{or}^2 - D^2)]$ where w_{or} is the diameter of the focusing orifice. By assuming the condition for droplet rupture to be $F_v / F_\gamma = Ca_{cri}$, Erb *et al.*¹⁵³ pointed out that droplet diameter $\bar{D}$ can be predicted by solving the following polynomial:

$$Ca_{cri}\bar{D}^4 + (\bar{Q}_c + \bar{Q}_d)\bar{D}^3 - (Ca_{cri}\bar{w}_{or}^2 + \bar{Q}_c + \bar{Q}_d)\bar{D}^2 - \bar{Q}_d\bar{w}_{or}^2\bar{D} + \bar{Q}_d\bar{w}_{or}^2 = 0, \quad (17)$$

where $\bar{Q}_c = Q_c / Q_0$, $\bar{Q}_d = Q_d / Q_0$ and $\bar{w}_{or} = w_{or} / w_d$ with $Q_0 = \pi w_d^2 \gamma / (12\eta_c)$. Similarly, when $Q_d \ll Q_c$, all terms in Eq. (17) involving $\bar{Q}_d$ can be neglected. An analytical solution can, therefore, be found by solving the quadratic equation $Ca_{cri}\bar{D}^2 + \bar{Q}_c\bar{D} - Ca_{cri}\bar{w}_{or}^2 - \bar{Q}_c = 0$ to determine the physically meaningful root:

$$\bar{D} = \frac{-\bar{Q}_c + \sqrt{\bar{Q}_c^2 + 4Ca_{cri}(Ca_{cri}\bar{w}_{or}^2 + \bar{Q}_c)}}{2Ca_{cri}}. \quad (18)$$

Note that $\bar{Q}_c = Q_c / Q_0 = 12\eta_c Q_c / (\pi w_d^2 \gamma)$ can be understood as the continuous-phase capillary number Ca_c at the dispersed nozzle. A single universal value of the fitting parameter $Ca_{cri} = 0.1$ is applicable to a wide range of flow rates, channel dimensions and viscosity ratios¹⁵³.

Eq. (18) can be generalized to predict droplet diameter in flow-focusing geometries with an arbitrary shape of collection microchannel of cross-sectional area Ar . In this situation, $\bar{Q}_c$ and $\bar{w}_{or}$ are redefined to be $\bar{Q}_c = 3\eta_c Q_c / (w_d^2 \gamma)$ and $\bar{w}_{or} = \sqrt{Ar / w_d^2}$, respectively. Liu *et al.*¹⁵⁴ applied the generalized Eq. (18) to successfully predict the absolute droplet size in a flow-focusing geometry with a square collection microchannel. Considering the shielding effect in the expression of F_v , the droplet diameter D is assumed to be larger than the nozzle diameter w_d in Eqs. (15)–(18), e.g. $\bar{D} > 1$. This assumption is however not always valid in flow-focusing configurations when Q_c is sufficiently large (see Fig. 3b, flow-focusing). Moreover, the influence of viscosity ratio on droplet diameter is neither considered in Eqs. (15)–(18). Nevertheless, Eqs. (15)–(18) provide a fast and efficient preliminary determination of the drop size once a two-phase microflow is known.

In cross-flow geometries, viscous and interfacial tension force balance is still applicable to the prediction of droplet size. Husny and Cooper-White¹⁵⁵ estimated $F_\gamma = \pi w_d^2 \gamma / D$, and, by taking both dispersed and continuous phase viscosities into account, $F_v = C_\lambda \pi \eta_c D \{2u_c[1 - (D_c - D)^2 / D_c^2] - u_d\}$ in a cross-flow geometry, where $C_\lambda = (3 + 2\lambda) / (1 + \lambda)$, $u_c = Q_c / (w_d h)$ and $u_d = \beta u_c$, with λ being the viscosity ratio, $D_c = 2w_c h / (w_c + h)$ the hydraulic diameter of the continuous-phase microchannel, and β the ratio of the droplet velocity u_d to the average velocity of continuous flow u_c ($0 < \beta < 2$). By equating F_γ and F_v , a fourth-order polynomial that determines the dimensionless droplet diameter $\bar{D} = D / D_c$ reads²¹,

$$\bar{D}^4 - 2\bar{D}^3 + \frac{\beta}{2}\bar{D}^2 + \frac{\bar{D}_d^2}{2C_\lambda Ca_c} = 0, \quad (19)$$

where $\bar{D}_d = w_d / D_c$, and $Ca_c = \eta_c Q_c / (w_c h \gamma)$. Husny and Cooper-White¹⁵⁵ used a fitting value of $\beta = 0.6$ to collapse all the experimental data onto the predicted curve, which, within the accuracy of measurements, represented good agreement for all investigated capillary number Ca_c and viscosity ratio λ . The proposed model Eq. (19) suggests a decreasing of droplet diameter D as viscosity ratio λ increases at a fixed capillary number Ca_c , which was however not observed¹⁵⁵. Instead, the measured droplet diameter appears to depend weakly on the viscosity ratio. In all cases, droplet generation frequency fre_{drop} can then be easily obtained based on mass conservation $fre_{drop} = 6Q_d / (\pi D^3)$ provided that droplet diameter D is predicted.

4.1.3 Squeezing-dripping transition. A transitional regime occurs between squeezing and dripping, where the emerging droplet blocks the continuous-phase fluid partially and temporarily during its formation. Due to the partial blocking, the breakup dynamics is dominated by both viscous shear and squeezing pressure. The pressure drop across the droplet is studied by direct measurements of the pressure difference across the droplet^{130, 136, 137, 149} and the distribution of local flow field¹⁴⁸. The condition for the existence of transitional regime was found to be $Ca_c \approx O(10^{-2})$. For examples, De Menech *et al.*¹³⁰ reported a value of $Ca_c = 0.015$ for squeezing-dripping transition in their numerical simulations; Liu and Zhang¹⁴³ identified a transition criterion of $Ca_c = 0.018$; Xu *et al.*¹³⁸ experimentally observed the transitional regime with $0.002 < Ca_c < 0.01$.

Arising from the co-existence of shear stress and squeezing pressure, the droplet size depends on both capillary number Ca_c and flow rate ratio ϕ in transitional regime^{138, 151, 156}, different from squeezing where droplet size is purely determined by flow rate ratio, and dripping where droplet size solely depends on the capillary number. Semi-empirical models were developed to predict droplet size by fitting experimental and numerical data respectively with scaling law of $L_p / w_c = \varepsilon + \delta \phi^n Ca_c^m$ ¹³⁸ or $L_p / w_c = (\varepsilon + \delta \phi) Ca_c^m$ ¹⁵¹ to determine the empirical parameters ε , δ , n , and m . Applying the balance of capillarity force ($F_\gamma \approx \gamma h$), viscous force ($F_v \approx \eta_c Q_c b / (w_c - b)^2$ with b being the depth of droplet penetration into the continuous channel) and squeezing pressure ($F_p \approx \eta_c Q_c b^2 / (w_c - b)^3$), Christopher *et al.*¹⁵² derived an analytical formula for the prediction of droplet volume in transitional regime,

$$V / (hw_c^2) = \bar{b}^2 + \Gamma \phi, \quad (20)$$

where $\Gamma = w_d / w_c$ is the channel width ratio, and $\bar{b} = b / w_c$ is expressed as

$$(1 - \bar{b})^3 = \bar{b} Ca_c, \quad (21)$$

Eq. (20) includes the influence of channel geometry Γ , in addition to flow rate ratio and capillary number, on droplet volume. This approximate model agrees reasonably well with numerical^{141, 157} and experimental¹⁵⁸ results, but underpredicts the droplet volume and does not consider the influence of viscosity ratio¹⁵².

A more robust model was later developed by Glawdel *et al.*¹⁵⁹, who experimentally identified three sequential stages of droplet formation in the transitional regime: lagging, filling, and necking¹⁶⁰. The last two stages are similar to those observed in squeezing regime¹²⁹, while the initial lagging stage is featured by a receding back of the interface into the injection channel of dispersed fluid after the previous droplet pinch-off. The retraction distance of the interface L_{lag} is evaluated by a balance between the viscocapillary velocity in a two-fluid system³¹ ($u_{v-c} \sim \gamma \lambda^{1/2} / \eta_d$) and the injection velocity of dispersed phase ($u_d \sim Q_d / (hw_d)$), $L_{lag} / w_c \sim u_{v-c} / u_d$:

$$L_{lag} / w_c \sim \lambda^{1/2} / Ca_d. \quad (22)$$

The lagging affects not only the downstream droplet spacing, but also the droplet volume¹⁶⁰. Taking the three stages into consideration, Glawdel *et al.*¹⁵⁹ developed a predictive model in T-junctions

$$V / (hw_c^2) = \alpha_{lag} + \alpha_{fill} + \zeta_{neck} \phi, \quad (23)$$

$$fre_{drop}^* = \phi / (\alpha_{lag} + \alpha_{fill} + \zeta_{neck} \phi), \quad (24)$$

by a detailed analysis of the 3D droplet shape and a force balance similar to that used by Garstecki *et al.*¹²⁹ and Christopher *et al.*¹⁵². In Eqs. (23) and (24), α_{lag} , α_{fill} , and ζ_{neck} account for contributions from the lagging, filling and necking, respectively, with their mathematical expressions available in ref. 159; $fre_{drop}^* = fre_{drop} / hw_c^2 / Q_c$ is the dimensionless droplet generation frequency. This model has considered the influences of channel width ratio, channel aspect ratio (h / w_c), flow rate ratio, viscosity ratio, and capillary number on droplet volume and droplet generation frequency, and is with a $\pm 10\%$ error compared with experimental data¹⁵⁹.

4.1.4 Jetting. By increasing either the continuous-fluid flow rate Q_c or dispersed-fluid flow rate Q_d , dripping to jetting transition can occur¹⁵, where an extended liquid jet emits from the dispersed channel, and ultimately breaks up into droplets at the end of the jet due to Rayleigh-Plateau instability (Fig. 3c). Subjected to capillary perturbations, jetting results in more polydisperse droplets compared with squeezing and dripping. For co-flowing liquid streams, jetting occurs when the sum of viscous forces exerted by continuous-fluid and dispersed-fluid inertia overcomes interfacial tension forces, summarized by $Ca_c + We_d \geq O(1)$ ¹⁵. According to this force balance argument, jetting in co-flow geometries can be divided into two groups: narrowing jets that occur when $Ca_c \geq O(1)$ (upper image in

Fig. 3c, co-flow) and widening jets that arise from $We_d \geq O(1)$ (lower image in Fig. 3c, co-flow). In narrowing jetting regime, the viscous-drag force overwhelms the capillary force, thus the jet stretching downstream with a thinner diameter. The widening jet is, however, the opposite, where the dispersed-fluid velocity u_d is larger than the continuous-fluid velocity u_c ; consequently, the jet is decelerated as moving downstream due to the velocity difference induced viscous shear at the interface, which renders the shape of jets widening.

The characteristics of jetting mode include the jet diameter d_{jet} , intact jet length L_{jet} (the length from the dispersed nozzle to the end of the jet), and droplet diameter D . In narrowing jetting regime, the jet diameter d_{jet} can be approximated as the diameter of an unperturbed jet with constant shape governed by Stokes equations for both continuous and dispersed fluids at low Reynolds number: $\eta_s \nabla^2 \mathbf{u}_s = \nabla p_s$. By solving Stokes equations, the jet diameter is found to be¹⁶¹,

$$d_{jet} / w_o = ((1 + \phi \lambda)^{1/2} - 1 - \phi) / (\lambda - 2 - \phi)^{1/2}, \quad (25)$$

where w_o is the diameter of the outlet channel with a circular cross-section. In case that flow rate ratio $\phi \ll 1$, Eq. (25) reduces into an asymptotic scaling law,

$$d_{jet} / w_o = (\phi / 2)^{1/2}, \quad (26)$$

that has been validated both experimentally^{15, 162} and numerically¹⁶¹.

The intact jet length L_{jet} in narrowing jetting is simply estimated as the product of the jet velocity $u_d = 4Q_d / (\pi d_{jet}^2) = 8Q_c / (\pi w_o^2)$ (by applying Eq. (26)) and the viscous-capillary time $t_{v-c} = d_{jet} \eta_d / \gamma$ ¹⁶². Cubaud & Mason¹⁶², therefore, derived the formula for the jet length L_{jet} to be

$$L_{jet} = C_{jet} \frac{\eta_d}{\gamma} \frac{8}{\pi w_o} \left(\frac{Q_d Q_c}{2} \right)^{1/2}, \quad (27)$$

where C_{jet} is a fitting parameter. The jet length L_{jet} thus depends on flow rates, device geometry, and fluid properties. By experimentally determining C_{jet} , the prediction shows satisfactory agreement with the measurements¹⁶². Cubaud and Mason¹⁶² proposed that viscosity ratio barely affects the jet length within their experimental operation.

The droplet size in narrowing jetting regime can be determined by mass balance. Subjected to Rayleigh-Plateau instability, the jet breaks up due to the growth of capillary perturbations with the maximum growth rate and wavelength λ_w . To estimate droplet diameter D , the volume of the droplet is assumed to be equal to the volume of the jet with wavelength λ_w : $\pi D^3 / 6 = \pi d_{jet}^2 \lambda_w / 4$. Since $\lambda_w = \pi d_{jet} / k^*$, with k^* being the dimensionless wave number ($k^* = k d_{jet} / 2$, with k being the wave number), the droplet diameter D is scaled with d_{jet} as $D / d_{jet} = [3\pi / (2k^*)]^{1/3}$. If d_{jet} is replaced by D , Eq. (26) then leads to

$$D / w_o = g(k^*) \phi^{1/2}, \quad (28)$$

where the prefactor $g(k^*) = (9\pi^2 / 32)^{1/6} (k^*)^{-1/3}$. The dimensionless wavenumber k^* depends on the viscosity ratio, and can be obtained by solving the dispersion relation deduced by Tomotika¹⁶³. For instance, the calculation gives $\lambda_w = 5.48 d_{jet}$ for viscosity ratio $\lambda = 0.1$, which results in $D / d_{jet} \approx 2$ and $g(k^*) \approx 1.43$, consistent with experimental results¹⁵. Cubaud & Mason¹⁶² obtained from experiments that $D / d_{jet} \approx 2.9 \pm 0.3$ for viscosity ratio in the range of 22.4 to 439, while the corresponding calculation of k^* gives $2.4 < D / d_{jet} < 3.1$, in a good agreement with experiments.

The characteristics of widening jetting differ from those of narrowing jetting as a result of different dominant effects. With a widening shape, the jet velocity is not fully developed prior to droplet pinch-off. Therefore, the steady-state Stokes equations fail to predict the jet diameter d_{jet} . Castro-Hernández *et al.*⁵⁹ proposed a formula to find d_{jet} by using linear stability analysis:

$$d_{jet} = \sqrt[3]{\frac{4Q_d}{\pi\omega_{i,max}}}, \quad (29)$$

where $\omega_{i,max}$ is the maximum growth rate of the capillary perturbations and can be determined from Tomotika's dispersion relation¹⁶³. Eq. (29) should be solved iteratively since $\omega_{i,max}$ also depends on jet diameter d_{jet} ⁵⁹. The droplet volume V is approximated as $V = \tau Q_d$, with τ being the droplet formation time and roughly equal to the advection time of the growing droplet $\tau \sim D/u_c$ ¹⁵. Consequently, the droplet diameter in widening jetting regime scales as

$$D \sim (Q_d/u_c)^{1/2}. \quad (30)$$

This simple scaling argument has been validated by experiments for viscosity ratio $\lambda = 0.1$. However, for $\lambda \gg 1$, its accuracy might lose¹⁵.

Despite the distinct dominant effects, droplet sizes in both narrowing and widening jetting regimes can be unified⁵⁹,

$$\frac{D}{w_d} = \frac{1}{w_d} \left(\frac{6Q_d d_{jet}}{k^* u_{jet}} \right)^{1/3}, \quad (31)$$

where u_{jet} is the advection velocity of the jet at its most downstream position. The difference between narrowing and widening jetting regimes lies in the different expressions for the jet diameter d_{jet} , and advection velocity of the jet u_{jet} . For narrowing jets, d_{jet} is given by Eq. (26), or more simply estimated as $d_{jet} = [4Q_d/(\pi u_c)]^{1/2}$; $u_{jet} = u_c$. In this case, Eq. (31) recovers the same scaling of Eq. (28): $D \sim \phi^{1/2}$. In widening jetting regime, d_{jet} is found by solving Eq. (29), and u_{jet} is determined experimentally. The dimensionless wavenumber k^* , corresponding to the fastest growth of the perturbations, can be determined by solving Tomotika's dispersion relation¹⁶³ for both narrowing and widening jets. Eq. (31) also includes the influence of viscosity ratio λ on the droplet diameter, as k^* depends on λ . Eq. (31) was validated for both narrowing and widening regimes under various Q_c , Q_d and λ with the relative errors being $\pm 30\%$ ⁵⁹.

4.1.5 Tip-streaming. In the pioneering work by Taylor¹⁶⁴, an isolated droplet was stretched to have pointed tips when subjected to an extensional or shear flow. Fine jets or tiny daughter droplets were seen to issue from the pointed tips, a phenomenon now commonly known as tip-streaming¹⁶⁵. Similarly, in microfluidics, droplets are generated in tip-streaming mode (Fig. 3d) when the dispersed liquid tip is deformed into a conical shape from whose apex a fine jet emanates and subsequently breaks into exceptionally small droplets^{11, 12, 16, 58, 166-170}. The thin jet is two to three orders of magnitude smaller than the injection nozzle in radius. In such a way, micrometer and even submicron-sized droplets can be readily produced^{11, 12, 166, 168-170}.

Two distinct mechanisms account for the occurrence of tip-streaming: surfactant-mediated and surfactant-free tip-streaming. The former was proposed by De Bruijn¹⁷¹, who ascribed tip-streaming phenomenon to the presence of surfactants on the liquid-liquid interface, and identified an optimal range of surfactant concentration for tip-streaming to occur. These findings were consistent with later experiments in both simple shear¹⁷² and plane hyperbolic¹⁷³ flows. Subsequent numerical simulations further unveiled the mechanism of surfactant-mediated tip-streaming from free drops¹⁷⁴ and bubbles¹⁷⁵ in extensional flows. More recently, Anna *et al.* reported the occurrence of tip-streaming, which was called thread formation by them too, in planar microfluidic flow-focusing devices^{24, 127, 176, 177}. They argued that the coupling between surfactant transport and bulk flow plays a critical role²⁴, and developed a semi-analytical model to predict the conditions for the occurrence of tip-streaming¹⁷⁷. Tip-streaming phenomenon was also observed in the presence of interfacial chemical reaction that produces a surfactant¹⁷⁸. On the other hand, no evidence suggested the possibility of surfactant-free tip-streaming until Zhang¹⁷⁹

predicted the elegant steady-state solution for the vanishing thin spout ejecting from the perfectly conical tip when the capillary number of external straining fluid is above a critical value. Then more conclusive proof was provided by simulations¹⁶ and experiments¹⁶⁶ which demonstrated surfactant-free tip-streaming in a double flow-focusing arrangement. In what follows, we focus only on the type of surfactant-free tip-streaming, which is referred to as tip-streaming for short hereafter.

Tip-streaming is featured by the steady con-jet structure (Fig. 3d) of the dispersed fluid, which very much resembles the Taylor cone in charged liquids during electrospraying^{180, 181}. In the cone region, the continuous viscous stress induces the recirculating flow pattern within the dispersed phase¹⁶. Then a speeding up of the dispersed flow is observed upon approaching the apex of the cone, which is the cone-jet transition region¹⁶. Sufficiently far downstream from the apex, the dispersed flow is nearly plug flow and the jet takes a cylindrical shape with an almost constant diameter of d_{jet} ¹⁶. The cone-jet transition is affected by both the flow process and fluid properties. For larger continuous-phase capillary number Ca_c and smaller viscosity ratio λ , the transition region becomes less slender^{168, 182}, and the local jet slope decreases. Experimental observations reveal that the cone-jet transition is stable for liquid-liquid systems, but not for gas-liquid^{168, 169, 182}. This can be qualitatively understood as that the very small viscosity ratio λ for gas-liquid systems requires a quite low jet slope¹⁸², which is, however, difficult to achieve in experiments. The cone-jet geometrical profile has been deduced by Zhang¹⁷⁹ in the extreme limit $Q_d \rightarrow 0$, and by Castro-Hernández *et al.*¹⁶⁸ as a function of Ca_c , ϕ , λ and device geometry in co-flow configurations. The resultant droplet size can also be precisely predicted by Eq. (28)^{59, 169}, because tip-streaming can be regarded as narrowing jetting with extremely thin jet diameter^{169, 182}. However, an accurate prediction for the intact jet length has yet to be developed.

Several conditions should be met for the occurrence of tip-streaming. Note that device geometry has a significant effect. No tip-streaming has been observed yet in cross-flow geometry. According to Tseng and Prosperetti¹⁸³, tip-streaming occurs when the local streamlines converge near the interface. This is easy to achieve in co-flow and flow-focusing, but difficult in cross-flow configuration (Fig. 3d). To generate tip-streaming, no flow separation should exist between the continuous flow and the drop interface, which requires creeping flow condition, $Re_c \ll 1$. Then, to stretch the dispersed fluid into a conical shape, the continuous-fluid viscous stress must overcome the capillary pressure at the injection nozzle, therefore requiring $Ca_c > Ca_{cri}$ (Ca_{cri} is a critical constant of the order unity). For the extremely thin jet to issue from the cone, $\phi \ll 1$ should also be maintained. The final condition $Re_d \ll 1$ ensures that the average dispersed velocity equals continuous velocity in the thin jet region due to large momentum diffusion, and the fine jet is thus almost a cylinder far downstream from the cone. These four conditions are sufficient to generate the tip-streaming¹⁶⁹.

Some effort has been devoted to exploring the value of Ca_{cri} . Suryo and Basaran¹⁶ found that the critical capillary number Ca_{cri} is a function of viscosity ratio λ and flow rate ratio ϕ . Gañán-Calvo *et al.*^{166, 182} then theoretically determined Ca_{cri} as a function of λ by using a spatiotemporal stability analysis, and concluded that once the jet surface velocity u_{js} is above a critical value, $u_{js} > \gamma(\eta_d\eta_c)^{1/2}Ca_{cri}$, the jet can be made arbitrarily thin (down to the continuum limit) without transition to dripping as the dispersed flow rate Q_d vanishes. More recently, Gordillo *et al.*⁵⁸ found that Ca_{cri} is less sensitive to flow rate ratio ϕ than to viscosity ratio λ , who also applied a global stability analysis to obtain the λ -dependent Ca_{cri} , below which the cone-jet structure is unsteady and the resultant droplet size is not uniform⁵⁸.

4.1.6 Tip-multi-breaking. Unlike the aforementioned four

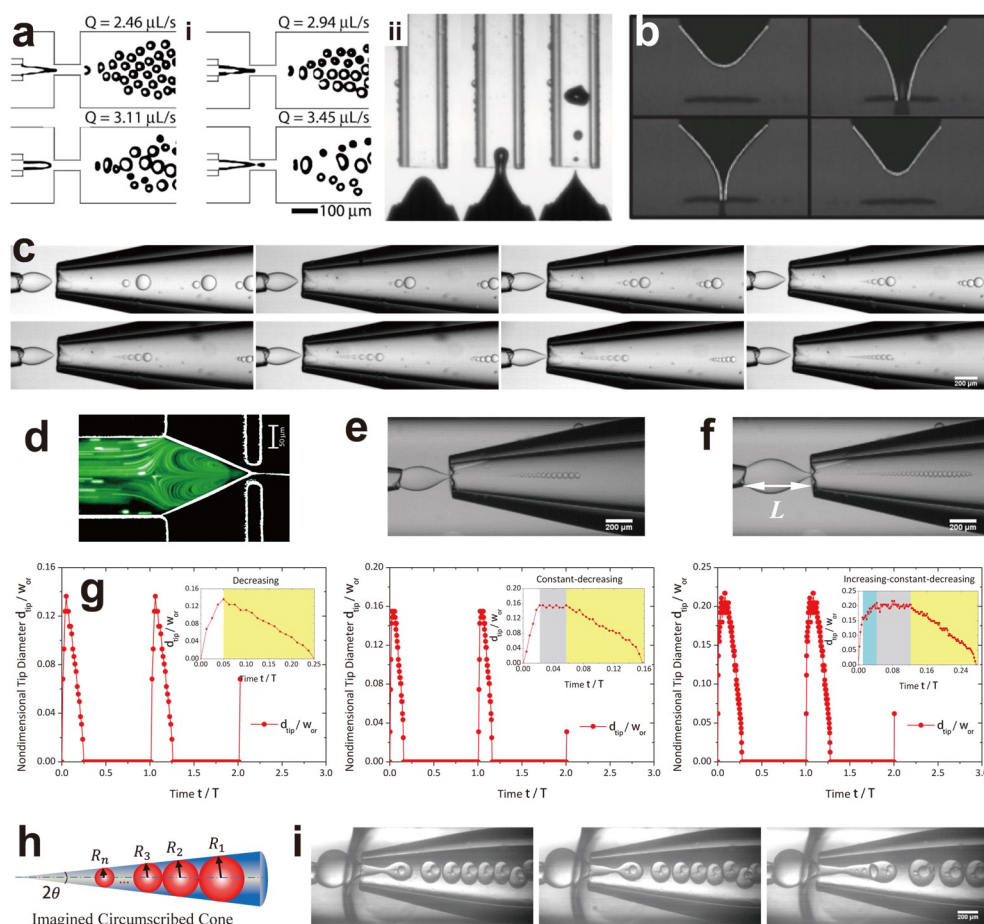


Fig. 4 Tip-multi-breaking mode in microfluidic devices. (a) Unsteady gas meniscus during bubble generation in gas-liquid microsystem. (i) Period-1, period-2, period-4 and period-3 gas thread behaviour with increasing continuous flow rate, respectively (from left to right, up to down). (ii) Irregular bubble generation when the continuous-fluid velocity u_c is lower than the critical threshold. (b) Unsteady liquid meniscus when a liquid jet is focused by a gas stream. (c) Tip-multi-breaking mode in liquid-liquid microfluidic flows by which droplets of decreasing size are generated in periodic sequences. The number of droplets in one sequence increases as the continuous-phase capillary number increases (from left to right, up to down). (d) Recirculation cell in the dispersed meniscus. When Q_d is smaller than the minimum flow rate Q_{min} , the recirculation cell is large and penetrates into the dispersed nozzle, in which case tip-multi-breaking takes place. (e) Droplet size distribution in the form of “constant-decreasing”. (f) Droplet size distribution in the form of “increasing-constant-decreasing”. (g) Temporal evolution of the tip diameter at the focusing orifice in three situations. (h) Schematic of the decreasing droplet size distribution in a geometrical progression. (i) Tip-multi-breaking mode in three-phase microfluidics. Snapshots from left to right are one four-droplet train, one five-droplet train and two droplet trains encapsulated in one middle phase droplet, respectively. (a,i) is adapted with permission from ref. 30. Copyright 2005, American Physical Society. (a,ii) is reprinted with permission from ref. 187. Copyright 2015, Cambridge University Press. (b) is reprinted with permission from ref. 189. Copyright 2011, American Physical Society. (d) is adapted with permission from ref. 127. Copyright 2006, American Institute of Physics.

breakup modes (Figs. 2a-d), in which continuous droplet streams are generated, tip-multi-breaking^{17, 184} generates droplets intermittently in periodic sequences (Fig. 3e). Similar to tip-streaming, a conical dispersed tip is observed in tip-multi-breaking mode (Figs. 2d-e). However, the dispersed cone structure is unsteady in tip-multi-breaking, in sharp contrast with the steady cone-jet in tip-streaming. Discrete droplets are ejected from the apex of the conical tip as a result of Rayleigh-Plateau instability. Oscillation of the conical meniscus then cuts off the continuous generation of droplet stream, and leads to periodic trains of droplets with non-uniform size distribution (Fig. 3e). Note that the periodic generation of droplet-train is synchronized with the oscillation of the tip.

The unsteady feature of dispersed tip has been widely observed in gas-liquid, liquid-gas and liquid-liquid systems. Garstecki *et al.*^{30, 126} found that the gas-liquid system destabilized and transitioned from period-1 (dripping, one bubble generation in one period) state to higher-order periodic or chaotic behaviour (as exemplified by period-2, period-3 and period-4 behaviour in Fig. 4a,i) when Q_c is above some critical value. The transition was ascribed to the inertial-

dominated dynamics of the gas-liquid interface³⁰. Similar phenomena were identified in a multi-orifice system¹⁸⁵ and later in a multi-section flow-focusing junction¹⁸⁶. More recently, Evangelio *et al.*¹⁸⁷ observed the unsteady gas meniscus when the continuous liquid velocity u_c is lower than the threshold u^* (Fig. 4a,ii). In parallel, for a liquid jet focused by a gas, the liquid meniscus is unsteady when the dispersed flow Q_d is below the minimum value Q_{min} ^{181, 188-191} (Fig. 4b). These observations suggest that dynamics of both dispersed and continuous fluids plays a role in rendering the meniscus unsteady. In a liquid-liquid co-flow system, Gordillo *et al.*⁵⁸ determined the critical continuous-phase capillary number Ca_{crit} as a function of viscosity ratio λ , which sets an upper limit for the occurrence of the unsteady conical tip. Zhu *et al.*¹⁷ reported that the number of the droplet in one droplet-train increases with the increasing continuous-phase capillary number Ca_c in a microcapillary flow-focusing device (Fig. 4c). Attributed to the feature that the unsteady dispersed fluid tip breaks off repeatedly into multiple droplets in one cycle, Zhu *et al.*¹⁷ named this mode of droplet breakup as tip-multi-breaking.

Since the unsteady meniscus gives rise to the generation of droplet-train, the condition of destabilizing the dispersed meniscus determines the occurrence of tip-multi-breaking mode. While higher-order periodic and chaotic behavior of the gas thread was ascribed to the dominant inertial effects³⁰, the mechanism responsible for the unsteady liquid meniscus seems to be different. From the perspective of energy balance, once the kinetic energy of the liquid meniscus is overwhelmed by the viscous dissipation and/or free surface energy, the steady emission of continuous droplets are halted¹⁸¹. Corresponding to the above three energy terms, the dispersed-phase inertial, viscous and capillary forces scale respectively as $\sim \rho_d u_d^2$, $\sim \eta_d u_d / d_{tip}$ and $\sim \gamma / d_{tip}$, where d_{tip} is the liquid tip diameter at the focusing orifice. With the assumption of $Q_d \sim u_d d_{tip}^2$, and $Q_d / Q_c \approx (d_{tip} / w_{or})^2$ at the focusing orifice (w_{or} is the diameter of the focusing orifice), inertia dominated by viscous forces ($\rho_d u_d^2 < \eta_d u_d / d_{tip}$) gives $(Q_d Q_c)^{1/2} < \eta_d / (\rho_d w_{or})$, and inertia overwhelmed by capillary forces ($\rho_d u_d^2 < \gamma / d_{tip}$) leads to $Q_d^{1/2} Q_c^{3/2} < \gamma / (\rho_d w_{or}^3)$ for the dispersed meniscus to be unsteady. Both of the criteria suggest a minimum dispersed flow rate Q_{min} below which a steady liquid tip cannot be formed when Q_c is fixed.

In liquid-liquid systems, the Q_{min} yields the condition that $\phi < \phi_{min} \ll 1$ for the occurrence of tip-multi-breaking¹⁷. Actually, for the dispersed fluid with low viscosity and sufficiently small flow rate, the flow displays a recirculation cell whose size increases as Q_d decreases (Fig. 4d). When $Q_d \leq Q_{min}$, the recirculation cell is large and penetrates into the injection capillary, consuming a large amount of momentum due to viscous friction between the liquid and the solid wall^{181, 188, 190}. Simultaneously, as Q_d decreases, the decreasing d_{tip} finally make the capillary pressure too large to be overcome in generating a new free surface for droplet production, thus destabilizing the dispersed tip¹⁸¹.

Similarly, the continuous fluid also reserves a critical flow rate Q_{cri} lower than which the shear force exerted by the continuous phase is not sufficient to hold a steady liquid meniscus. To have unsteady meniscus at the focusing orifice, the shear stress $\sim \eta_c u_c / w_{or}$ should be dominated by the capillary pressure $\sim \gamma / w_{or}$. This criterion then leads to the existence of a critical capillary number Ca_{cri} which sets an upper limit for the liquid tip to be unsteady, $Ca_c < Ca_{cri}$. In liquid-liquid systems, Ca_{cri} is a function of viscosity ratio, and weakly depends on flow rate ratio ϕ ⁵⁸. Apparently, Ca_c cannot be infinitely small, otherwise, the system would transfer into other modes, such as dripping and squeezing¹⁷. It is worth noting that the two conditions, $\phi < \phi_{min} \ll 1$ and $Ca_c < Ca_{cri}$, should be met simultaneously for the occurrence of tip-multi-breaking mode in liquid-liquid systems. Experiments revealed that ϕ_{min} , to some extent, depends on and decreases with Ca_c ¹⁷.

The droplet size distribution in tip-multi-breaking mode is highly synchronized with the dynamics of the tip evolution at the focusing orifice. The normalized droplet diameter D / d_{tip} scales as $D / d_{tip} \sim Ca_c^{-1}$ in tip-multi-breaking mode¹⁸⁴. Depending on device geometry, the droplet size distribution can be either monotonically decreasing (Fig. 4c), or firstly keeping constant and then decreasing ("constant-decreasing" in Fig. 4e), or firstly increasing, followed by keeping constant and finally decreasing ("increasing-constant-decreasing" in Fig. 4f), as the spacing L (Fig. 4f) between the left injection and right focusing orifices increases¹⁸⁴. Figure 4g shows the temporal evolution of d_{tip} / w_{or} for the three cases of droplet size distribution. The very apparent oscillation of the liquid tip is observed with period T . During one cycle of oscillation, d_{tip} can undergo any one of the three evolutionary dynamics (insets in Fig. 4g), and simultaneously droplets are ejected from the tip in sequences. This verifies that the non-uniform droplet and unique droplet size distribution is a consequence of the d_{tip} oscillation. It is worth clarifying that an initial very-fast-growing stage of d_{tip} in all three cases (non-shadow areas in insets of Fig. 4g) is due to the penetration of the liquid tip into the focusing orifice, and no droplet

is generated at this stage.

The size distribution in one droplet-train is predictable. Since the thinning rate of d_{tip} is nearly linear for all the cases (yellow region in insets of Fig. 4g), the decreasing size of droplet, very interestingly, is in the distribution of geometric progression¹⁷,

$$R_i = R_1 \chi^{i-1}, \quad i = 1, 2, \dots, n, \quad (32)$$

where R_i is the radius of the i^{th} droplet (Fig. 4h), and $\chi < 1$ is the common ratio. The common ratio χ can be determined either geometrically (Fig. 4h) as $\chi = (1 - \sin\theta) / (1 + \sin\theta)$ or dynamically once the linear thinning rate of d_{tip} and material properties are known¹⁷. As a first-order approximation, χ has a correlation with the number of droplets n in the form of $1 - \chi^{n-1} \approx \chi$, for the droplet train with decreasing size in Fig. 4c¹⁷. Eq. (32) is deduced for, but is not limited to, the decreasing size distribution. It can also be applied to the increasing size distribution in Fig. 4f, because the increasing rate of d_{tip} is also almost constant (inset in Fig. 4g for "increasing-constant-decreasing"). More effort should be devoted to further characterize the tip-multi-breaking mode, for instance, the condition that distinguishes the three types of droplet size distribution in Figs. 3c, e and f, respectively.

To date, studies on the unsteady dispersed meniscus have mainly focused on its hydrodynamic interest, but underestimate its potential in applications. This is because droplets generated in tip-multi-breaking mode are non-uniform, while monodisperse droplets are usually in favor for most applications. Nevertheless, some applications prefer non-uniform droplets. For example, in multi-volume droplet digital PCR (MV-dPCR)¹³, droplets with various volumes provide simultaneous measurements of a sample at different copies per droplet. Compared with single-volume digital PCR, MV-dPCR retains higher detection reproducibility, wider dynamic range, and better resolution while reducing the total number of droplets/wells required for the measurements^{13, 192, 193}. Tip-multi-breaking mode can, therefore, fit well in the application of MV-dPCR, due to its nature in producing identical droplet trains with a tunable number of droplets and predictable size distribution. Besides, the droplet trains can be successfully encapsulated into middle-phase droplets (Fig. 4i), which would be beneficial for chemical and biomedical applications, for example, in the design of chemically-communicating droplet networks¹⁹⁴.

4.1.7 Droplet generation in step emulsification. In droplet generators with the step emulsification, the inlet channel with strong confinement suppresses the interfacial instability, thus stabilizing the interface between the two immiscible liquids^{195, 196}. On reaching the reservoir, the relaxed confinement destabilizes the interface and triggers droplet pinch-off as a result of minimizing interfacial energy. Three regimes of droplet generation have been identified with the increasing flow rate ratio ϕ ⁶⁷: breakup at the junction where the two fluids meet, step-regime where fluid filaments break up upon reaching the step, and jet-regime where droplets are generated downstream the step. The first is the shear-induced breakup^{66, 67}, while the latter two are unique to the step emulsification and thus attract recent studies^{65, 70, 197-199}.

Dangla *et al.*¹⁹⁸ investigated the step-regime in a vertical step. They ascribed the breakup to the quasi-static balance between the curvature of the dispersed thread in the inlet channel and that of the droplet protrusion in the reservoir, based on which a lower bound of droplet size is determined. Recently, the closed-form expression for predicting droplet size in step-regime is theoretically derived by Li *et al.*¹⁹⁷,

$$D / h_c = 3C \left[1 + \frac{1 + (\lambda\phi)^{-1}}{w_c / h_c} \right]^{-1}, \quad (33)$$

where C is the fitting parameter of a value about 2, and w_c and h_c are the width and height of the inlet channel, respectively. Eq. (33) suggests a constant droplet size by keeping the flow rate ratio ϕ

constant when the device geometry and material properties are fixed. As such, droplet generation frequency fre_{drop} ($fre_{drop} = 6Q_d / (\pi D^3)$) is proportional to the dispersed-phase capillary number Ca_d , $fre_{drop} \sim Ca_d$, when ϕ is constant⁶⁷.

With the increase of dispersed phase flow rate, the step-regime gives its way to the jet-regime at a critical dispersed-phase capillary number^{67, 199}. The condition for such a transition is experimentally determined to be¹⁹⁷

$$Ca_d \frac{w_c}{h_c} \approx 0.38. \quad (34)$$

In jet-regime, the dispersed thread shrinks in its width and results in a tongue-like tip upstream near the step, which was ascribed to the capillary focusing effect⁷⁰. The quasi-static shape of the tongue has been determined by numerical simulation¹⁹⁹ and simplified theoretical modeling¹⁹⁷, respectively, both of which were confirmed by experiments. Aside from single step-regime or jet-regime, the coexistence of different regimes on the same dispersed-phase filament was observed recently by Hein *et al.*⁶⁵, which implies a symmetry breaking in filament breakup even under symmetric flow conditions.

4.1.8 Effects of surfactants. Surfactants are routinely used in droplet microfluidics to reduce the interfacial tension and prevent the droplet coalescence²⁰⁰. In the presence of surfactants, the interfacial tension is determined by the competition between interfacial deformation and surfactant convection, diffusion and adsorption-desorption kinetics during droplet generation²⁴. A faster interfacial deformation and slower mass transfer process render the surface coverage of surfactants smaller, thereby interfacial tension larger. In microfluidic flows, convection enhances the mass transportation of surfactants²⁰¹⁻²⁰³. Owing to internal convection, the mass transport inside the droplet is faster than that outside²⁰³. Increasing surfactant concentration would increase the rate of mass transport²⁰⁴. Droplet generation is subjected to a constant interfacial tension provided that surfactant concentration is high enough^{202, 205}, or the droplet generation time t_{drop} is either much longer²⁰⁶ or much shorter²⁰⁷ than surfactant mass transfer time t_m , where surfactants are at saturated adsorption in the first case, at equilibrium adsorption in the second case, and nearly absent on the interface in the last case. Contrary to being constant, the interfacial tension is varying with time, known as dynamic interfacial tension, when the surfactant is at intermediate concentrations and the two timescales (t_{drop} and t_m) are comparable.

Variations of interfacial tension dramatically alter the dynamics of liquid thread pinch-off. During pinch-off, the presence of surfactants lowers the interfacial tension and may induce the depletion of surfactants from the pinch region²⁰⁸. The reduction in interfacial tension gives rise to a decrease in capillary pressure ($\sim \gamma / L$) that drives surfactants out of the pinching neck; surfactant depletion induces an interfacial tension gradient and a Marangoni stress counteracting the outflow. Considering the pinch-off of a surfactant-covered fluid thread at a small surfactant Peclet number, Pe (the parameter characterizing the relative strength of the surfactant convection over the diffusion), the interfacial tension starts at the equilibrium value and then increases due to surfactant depletion as pinch-off approaches²⁰⁹⁻²¹¹. Provided a strong Marangoni stress, a third stage of surface tension reduction can occur²¹¹, which was also observed for a two-fluid thread pinch-off in microchannels²¹⁰. The temporal variation of dynamic interfacial tension, therefore, changes the thinning rate of the pinching thread^{209, 210}. Besides, the thread shape near the pinch-off is also altered by surfactants²¹⁰⁻²¹², which induces variations in pinch-off location^{211, 212} and influences the size of satellite droplets²¹³.

Several models have been developed to predict the dynamic interfacial tension during droplet formation. The experimental measurements of dynamic interfacial tension are carried out by first

Table 4 Relationship between droplet breakup mode and hydrodynamic instabilities.

Droplet breakup mode	Interfacial instability	Capillary instability		
		Absolute instability	Convective instability	Global instability
Squeezing	✓			
Dripping		✓		
Widening jet ^a		✓		
Narrowing jet			✓	
Tip-streaming			✓	
Tip-multi-breaking ^b				✓

^a Fine balance of forces is required to generate absolute instability²²⁵.

^b Dripping, jetting, and tip-streaming modes are globally stable.

determining the relationship between interfacial tension and variables such as the droplet generation frequency²¹⁴, droplet size^{201-206, 215}, droplet deformation²¹⁶, and the pressure drop of the continuous phase^{217, 218}, and then measuring these variables. Semi-empirical models are developed by fitting the model to experimental data to determine the empirical parameters, such as those in refs. 201-206, 215, 217, 218. However, semi-empirical models are highly sensitive to the fluidic system, and the results are normally not transferable. On the other hand, Glawdel and Ren²¹⁶ developed a theoretical model to predict the dynamic interfacial tension during droplet generation in squeezing-dripping transitional regime in T-junctions, based on which droplet volume is well predicted with surfactants. Nevertheless, no experimental method is available for capturing the spatial distribution of surfactants on the interface, which is usually uneven during droplet formation²¹⁹.

4.2 Fluid instabilities in microfluidic biphasic flows

The fragmentation of fluid threads into droplets is initiated by fluid instabilities. The currently identified five fundamental modes of droplet formation are related to different types of instabilities, of which droplet pinch-off in squeezing mode is imposed by pressure buildup in the continuous flow, while dripping, jetting, tip-streaming and tip-multi-breaking are associated with capillary instability. Such a difference originates from the distinct degree of channel confinement between squeezing and the other four modes. In this subsection, we briefly summarize the characteristics of hydrodynamic instabilities for each of the five breakup modes in microfluidic biphasic flows (Table 4).

4.2.1 Interfacial instability of squeezing. Geometrical confinement dramatically suppresses capillary instability, thereby stabilizing the fluid-fluid interface during thread pinch-off in squeezing mode. Basically, the collapse of a gas thread surrounded by an outer fluid and confined in a microchannel undergoes two stages¹⁴: the neck, with the minimal width of w_m , shrinks at a constant rate, then followed by a nonlinear rapid collapse (Fig. 5a). In the first linear collapse stage, the shrinkage speed dw_m / dt is proportional to the continuous flow rate Q_c , but independent of the gas pressure P_d , liquid viscosity η_c , and interfacial tension γ . Besides, dw_m / dt is one to three orders of magnitude smaller than the speed of capillary waves. These observations signify that the collapse is not driven by surface tension. Instead, Garstecki *et al.*¹⁴ argued that the confined collapse proceeds a series of equilibria of minimal surface energy (Fig. 5b), in contrast to the non-equilibrium capillary instability. Similar linear collapse process was identified by other groups in flow-focusing and T-junction, as well^{32, 144}.

The mechanism responsible for the second nonlinear rapid collapsing stage is controversial. Dollet *et al.*³² argued that the onset of rapid collapse is initiated from capillary instability, coinciding with the detachment of the shrinking thread from the wall (Fig. 5c). Then the shrinkage of the gas thread is driven solely by liquid and gas inertia, with the minimal neck width scaling as $w_m \sim (t_c - t)^{1/3}$,

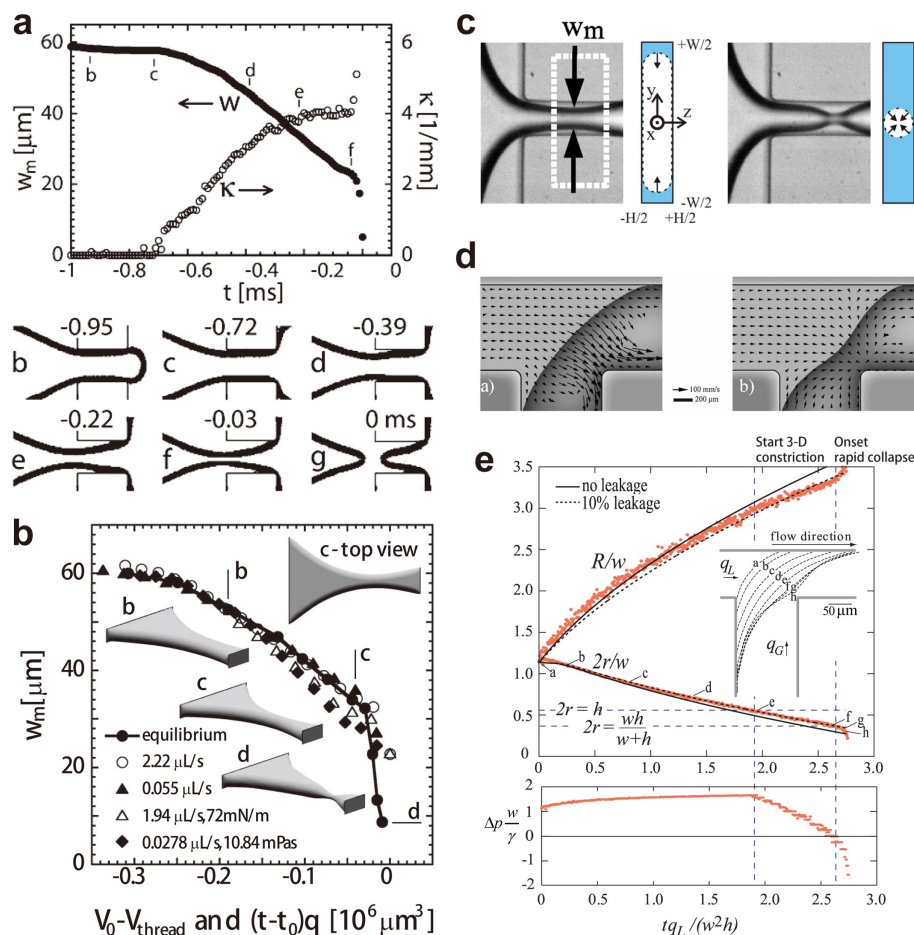


Fig. 5 Fluid instability of squeezing mode. (a) Upper diagram: evolution of the minimal width w_m and the axial curvature of the gas-liquid interface in a typical breakup event. Lower images: optical micrographs of the gas-liquid interface along the breakup trajectory. (b) Comparison of the minimal width w_m of the simulated (equilibrium) interface ($V_0 - V_{\text{thread}}$, solid dots connected by solid lines) and the experimentally measured minimal widths w_m ($(t - t_0)q$, unconnected symbols). Insets of “b–d” represent the equilibrium shape of the gas-liquid interface along the trajectory leading to final breakup. The interface shown in inset “d” (rapid collapse stage) is also in equilibrium. (c) The gas thread of minimal width w_m is first squeezed inwards from the sides by the surrounding liquid (2D collapse, left two graphs). The gas thread experiences finally a fast pinch-off, where it is squeezed radially (3D collapse, right two graphs). (d) In-plane velocity at $0.1h$ from the top wall. Left: before the rapid collapse, flow in the gutters runs towards the tip end. Right: the thread starting to collapse, liquid now runs from the tip to the neck through all four gutters. (e) Upper diagram: evolution of the radii of curvature r and R . Lower diagram: evolution of the pressure drop over the gutters, indicating that the collapse coincides with reversal of pressure drop in the gutters. Panels (a) and (b) are reprinted with permission from ref. 14. Copyright 2005, American Physical Society. Panel (c) is reprinted with permission from ref. 32. Copyright 2008, American Physical Society. Panels (d) and (e) are reprinted with permission from ref. 144. Copyright 2009, American Physical Society.

where t_c is the critical time for thread breakup. In contrast to the above explanation, comparisons of experimental and simulated static interfaces reveal a quasi-static pinch-off mechanism until the last stage of breakup (Fig. 5b)¹⁴. van Steijn *et al.*¹⁴⁴ later demonstrated that the reversal of liquid flow in corner gutters, from the end tip of the gas bubble to the shrinking neck (Fig. 5d), triggers the rapid collapse of gas thread. The flow reversal was attributed to the onset of an adverse liquid pressure drop over the gutters, which determines a critical thread radius r_{cri} for the onset to occur $r_{\text{cri}} = hw_c / [2(h + w_c)]$ in straight microchannels (Fig. 5e), incompatible with $r_{\text{cri}} = \min[w_c / 2, h / 2]$ as predicted by capillary instability. Combining all the evidence together, one concludes that the interfacial instability of a squeezing fluid thread does not originate from capillary instability. Consequently, highly monodisperse droplets can be produced in squeezing regime due to the elimination of satellite droplets that are almost inevitable for breakups governed by capillarity instability.

4.2.2 Capillary instability. Different from squeezing mechanism, dripping, jetting, tip-streaming and tip-multi-breaking are initiated by capillary instability. These four modes differ from each other in their distinct instability natures, of which jetting-dripping transition

is related to the transition from local convective to absolute instability, and tip-streaming to tip-multi-breaking transition corresponds to the transition from global stability to global instability. Here, terms “local” and “global” describe the stability/instability of the local velocity profile and of the entire flow field, respectively²²⁰. If growing disturbances are advected downstream from the source, the velocity profile is recognized as convective instability. If by contrast disturbances propagate both upstream and downstream and contaminate the entire domain when amplifying, the velocity profile is said to be absolute instability²²⁰. In this subsection, we briefly summarize the basic principles and concepts in hydrodynamic instability and their applications to biphasic flows in microfluidic devices. The reader is referred to Huerre and Monkewitz²²⁰ and references therein for a detailed review on the absolute/convective and local/global instability concepts.

A linear stability analysis²²¹ is usually performed to assess the local instability. In doing this, initial infinitesimal perturbations $\psi(z, y, t)$ that develop in space and time are applied to a given parallel basic flow, where z , y , and t represent streamwise, cross-stream and time coordinates, respectively. Perturbations are then decomposed

into elementary solutions $\psi(z, y, t) = \zeta(y; k)\exp[i(kz - \omega t)]$ after linearization around the base state, with complex wavenumber $k = k_r + ik_i$, complex frequency $\omega = \omega_r + i\omega_i$ and eigenfunction $\zeta(y; k)$ that describes the cross-stream distribution of perturbations. In the eigenvalue problem, non-trivial solutions for $\zeta(y; k)$ exist if and only if k and ω satisfy a dispersion relation in a manner:

$$D(k, \omega; S) = 0, \quad (35)$$

where S is the control parameter, for example, the dimensionless parameters discussed in Section 2. The following analysis is denoted as temporal stability analysis if k is real and ω is complex, as spatial stability analysis if k is complex and ω is real, and as spatiotemporal stability analysis if both k and ω are complex.

Typically, a temporal stability analysis is adopted to determine whether a system is stable or not by the following criterion:

$$\begin{aligned} \omega_{i, \max} > 0, & \text{ linearly unstable flow;} \\ \omega_{i, \max} < 0, & \text{ linearly stable flow,} \end{aligned} \quad (36)$$

where $\omega_{i, \max}$ is the maximum temporal growth rate determined by $\omega_{i, \max} = \omega_i(k_{\max})$ when $\partial\omega_i / \partial k(k_{\max}) = 0$. For example, if a co-flowing jet is stable in microfluidic devices, pinch-off will not occur, thus no generation of droplets. Conversely, if a jet is unstable, it ultimately breaks up into discrete droplets. As such, flows with dripping, jetting, tip-streaming and tip-multi-breaking are temporally unstable. For temporally unstable capillary jets, an additional spatiotemporal stability analysis is applied to distinguish convective from absolute instability. To that end, one seeks complex pair (ω_0, k_0) satisfying Eq. (35), $D(k_0, \omega_0; S) = 0$, and zero group velocity $\partial\omega / \partial k(k_0) = 0$ at a fixed spatial location. The corresponding $\omega_0 = \omega(k_0)$ is called the absolute frequency, and the absolute growth rate gives $\omega_{0,i} = \omega_i(k_0)$. The following criterion determines the

absolute/convective nature of the instability:

$$\begin{aligned} \omega_{0,i} > 0, & \text{ absolutely unstable flow;} \\ \omega_{0,i} < 0, & \text{ convectively unstable flow.} \end{aligned} \quad (37)$$

In general, ω_0 are branch-point singularities of $k(\omega)$ with two spatial branches $k^+(\omega)$ and $k^-(\omega)$. As an additional requirement, the physically relevant complex pair (ω_0, k_0) must satisfy the Briggs-Bers criterion²²⁰, in which branches $k^+(\omega)$ and $k^-(\omega)$ originate respectively from the upper and lower half of the complex k -plane when ω_i is decreased from positive values to zero. Convective instability, such as narrowing jetting, displays extrinsic dynamics as a spatial amplifier, where external noises are amplified when advected downstream, while absolute instability, such as dripping, exhibits intrinsic dynamics that induces breakup at a fixed spatial location and at a self-sustained frequency intrinsic to the system. This explains experimental observations that droplets produced by dripping are highly monodisperse, but by jetting are more polydisperse.

Guillot *et al.*¹³³ performed a linear spatiotemporal stability analysis to parametrically map jetting and dripping regions for confined capillary jets in cylindrical co-flowing microchannels. Consistent with criterion Eq. (37), a temporally unstable flow is convectively unstable (jetting) if the front velocity of the trailing edge of the wave packet v_-^* is positive (perturbations convected downstream), and is otherwise absolutely unstable (dripping) if $v_-^* < 0$ (upstream propagation of perturbations). Accordingly, a marginal stability criterion was obtained for the jetting-dripping transition¹³³:

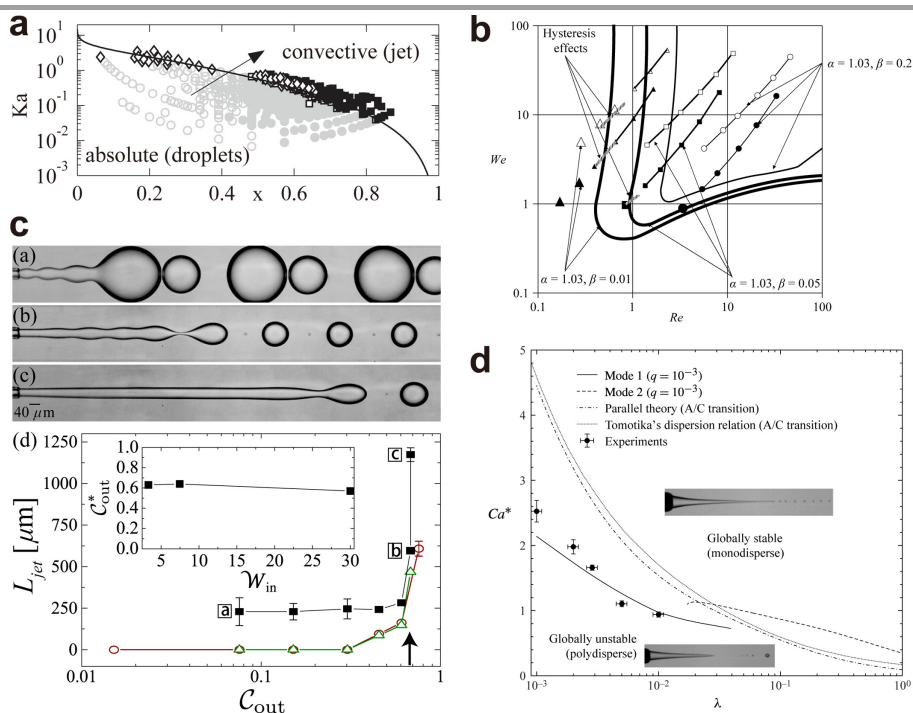


Fig. 6 Capillary instability in microfluidic biphasic flows. (a) Flow behaviour in the (x, Ka) plane for a given value of the viscosity ratio $\lambda = 0.23$. Symbols of light colour correspond to droplets (absolute instability), and those of dark colour represent jet (convective instability), both of which are experimental data. The line is the theoretical prediction of absolute-to-convective instability transition. (b) Theoretical jetting-dripping transition, for various (α, β, Re, We) . α and β stand respectively for density ratio and viscosity ratio of continuous to dispersed phase liquids²²⁴. Re and We are Reynolds number and Weber number of the dispersed liquid, respectively. (c) Upper images: jet length increases with increasing continuous capillary number C_{out} (C_{out} defined in ref. 225). The oscillations on the jet gradually die out as the length increases at the critical value, C_{out}^* . Lower diagram: plot of the jet length as a function of C_{out} . The arrow marks C_{out}^* obtained from the linear stability analysis. Inset: linear stability analysis. Below C_{out}^* the jet breaks due to an absolute instability, while above, it breaks due to a convective instability. For these experiments $\lambda = 0.01$. (d) The critical capillary number, Ca^* (defined in ref. 58) as a function of the viscosity ratio λ , for the transition between globally stable to globally unstable flows. (a) is reprinted with permission from ref. 133. Copyright 2007, American Physical Society. (b) is reprinted with permission from ref. 224. Copyright 2006, Cambridge University Press. (c) is reprinted with permission from ref. 225. Copyright 2008, American Physical Society. (d) is reprinted with permission from ref. 58. Copyright 2013, Cambridge University Press.

$$v_-^* = \frac{Ca_c x^3 E(x, \lambda) - C_1 F(x, \lambda)}{x^9 (1 - \lambda^{-1}) - x^5} = 0, \quad (38)$$

with

$$C_1 = \frac{5 + \sqrt{7}}{18} \sqrt{\frac{24}{\sqrt{7} - 1}}, \quad (39)$$

$$E(x, \lambda) = -4x + (8 - 4\lambda^{-1})x^3 + 4(\lambda^{-1} - 1)x^5, \quad (40)$$

$$F(x, \lambda) = x^4 [4 - \lambda^{-1} + 4 \ln(x)] + x^6 (-8 + 4\lambda^{-1}) + x^8 [4 - 3\lambda^{-1} - (4 - 4\lambda^{-1}) \ln(x)] \quad (41)$$

where viscosity ratio $\lambda = \eta_d / \eta_c$, the degree of confinement of the unperturbed jet $x = \sqrt{(\alpha - 1) / (\lambda^{-1} + \alpha - 1)}$ with $\alpha = \sqrt{1 + \varphi / \lambda}$, and capillary number $Ca_c = 8\eta_c Q_c [1 + \lambda(\alpha - 1)]^2 / (\pi R_c^2 \gamma)$ (R_c , the inner radius of the cylindrical channel wall). Criterion Eq. (38) is plotted in the (x, Ca_c) plane as operational conditions for the jetting-dripping transition with fixed λ (Fig. 6a, where Ka denotes Ca_c). Guillot *et al.*²²² then extended the analysis to square and rectangular microchannels in which dripping is found to be promoted in comparison to cylindrical channels. Adopting lubrication approximation at low Reynolds numbers (neglecting inertia), Eq. (38) is simple and robust in predicting jetting-dripping transition for highly confined jets (Fig. 6a), but fails for the low degree of confinement. Herrada *et al.*²²³ proposed axisymmetric models based on the analysis by Guillot *et al.*¹³³ to improve the accuracy of prediction in low confinement cases. In addition, jetting-dripping transition in an open flow-focusing configuration (Fig. 2c,i) was studied by considering a viscous jet flowing in an unbounded co-flowing viscous liquid domain²²⁴. The transition is mapped in the parametrical space of (Re, We) and shows good agreement with experiments (Fig. 6b).

Absolute instability is also observed in widening jets breakup (Fig. 6c)²²⁵. This is supported by several experimental identifications: (i) the neck diameter oscillates with exponentially growing amplitude until pinch-off occurs, implying a positive temporal growth rate ω_i ; (ii) the temporal oscillations of the widening jet remain nearly stationary in space, suggesting the zero group velocity at a fixed spatial location; (iii) despite that $We_d > O(1)$ of a widening jet at injection, droplet pinch-off at the end of the jet occurs only when the neck diameter is sufficiently widened such that $We_d \lesssim O(1)$, analogous to droplet formation in dripping mode due to absolute instability. Utada *et al.*²²⁵ predicted the critical capillary number $Ca_{cri} = 0.69$ (with $\lambda = 0.01$) of continuous phase by performing a spatial stability analysis. Above Ca_{cri} , the absolute instability of widening jets transitions to convective instability together with a significant increase in the jet length (Fig. 6c). The theoretical prediction is in good agreement with the experimental result of $Ca_{cri} \approx 0.65$, and independent of dispersed Weber number We_d (Fig. 6c).

Spatially-developing flows, for example, capillary jets emanating from a nozzle, may exhibit self-sustained global modes when flows have a sufficient region of absolute instability^{220, 226}. In flow-focusing microfluidic devices, the tapering liquid meniscus of the dispersed phase oscillates periodically with time and spatially between the injection and focusing nozzle, emitting droplets in sequence by tip-multi-braking mode^{17, 184}, which displays global instability^{189, 190}. In comparison, flows in dripping, jetting and tip-streaming modes are globally stable because of their stable menisci (Figs. 3a-d, flow-focusing). Similar to flows in flow-focusing, highly stretched jets in co-flowing streams set off global instability with an unsteady conical meniscus when the continuous-phase capillary number is below the critical value Ca_{cri} ⁵⁸. To determine Ca_{cri} that separates tip-streaming from tip-multi-breaking mode, Gordillo *et al.*⁵⁸ performed a global stability analysis by taking the real shape of the stretched jets into consideration with no simplification of parallel

flows. It was found that Ca_{cri} is a decreasing function of viscosity ratio λ but insensitive to flow rate ratio φ provided that $\lambda \lesssim 0.1$ and $\varphi \ll 1$. The theoretical predictions by global stability analysis are in fair agreement with experiments, and more accurate than predictions by local stability analysis of absolute/convective transition (Fig. 6d).

5. Active droplet generation

Modulating droplet generation with the aid of additional energy input by active controls has several advantages over passive droplet generation, for example, (i) active droplet generation offers additional handles and higher flexibility in controlling droplet size and production rate, and in some cases enables on-demand droplet generation (e.g., droplet with prescribed volume produced at a required frequency), therefore largely facilitating practical applications of microfluidic droplets; it is however almost impossible, in passive droplet generation, to independently control droplet size and generation frequency, as they correlate with each other by mass conservation, (ii) system response time, the time needed for stabilized droplet production, is much shorter in active control than in passive method. For example, the response time can be down to several milliseconds in active approach compared with several seconds or even minutes in passive method.

The energy imbalance modulates the nature of force balance on the interface for droplet generation. According to the nature of force balance variation, the active droplet generation can be categorized into two groups: introducing additional forces, and modifying intrinsic forces. As discussed in Section 2, microfluidic droplet formation is controlled by external forces and intrinsic ones of inertial, viscous, and capillary effects scaling as $f_i \sim \rho_s u_s^2$, $f_v \sim \eta_s u_s / L$, and $f_\gamma \sim \gamma / L$, respectively. An active control is effective in modulating droplet generation if it affects at least one type of the aforementioned four forces. Normally, additional forces are exploited by applying external electric, magnetic and centrifugal fields, and modifying the intrinsic inertial, viscous and capillary forces is realized by manipulating the dynamic velocity u and material properties, including viscosity, interfacial tension, channel wettability, and fluid density. A recent review²⁵ classified active methods according to the energy type: electrical, magnetic, thermal and mechanical. We note that the mechanism of active control could be different even for the same energy input (Table 5). Taking electrical control for example, dielectrophoresis, electrowetting (EW), electrochemical, electrocapillary and electrorheological effects can be utilized. In dielectrophoresis an external dielectric force is applied on the liquid-liquid interface; in EW the wettability of the microchannel substrate is changed; in electrochemical reaction and electrocapillarity, the interfacial tension is modified; by using electrorheological fluids the viscosity is actively controlled (Table 5). As such, we organize this section based on the force-balance argument of active droplet generation.

5.1 Additional forces in electrical control

Similar to the cases of electrospray and electrohydrodynamic atomization¹⁸⁰, electric fields can be employed to manipulate droplet generation in microfluidics. To apply an electric field, a microfluidic device is embedded with electrodes, one²²⁷ or pairs of²²⁸ which are connected to the high voltage and the others are grounded. Taking water-oil systems for example, high voltage U_e is applied to the water phase, and the grounded oil phase behaves as an insulator (Fig. 7a). Application of electric fields induces the migration and accumulation of charges on water-oil interface. With appropriate distribution of electric field, the interaction between the surface charges and electric field offers an additional handle to control droplet generation. Electrical control is categorized into two

Table 5 Comparison of different active droplet generation techniques.

Type of control	Mechanism	Implementation	Device geometry	Type of droplet generation ^a	Reference
Electrical	Electric force	Constant DC	Flow-focusing	Tunable	19, 227, 231, 232
		DC pulse	Step emulsification	On-demand	229, 233
		Low-frequency AC	Flow-focusing	Tunable	230
		High-frequency AC	Flow-focusing	Tunable	228, 234-236
		DC pulse	T-junction	On-demand	295
	Electrorheology	High-frequency AC	Flow-focusing	On-demand	295, 299
			Flow-focusing	Tunable	316-319
	EW effect	High-frequency AC	Flow-focusing	On-demand	320
			Flow-focusing	Tunable	313
			Flow-focusing	Tunable	313
Magnetic	Electrochemical and electrocapillary effects	Constant DC and low-frequency AC	Flow-focusing	Tunable	238-241, 243
		Magnetic force	Uniform field, in-plane	Tunable	240
			Uniform field, out-of-plane	Tunable	242
			Non-uniform field	Tunable	237
			Step emulsification	On-demand	85
Centrifugal	Centrifugal/Coriolis/Euler forces	Rotation	Dispenser nozzle	Tunable	247-249
			Flow-focusing	Tunable	250, 251
			Cross-flow	Tunable	252, 253
			Step emulsification	Tunable	82-84
			T-junction	On-demand	254
Optical	Modifying fluid pressure Marangoni effect	Laser pulse	T-junction	Tunable	308
		Laser heating	Flow-focusing	Tunable	307, 309, 310
		UV/blue light Heating	Co-flow	Tunable	311
			Flow-focusing	Tunable	315
			T-junction	Tunable	302, 303
Thermal	Photosensitive surfactant Temperature-dependence of viscosity and interfacial tension	Heating	Flow-focusing	Tunable	297, 300, 301, 304
			T-junction	Tunable	305, 306
			Flow-focusing	Tunable	321
			Co-flow	Tunable	255-260
			T-junction	On-demand	261-265
Mechanical	Phase change Modifying fluid pressure	Mechanical vibration Off-chip valve	Flow-focusing	On-demand	18
			Step emulsification	On-demand	266
			T-junction	Tunable	276
			On-demand	On-demand	273, 274
			Flow-focusing	Tunable	275
	Channel deformation/blocking	Piezoelectric actuator	Flow-focusing	Tunable	269-271
			T-junction	On-demand	267, 268, 272
			Flow-focusing	On-demand	268
			T-junction	Tunable	13, 279, 282, 287, 291
			On-demand	On-demand	283-286, 289, 290
		On-chip microvalve	Flow-focusing	Tunable	277, 278, 280, 281, 288, 292-294
			On-demand	On-demand	284

^a Tunable – droplet generation with tunable size and production rate; On-demand – on-demand droplet generation.

groups based on the type of current: direct current (DC)²²⁷ and alternating current (AC) control²²⁸. In DC control the voltage is only in one direction, whereas the polarity of voltage reverses periodically in AC control. DC control can be applied at constant²²⁷ or pulsatile voltage²²⁹, while AC control is subcategorized into low-frequency (on the order of Hz)²³⁰ and high-frequency (on the order of kHz)²²⁸ control, depending on whether the frequency of applied voltage is lower (low-frequency) or higher (high-frequency) than that of droplet generation.

In DC control where constant high voltage is applied to the dispersed phase (Fig. 7a), the electric stress on the water-oil interface scales as²²⁷

$$f_e \sim \epsilon_c E^2, \quad (42)$$

where $E \sim U_e / L$ is the electric field strength, ϵ_c the permittivity of continuous liquid (usually oil). The electric stress f_e increases with U_e and competes with surface tension $\sim \gamma / L$. When U_e is sufficiently large, the electric stress decreases the droplet size, consistent with experimental observations^{227, 231, 232}. Flow rate ratio and surfactant concentration play roles in electrical control. At high flow rate ratio ϕ and high voltage U_e , jetting appears and bridges the

dispersed conducting liquid and the grounded electrode, inducing no breakup (Fig. 7b,i)²³¹. At low ϕ , increasing the voltage above a threshold otherwise gives rise to the formation of a Taylor cone, where droplets with a diameter less than 1 μm are generated in electrospray regime (Fig. 7b,ii)²³¹. The threshold voltage for electrospray decreases with surfactant concentration²³². At constant surfactant concentration, the semiangle of Taylor cone in microfluidic chips was found to decrease as the flow rate ratio decreases, and always smaller than that in air²³².

Variation in DC implementation affects the process of droplet formation. In a flow-focusing geometry where two continuous inlets are connected respectively to high DC power supplier and ground (Fig. 7c), the electric force points from the continuous phase to the dispersed phase due to the smaller ϵ_d (ϵ_d , the permittivity of dispersed phase) than ϵ_c ¹⁹. Numerical simulation¹⁹ demonstrates three distinct stages of droplet size variation with the increase of voltage: decreasing, increasing and decreasing again. This trend is explained by the counterbalance between the x and y components of electric forces (Fig. 7c). The x component, perpendicular to the flow direction, accelerates the squeezing rate of viscous droplet, whereas

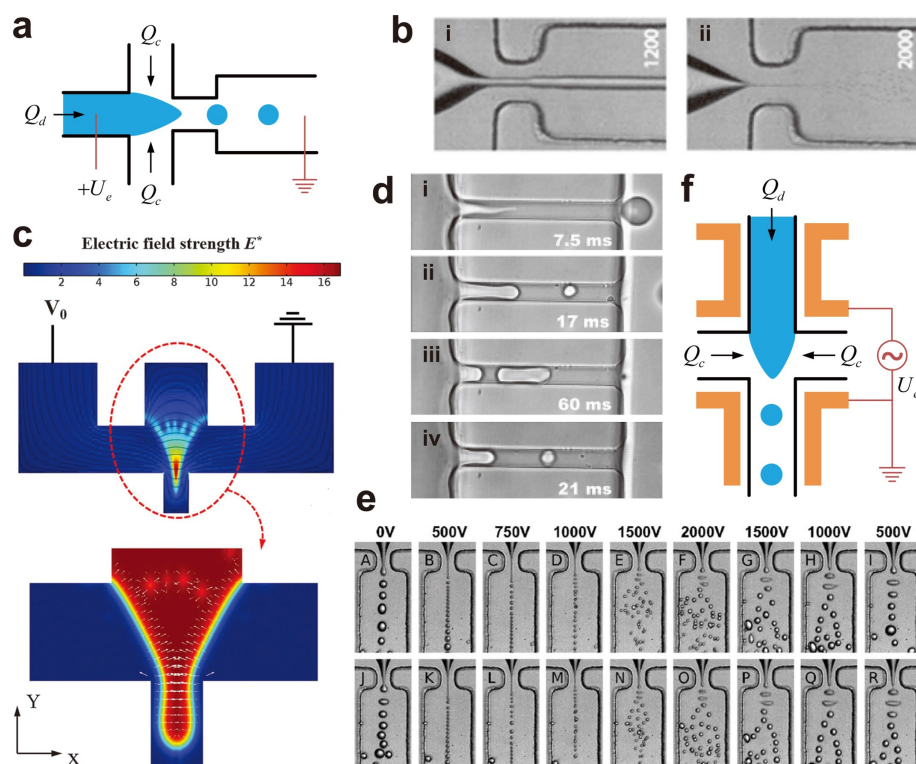


Fig. 7 Electrical control of droplet generation. (a) A high DC voltage U_e is applied to the dispersed phase, and the continuous phase is grounded. Two methods of implementation have been demonstrated: inserting electrodes into the liquids¹⁹⁴, or patterning indium tin oxide (ITO) electrodes on the substrate¹⁹⁰. (b) Droplet generation in a DC electric field. (i) Stable jet. Flow rate ratio $\phi = 80 / 350$. Voltage $U_e = 1200$ V. (ii) Electrospray. Flow rate ratio $\phi = 5 / 350$. Voltage $U_e = 2000$ V. (c) Upper: contour plot of the electric field strength at one time-step during the droplet formation. The black lines indicate the electric field lines. High voltage is applied to the left side inlet. Lower: contour plot of the volume fraction of the dispersed phase in the regions enclosed by the red dashed curves. The white arrows indicate the vectors of the electric body force induced on the fluid interface. (d) A sequence of images showing the electrogeneration of water droplets with a DC voltage pulse. (i-iii) Images of two droplets formation in one pulse. (i) Advancing of the water jet. (ii) Primary droplet formation. (iii) Secondary droplet formation. (iv) Formation of one droplet per pulse. (e) Droplet series generated using an AC electric field of the triangular waveform. A-I: the first half period. J-R: the second half period. (f) Schematic of a flow-focusing device under an AC electric field. The electrodes are in orange and the dispersed phase is in blue. (b) is adapted with permission from ref. 231. Copyright 2007, American Institute of Physics. (c) is reproduced with permission from ref. 19. Copyright 2015, Royal Society of Chemistry. (d) is adapted with permission from ref. 233. Copyright 2006, American Chemical Society. (e) is adapted with permission from ref. 230. Copyright 2010, American Institute of Physics.

the y component, opposite to the flow direction of dispersed phase, exerts a retardation effect on the fluid interface.

DC pulse is capable of producing droplet on demand^{229, 233}. Initially, water-oil interface is stabilized by capillary pressure at the orifice. Upon pulse initiation, the interface is forced by electric effects into the narrow orifice channel (Fig. 7d,i) at a nearly constant speed before reaching its equilibrium displacement²²⁹. The maximum equilibrium displacement increases with the pulse amplitude. At high amplitude droplet is emitted at the end of the protruded jet (Fig. 7d,ii), then the jet either re-protrudes and issues a secondary droplet (Fig. 7d,iii) or directly retracts (Fig. 7d,iv) after the pulse²³³. The secondary droplet is only observed in narrow channel with a high axial aspect ratio (ratio of channel length L to equivalent radius $wh / (w + h)$). The secondary droplet formation was ascribed to Rayleigh instability, different from the primary one of electrohydrodynamic origin²³³.

In low-frequency AC control, droplet generation is asynchronous with applied voltage and displays hysteresis²³⁰. The droplet size is independent of the polarity of the electric field, but displays time dependence as electric fields vary with time (Fig. 7e). The irregular size distribution gives rise to uneven charges in individual droplets²³⁰. In addition, the dispersed flow rate Q_d issued from the Taylor cone exhibits relaxation oscillation that might be responsible for the hysteresis of droplet size variation²³⁰.

In high-frequency AC control, Tan *et al.*²²⁸ designed a microfluidic setup where electrodes are not in contact with fluids

(Fig. 7f). This method enables reliable droplet generation and manipulation with response time in the order of milliseconds, based on which a musical interpretation of droplet microfluidics was demonstrated²³⁴. In this design, dripping to jetting transition is observed when the voltage increases²²⁸. At a fixed high voltage, systematic variation of the voltage frequency f_{re} and electrical conductivity of the dispersed liquid k_e leads to three regimes: dripping, unstable production of droplets, and jetting²²⁸. The dripping and unstable production regimes are separated by the boundary of $f_{re} / k_e \sim 5 \times 10^5 \text{ mF}^{-1}$, while the transition from jetting to unstable production occurs when f_{re} is below a certain value $f_{re, mess}$ ^{228, 235}. In dripping regime, the droplet size is a function of electric field, and can be quantitatively related to an effective capillary number Ca_{eff} of the form $Ca_{eff} = Ca_c / (1 - B_e)$, where B_e is the electric bond number²²⁸. The unstable production of droplets is likely to be induced by nonaxisymmetric instabilities when the electric field at the jet surface is of a critical value²³⁵. In jetting regime^{235, 236}, the jet length L_{jet} increases with applied voltage and scales as $L_{jet} \sim (f_{re} / k_e)^{0.5}$ at a constant voltage. The electrical potential at the tip of jets was found to be about 550 V, which might confirm that jet breakup occurs for a given electric field distribution around the tip²³⁵.

5.2 Additional forces in magnetic control

Ferrofluid is a type of nanofluids containing nanometer-sized magnetic particles (≤ 10 nm), which is magnetized by applying a magnetic field, and demagnetized when the magnetic field is withdrawn. Consequently, ferrofluid is a good candidate for magnetic control of droplet generation in microfluidics. In the presence of a magnetic field, magnetic forces per unit volume f_m are expressed as,

$$f_m = \mu_0 M \nabla H, \quad (43)$$

where μ_0 is the magnetic permeability of free space, M represents the magnetization, and ∇H stands for the gradient of the magnetic field strength. Accordingly, droplet generation can be harnessed by varying either M or ∇H or both in magnetic control. In this subsection, we focus only on droplet generation actively controlled by the extrinsic magnetic forces where ferrofluid is used as the dispersed phase. We do not consider magnetorheology, because the strong Brownian motion of magnetic nanoparticles renders this effect very weak in ferrofluid.

Implementation of magnetic fields can be adjusted in various aspects: the type of magnet, the location, uniformity, direction, and polarity of the magnetic field. Both permanent magnets^{85, 237, 238} and electromagnets²³⁹⁻²⁴¹ have been used. The magnetic field can either cover the entire microfluidic device^{85, 237-240, 242} or be localized²⁴¹, and be either uniform²³⁸⁻²⁴² or non-uniform^{85, 237} (Fig. 8a). For the uniform magnetic field, it can be either in-plane or out-of-plane²⁴² of the microfluidic chip, and either parallel to or perpendicular to²³⁸ the main flow direction (Fig. 8a). Finally, the polarity of the uniform magnetic field can be inverted as well.

Droplet generation depends on the specific implementation of the magnetic field. First of all, the direction of the magnetic field matters. In T-junction, the droplet size increase when a permanent magnet is placed upstream the junction, whereas the size decreases when the magnet is positioned downstream²³⁷. Changing the position of the permanent magnet varies the direction of the non-

uniform magnetic field, as well as that of the magnetic forces f_m . In the upstream case, f_m pulls the droplet back and delays the breakup, resulting in larger droplets. In contrast, the downstream case has the opposite effect. In addition, by employing a uniform magnetic field in T-junction, the in-plane application causes a decrease in droplet size²⁴⁰, while the out-of-plane implementation induces the droplet size increase²⁴². Moreover, in flow-focusing configuration, magnetic fields parallel to and perpendicular to the main flow mainly affect the breakup (Fig. 8b,i) and expanding (Fig. 8b,ii) process of the dispersed thread, respectively²³⁸. Interestingly, polarity of the magnetic field has no influence on droplet size²⁴⁰.

Secondly, the device geometry plays a significant role in modulating droplet size. In the case of a uniform magnetic field parallel to the main flow, droplet size increases in flow-focusing but decreases in T-junction geometries as magnetic flux density increases²⁴⁰. This difference results from the orientation of the magnetic field relative to the dispersed flow direction. When subjected to an external magnetic field, the magnetic nanoparticles align with the magnetic field. In flow-focusing configuration, the alignment is in the same direction of the dispersed flow, elongating the fluid tip and delaying breakup (Fig. 8c,i)^{240, 243}. However, in T-junction, the alignment is perpendicular to the dispersed flow direction (Fig. 8c,ii), thereby accelerating breakup²⁴⁰. In step emulsification, droplet size is not affected by the magnetic field, but depends on the channel dimension⁸⁵. Kahkeshani and Di Carlo⁸⁵ recently showed that magnetic body forces, counteracted by interfacial and viscous forces, drives the dispersed fluid into the continuous reservoir. In a device with fixed channel dimension, increasing magnetic forces, or decreasing interfacial and viscous forces enables a larger rate of droplet generation, while droplet size keeps constant⁸⁵.

Finally, the location of magnetic field impacts droplet size variation. In flow-focusing geometry with a uniform magnetic field perpendicular to the main flow, the droplet size increases when

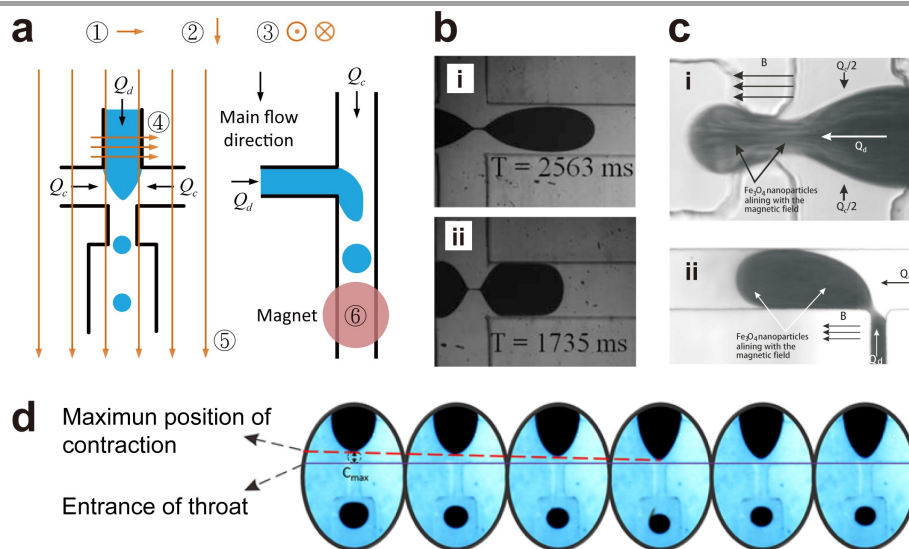


Fig. 8 Magnetic control of droplet generation. (a) Schematic of microfluidic devices under magnetic fields, as exemplified by flow-focusing (left) and cross-flow (right) geometries. In-plane magnetic field: ①, ②, ④, ⑤. The direction of magnetic field perpendicular to (①, ④) and parallel to (②, ⑤) the main flow direction. Out-of-plane magnetic field with two polarities: ③. Localized magnetic field: ④. The entire device exposed to a magnetic field: ⑤. Uniform magnetic field: ④, ⑤. Non-uniform magnetic field, exemplified by a downstream magnet: ⑥. Note that all types of magnetic fields (①–⑥) can be applied to any types of fluidic structures. In addition, ④ and ⑥ can be placed at any locations of microfluidic devices. (b) Formation of ferrofluid droplets by dripping in a flow-focusing device. (i) Magnetic field parallel to the main flow direction. During pinch-off, the droplet is stretched in the main flow direction. (ii) Magnetic field perpendicular to the main flow direction. During pinch-off, the droplet is expanded to block the outlet channel. (c) Alignments of the clustered magnetic nanoparticles in the direction of the magnetic field. (i) Alignment parallel to the dispersed flow direction in a flow-focusing device. (ii) Alignment perpendicular to the dispersed flow direction in a T-junction. (d) Maximum contraction position C_{max} (after droplet formation) and droplet size decreasing with magnetic flux density (from left to right). (b) is adapted with permission from ref. 238. Copyright 2013, Royal Society of Chemistry. (c) is adapted with permission from ref. 240. Copyright 2011, American Physical Society. (d) is adapted with permission from ref. 241. Copyright 2015, Springer.

magnetic field covers the entire device²³⁸, but decreases when the field locates at continuous inlet microchannel²⁴¹. In the former case, the magnetic field accelerates the expanding velocity of the dispersed tip and results in larger expanding tips as well as larger droplets (Fig. 8b,ii)²³⁸. In comparison, the latter observation is interpreted as a result of the magnetic drag²⁴¹. The drag drives the dispersed tip to move downstream and block the outlet channel (Fig. 8d), leading to an increased continuous-fluid pressure buildup that accelerates droplet breakup²⁴¹.

In general, the effectiveness of magnetic control is attenuated by the increase of flow rate^{237, 238, 240}, because magnetic forces, Eq. (43), are counteracted by the elevated viscous and/or inertial forces. To the best of our knowledge, a closed-form expression or scaling law for predicting the droplet size under external magnetic fields has yet to be developed.

5.3 Additional forces in centrifugal control

Centrifugal microfluidics has been developed to fulfill many unit fluidic functions, such as mixing, valving, separation, detection, metering and droplet generation, and various applications ranging from nucleic acid analysis, immunoassays, cell lysis, blood plasma separation to water, food and soil analyses^{244, 245}. We herein focus exclusively on the droplet generation in centrifugal microfluidics. To this end, several additional forces due to the presence of rotation are introduced first. In a rotating frame of reference, three pseudo forces may act on a fluid volume: centrifugal force f_Ω , Coriolis force f_{Co} and Euler force f_E , respectively (Fig. 9a). The centrifugal force is caused by the rotary motion of the microchannel network, always pointing radially outward. The Coriolis force is generated by the relative motion of the object, therefore perpendicular to the direction of the motion. The Euler force is proportional to the rotational

acceleration. Considering a fluid plug of density ρ at a radial position r rotating at an angular velocity Ω , the three forces scale as (in the form of forces per unit area)²⁴⁶:

$$f_\Omega \sim \rho \Omega^2 r L, \quad (44)$$

$$f_{Co} \sim \rho^2 \Omega^3 r L^3 / \eta, \quad (45)$$

$$f_E \sim \rho \dot{\Omega} r L, \quad (46)$$

where L , η and $\dot{\Omega}$ refer to the characteristic length scale of the fluid plug, dynamic viscosity of the fluid, and angular acceleration, respectively. During droplet generation, these forces counteract interfacial tension effects and facilitate the fluid breakup. Accordingly, two Bond numbers Bo_Ω and Bo_{Co} can be defined as the ratios of centrifugal and Coriolis forces to interfacial tension force, respectively:

$$Bo_\Omega \sim \rho \Omega^2 r L^2 / \gamma, \quad (47)$$

$$Bo_{Co} \sim \rho^2 \Omega^3 r L^4 / \eta \gamma. \quad (48)$$

To date, four different structures have been employed in centrifugal droplet generation: dispenser nozzle²⁴⁷⁻²⁴⁹, flow-focusing^{250, 251}, cross-flow^{252, 253} and step emulsification⁸²⁻⁸⁴. One scheme to generate droplet is to rotate a dispenser nozzle, by which individual droplets are issued into the intermediate air spacer before they are collected by the receiving liquid (Fig. 9b)²⁴⁷⁻²⁴⁹. To generate droplet, the pressure difference ($\Delta p = \rho_d \Omega^2 r^2 / 2$) of the dispersed liquid in the nozzle should exceed the capillary pressure ($f_\gamma = 4\gamma / w_d$). This gives a minimum rotation velocity²⁴⁷:

$$\Omega_{\min} = \frac{2}{r} \sqrt{\frac{2\gamma}{\rho_d w_d}}, \quad (49)$$

with w_d being the dispenser nozzle diameter. When the droplet detaches from the nozzle, the total surface tension force ($F_\gamma = \pi w_d \gamma$)

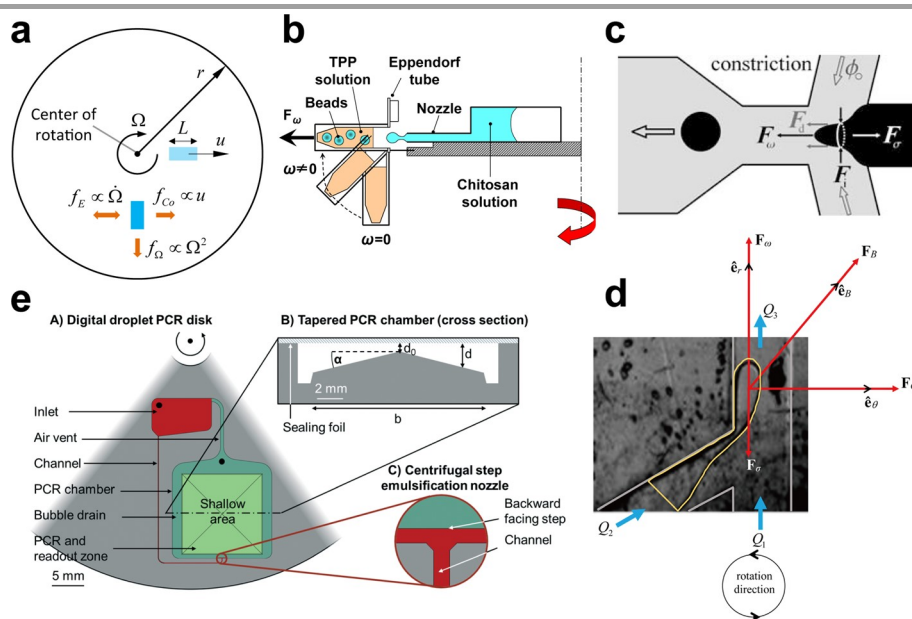


Fig. 9 Centrifugal control of droplet generation. (a) Three pseudo forces in centrifugal droplet generation. Centrifugal force f_Ω is proportional to Ω^2 , pointing radially outward. Coriolis force $f_{Co} \propto \Omega^3$, perpendicular to both Ω and fluid velocity u . Euler force is proportional to angular acceleration, $f_E \propto \dot{\Omega}$, perpendicular to both $\dot{\Omega}$ and radial coordinate r . (b) Schematic of the experimental setup consisting of a centrifugal platform with a micronezzle and Eppendorf tube in a swinging bucket. (c) Centrifugal flow-focusing configuration. The centrifugal force F_ω and viscous drag F_d of the sheath flow ϕ_0 support droplet formation. The emerging droplet is squeezed by the inertia (F_i) induced by the transversal component of the sheath flow. The surface tension counterforce F_γ prevents the droplet break-off until a critical mass is reached²⁵⁰. (d) Droplet formation in a centrifugal cross-flow configuration. While surface tension F_γ indulges in maintaining the symmetric shape of the droplet, centrifugal force F_ω and Coriolis force F_C continuously pull the liquid finger until pinch-off²⁵². (e) Centrifugal step emulsification device. The microfluidic device is located on a spinning disk (inset A) and consists of: an inlet chamber (red), a channel (red) which connects the inlet chamber to a step emulsification nozzle (inset C), and a droplet collection chamber (PCR chamber, inset B). (b) is reproduced with permission from ref. 247. Copyright 2009, Elsevier. (c) is reproduced with permission from ref. 250. Copyright 2006, Springer. (d) is reproduced with permission from ref. 252. Copyright 2015, American Institute of Physics. (e) is reproduced from ref. 84. Copyright 2016, Royal Society of Chemistry.

is balanced by the total centrifugal force ($F_{\Omega} = \pi \rho_d D^3 \Omega^2 r / 6$), thereby the droplet diameter D is expressed as^{247, 248},

$$D = \sqrt[3]{\frac{6w_d\gamma}{\rho_d\Omega^2 r}}. \quad (50)$$

Since flow rate Q_d obeys $Q_d \propto \Delta p \propto \Omega^2$ ²⁴⁵, the droplet generation frequency fre_{drop} for Newtonian fluids could be $fre_{drop} \propto \Omega^4$ based on mass conservation. Rotating the dispenser nozzle has been employed to produce alginate/chitosan microparticles after solidification of droplets by cross-linking solutions²⁴⁷⁻²⁴⁹. The abilities to tailor the particles into non-spherical shape and with 3D multi-compartment structure have also been demonstrated by varying the distance between the nozzle and the surface of the cross-linking solution, and by using a multi-barrelled capillary nozzle, respectively²⁴⁸.

In flow-focusing configuration, the dispersed liquid tip is pulled downstream by centrifugal and viscous drag forces. Once the tip is long enough, thread breakup is stimulated by the junction constriction and inertia coming from the transversal component of continuous flows (Fig. 9c)²⁵⁰. Depending on the channel geometry and Bond number Bo_{Ω} , squeezing, dripping and jetting modes of droplet generation are observed. In squeezing and dripping, highly monodisperse droplets (e.g., CV = 0.84% by dripping) are generated with the droplet size decreasing but generation frequency increasing as Ω increases²⁵⁰. In contrast to this trend, both the bubble length L_p and bubble generation frequency fre_{bubble} are proportional to Bo_{Ω} (equivalently, $L_p \propto \Omega^2$ and $fre_{bubble} \propto \Omega^2$) when $Bo_{\Omega} \lesssim 1$ in centrifugal bubble generation²⁵¹. By programming the rotational speed, it is feasible to tailor the inter-bubble spacing as well as bubble size²⁵¹.

In cross-flow structure, the force balance depends on the rotation speed Ω . According to Eqs. (44) and (45), $f_{\Omega} \propto \Omega^2$, whereas $f_{Co} \propto \Omega^3$, growing faster with Ω than f_{Ω} . As such, at high Ω , f_{Co} would probably dominate f_{Ω} . In a Y-shaped microchannel with the rotation speed ranging from 104.72 rad / s to 167.55 rad / s, the two Bond numbers are in the range of $Bo_{\Omega} = 0.22 - 0.2$ and $Bo_{Co} = 12.62 - 20.90$, respectively²⁵². The dominance of f_{Co} over f_{Ω} and f_{γ} indicates that the droplet generation is governed by the effective pulling of Coriolis force until the pinch-off (Fig. 9d). In this circumstance²⁵², the droplet volume V is ruled by $V \propto \Omega^{-1}$, and droplet generation frequency fre_{drop} is thus proportional to Ω^3 . In addition, by executing the rotation speed in alternate directions of clockwise and counterclockwise, Kar *et al.*²⁵² demonstrated the production of long serpentine threads, as a result of the repetitive alterations in the direction of f_{Co} , as well as the impact of Euler force at the momentary time stages of speed turnover.

Centrifugal step emulsification (Fig. 9e) takes the advantage of step emulsification that droplet size is less sensitive to pressure and flow rate fluctuations, but only depends on the nozzle geometry and interfacial tension⁸². Subsequently, uniform droplet generation (CV of 2 – 4%) could be achieved, which facilitates the afterward application of digital droplet recombinase polymerase amplification (RPA)⁸² and digital droplet PCR⁸⁴. At high rotation speed ($\Omega \approx 125.6$ rad / s)^{82, 83}, the setup enables in situ production of gel emulsions with very high internal volume fractions of > 97.2%. Parallelization of 72 nozzles improves the droplet production rate up to 3,700 Hz⁸³.

5.4 Modifying fluid velocity

Intrinsic forces such as inertial and viscous forces could be modified dynamically by manipulating fluid velocity u . In general, modulating the hydraulic pressure and flow resistance (e.g., changing channel dimension) leads to variations in velocity u , as implemented by off-

chip and on-chip actuation. The former implementation includes optical control, mechanical vibration and off-chip valve, while the latter involves piezoelectric control and on-chip microvalve. In contrast to on-chip actuations, off-chip modules are simple in assembly, and compatible with microfluidic devices.

5.4.1 Optical control. An intense laser pulse (Fig. 10a) is employed to actuate fluid pressure by inducing rapidly expanding cavitation bubbles, therefore enabling on-demand droplet generation²⁵⁴. To achieve this, initially, the immiscible interface between dispersed water and continuous oil phase is set to be stable at the junction by adjusting flow rates. Then an intense laser pulse is focused on the laser-absorption water phase to break down water molecules and generate hot plasma²⁵⁴. Arising from the dissipation of heat, laser-induced cavitation bubble expands rapidly, and destabilizes the oil-water interface. The expanding cavitation acts as a micropump to push water into oil phase and produces aqueous droplets. Due to controllable laser pulsing intervals, on-demand droplet generation of production frequency as high as 10 kHz can be achieved, meanwhile maintaining high monodispersity (droplet volume variations < 1%)²⁵⁴.

5.4.2 Mechanical vibration. Off-chip mechanical vibration can be achieved by perturbing the flexible dispersed-phase microtubing using a mechanical vibrator (Fig. 10b)²⁵⁵⁻²⁶⁰. This technique demonstrates the capability of inducing corrugations on water-water interfaces with ultra-low interfacial tension in APTS²⁵⁵, breaking jets into droplets^{255-257, 259}, and producing droplets with core-shell and multi-compartment structures^{257, 258}. Upon perturbing the microtubing, the dispersed-phase pressure pulsates and induces an oscillatory flow at the nozzle²⁵⁹. Experimental findings suggested that vibration-enhanced dispersed-fluid inertia counteracts the viscous stress, inducing faster pinch-off of dispersed fluid threads and producing smaller droplets²⁶⁰. In a certain range of frequency, the droplet generation was synchronized with the vibration, by which droplet with predefined size is achievable regardless of the continuous flow rate, channel dimension, and fluid properties, provided that the frequency of vibration and dispersed flow rate are fixed²⁵⁹. The lower limit of synchronization frequency is approximate the frequency of passive droplet generation, while the upper limit is determined by²⁵⁹

$$fre_{upper} = \frac{Q_d}{2V_c} \left(1 + \frac{2Y}{\pi}\right), \quad (51)$$

where V_c is the critical volume for a drop protrusion to break up, Y is the dimensionless amplitude of vibration of the form $Y = \pi R_{tubing}^4 \rho_d g Y_0 / (8 Q_d \eta_d l_d)$, with R_{tubing} being the inner wall diameter of microtubing, g the gravitational acceleration, Y_0 the amplitude of vibration in height, and l_d the distance between the vibration location and injection nozzle.

5.4.3 Off-chip valve. Mechanical valves have been externally connected to microfluidic devices for on-demand droplet generation (Fig. 10c)^{18, 261-265}. In doing this, the dispersed-phase pressure is pulsated by valves in a square-wave where open and close statuses are denoted by “on” (duration t_{on}) and “off” (duration t_{off}), respectively. The continuous phase is driven at either constant volumetric flow rate¹⁸, or constant pressure²⁶¹, or pulsatile pressure²⁶⁵. The droplet volume is proportional to t_{on} in the form of $V \sim t_{on} Q_d$. Due to laminar flows in microchannels, the dispersed-phase flow rate Q_d is well described by the Hagen-Poiseuille law $Q_d = \Delta P_d / Rre_d$, where ΔP_d is the pressure drop along the dispersed microchannel of length l_d , and Rre_d is the flow resistance, scaling as $Rre_d \sim \eta_d l_d / w_d h^3$ for rectangular channels. Consequently, prediction of droplet volume yields

$$V \sim \Delta P_d t_{on} / Rre_d. \quad (52)$$

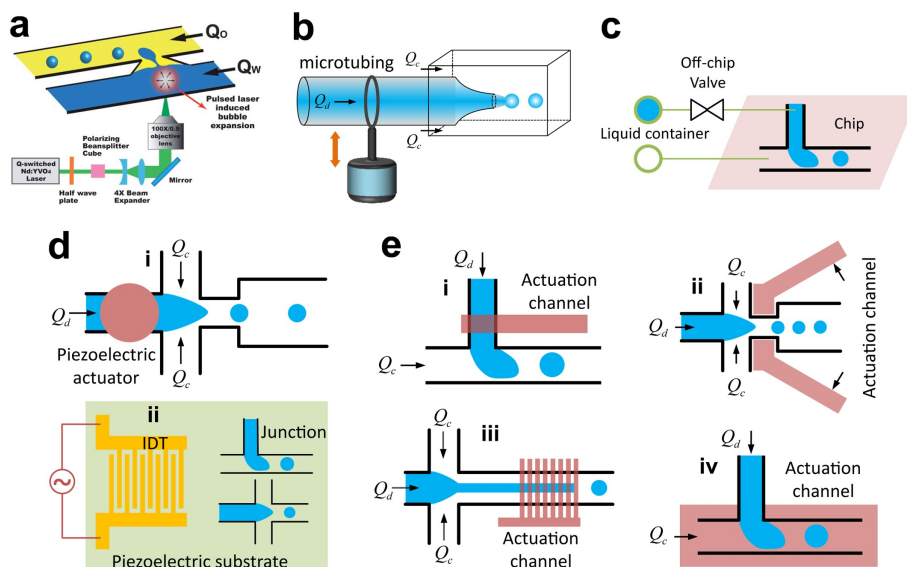


Fig. 10 Active control by modulating dynamic velocity of fluid flows. (a) Schematic of the pulse laser-driven droplet generation device that consists of two microfluidic channels connected by a nozzle-like opening. A highly focused intense laser pulse induces a rapidly expanding cavitation bubble to push the nearby water into the oil channel for droplet formation. (b) Schematic of dispersed fluid pressure perturbed by a mechanical vibrator by shocking the flexible microtubing. (c) Schematic of an off-chip valve regulating the dispersed fluid pressure. Besides, off-chip valve can also be applied to continuous phase fluid. (d) Schematic of piezoelectric actuation. (i) A piezoelectric actuator is incorporated into the dispersed fluid inlet channel. (ii) Droplet generation actuated by SAW. Implementation in both T-junction and flow-focusing devices has been demonstrated. (e) Schematic of on-chip valve actuation. (i) One actuation channel placed upstream the junction. (ii) Two actuation channels located aside the junction. (iii) Chopping of droplet generation by several actuation channels placed atop the downstream fluidic channel. The actuation channels can also be placed aside the fluidic channel. (iv) Deformable droplet generator with an underneath actuation channel covering the entire fluidic channel with a T-junction. This implementation is also applicable to flow-focusing devices. (a) is reproduced with permission from ref. 254. Copyright 2011, Royal Society of Chemistry.

Eq. (52) denotes that droplet volume is proportional to the amplitude of pressure pulse ΔP_d and valve “on” duration t_{on} , consistent with experiments^{18, 261, 265}. In addition, experiments revealed that, in ATPS with a constant continuous-phase flow rate Q_c , droplet volume is inversely proportional to the product of Q_c and t_{off} ¹⁸. As such, V scales as:

$$V \sim \Delta P_d t_{on} / Rre_d Q_c t_{off}. \quad (53)$$

By programming the on-off cycle, sequences of droplets with various volumes are produced, and independent control over droplet volume and generation frequency is attainable. Moreover, Jung *et al.*²⁶⁶ demonstrated on-demand control of droplet numbers per pulse by varying ΔP_d and t_{on} in a step emulsification device equipped with an outside pressure regulator. Compared with systems actuated by electromagnetic valve, those by piezoelectric valve have a higher droplet generation frequency. The frequency is lower than hundred-Hz for the former^{18, 261-264} and up to 400 Hz for the latter²⁶⁵.

5.4.4 Piezoelectric actuation. On-chip piezoelectric control is categorized into two groups: utilizing the pulse of piezoelectric actuator to deform microchannels/chambers (Fig. 10d,i), exploiting surface acoustic wave (SAW) (Fig. 10d,ii). For the former type, the piezoelectric actuator is placed atop the dispersed phase chamber (Fig. 10d,i). The devices are normally fabricated by polydimethylsiloxane (PDMS), and a flexible PDMS membrane separates the chamber and actuator²⁶⁷⁻²⁷². Piezoelectric pulsation can either have on-demand droplet generation^{267, 268} or speed up the droplet production^{270, 271}. In the former case, no droplet is generated in the absence of actuation. Upon actuation, the droplet is dispensed from the junction. The droplet size and generation frequency can be controlled independently. Experiments showed that droplet volume first increases linearly with the pulse amplitude and pulse duration before reaching a plateau²⁶⁸. In addition, the shape of the pulse signal affects the droplet size and its uniformity²⁶⁷. As for the speed-up of droplet generation, the droplet size and generation frequency is controlled by both of the actuation and hydrodynamic

forces. In squeezing regime, piezoelectric actuation accelerates the droplet pinch-off²⁶⁹, and reduces the droplet size²⁷⁰. A similar enhancement was observed in ATPS for jet breakup, where monodisperse hydrogel microspheres were produced under piezoelectric actuation²⁷¹.

Different from channel deformation by the pulse of piezoelectric actuator, SAW utilizes acoustic radiation force (ARF) to deform the liquid-liquid interface by advancing one liquid into another²⁷³. In SAW, ARF arises from the gradient in the density of a traveling acoustic wave such as at an interface between two fluids with different acoustic impedances. In the implementation of SAW, a microfluidic device is placed on top of an interdigitated transducer (IDT) that comprises a series of comb-like metal electrodes arrayed on a piezoelectric substrate (Fig. 10d,ii)²⁷⁴. Both on-demand drop generation^{273, 274} and drop size modulation^{275, 276} are achievable by steering SAW. In water-oil systems where the dispersed water phase is pressurized by focused SAW, on-demand emulsification was realized by manipulating the applied power and pulse duration^{273, 274}. Single or a batch of droplets is produced during SAW actuation. The number of droplets is determined by pulse duration, while droplet size is affected by the applied power, pulse duration, continuous-phase flow rate and channel geometry^{273, 274}. As the continuous-phase flow rate increases, squeezing, dripping and jetting regimes are successively observed²⁷³. Meanwhile, by using unfocused SAW, Schmid and Franke demonstrated the decrease of aqueous droplet size in both flow-focusing²⁷⁵ and T-junction²⁷⁶. The size decrease is explained by an SAW-induced pressure increase in the inlet channel of continuous phase^{275, 276}.

5.4.5 On-chip microvalve. On-chip valves are usually integrated into microfluidic devices, and activated pneumatically by compressed air or hydraulically by the pressurized liquid in actuation channels. Under valve actuation, the fluidic channel is deformed or even blocked, thereby modulating droplet generation. The actuation channel, usually separated by a membrane from the fluidic channel, can be placed upstream of, at, and downstream of

the junction, or covering the entire fluidic channel (Fig. 10e). In implementation, the actuation channel has been designed to locate aside^{277, 278}, atop²⁷⁹, underneath²⁸⁰, and around²⁸¹ the fluidic channel, in which fabrication of aside actuation channels is easy, while the other three arrangements require multi-layer chip fabrication.

When located upstream of the junction, actuation valves are normally used to perturb the dispersed-fluid velocity. Both blocking and deformation of the dispersed channel have been demonstrated by actuation channels^{13, 281-286}. In the blocking case²⁸³⁻²⁸⁶, the resultant droplet size can be well predicted by Eq. (52), in that underlying mechanisms are the same for both on-chip and off-chip valve-blocked systems. In the case of fluidic channel deformed by an actuation channel (Fig. 10e,i), monodisperse droplets are favored in synchronized regime, and polydisperse droplets are produced in quasiperiodic regime²⁸⁷. The synchronized regime occurs when the frequency of valve pulse fre_{pulse} is close to multiple times the frequency of valve-free droplet generation fre_0 , such as $fre_{drop} / fre_0 \approx 1, 3/2, 2, 3, 4, 5$. The under-forcing droplet generation frequency fre_{drop} is synchronized with fre_{pulse} in a manner, $fre_{drop} / fre_{pulse} = 1, 2/3, 1/2, 1/4, 1/5$ ²⁸⁷. Apart from above normally-open valves, normally-closed micro-valve is used where the dispersed channel is initially closed by a PDMS membrane²⁸⁸. Upon valve actuation, the PDMS membrane is deflected to open the dispersed channel. The degree of membrane deflection maneuvers the dispersed flow velocity, so that droplet size is tunable.

Valves located at the junction modulate drop generation by directly affecting the liquid-liquid interface through fluidic channel deformation or blocking. In the deformation case, variations in junction dimension (Fig. 10e,ii) make the continuous-phase shear stress unsteady and the droplet size tunable²⁷⁷⁻²⁷⁹. In the case of channel blocking, the volume of the droplet V was found to increase, either linearly²⁸⁹ or nonlinearly²⁹⁰, with the valve "on" time t_{on} :

$$V = m_1 t_{on}^{m_2}, \quad (54)$$

where m_1 and m_2 ($m_2 \geq 1$) are constants mainly depending on fluid driving pressures. The nonlinearity ($m_2 > 1$) might come from the transient dispersed-phase flow rate during valve actuation²⁹⁰. Increasing the dispersed driving pressure leads m_2 close to 1, probably because the transient effect is depressed²⁹⁰.

The downstream microvalve (Fig. 10e,iii) can break discrete droplets secondarily into daughter droplets²⁹¹ and chop the prefocusing stable stream into individual droplets²⁹²⁻²⁹⁴. In both situations, the resultant droplet sizes decrease with increased pressure that activates the valve. The stable stream is fragmented through a chopping method²⁹²⁻²⁹⁴ where two or more choppers deflect the fluidic channel. The resultant droplets are monodisperse ($CV < 3\%$)²⁹², and the size depends on flow rate ratio and pulse frequency of the chopper. However, the working frequency was relatively low (< 17.4 Hz)^{293, 294}.

Raj *et al.*²⁸⁰ recently proposed a deformable droplet generator (Fig. 10e,iv) where the actuation channel lay underneath the entire fluidic channel. By characterizing the deformation of fluidic channel in response to the actuation pressure, Raj *et al.*²⁸⁰ determined the relationship between droplet size and channel deformation, flow rate ratio, and viscosity ratio. The results show that monodisperse droplets are generated in dripping regime for a wider range of capillary number Ca_c in comparison to passive method. It is also possible to produce droplets of size slightly smaller than the junction size in this deformable device.

5.5 Modifying material properties

5.5.1 Viscosity. Three methods have been adopted to actively manipulate the viscosity of liquids: electrorheological (ER) fluid by electrical control²⁹⁵, magnetorheological (MR) fluid by magnetic control²⁹⁶, and heating the liquids by thermal control²⁹⁷. In

electrical control, ER fluid consists of non-conducting but electrically active particles dispersed in a carrier insulating liquid (usually oil)²⁹⁸. When subjected to an electric field, the suspended particles are electrically polarized, and aggregate into chains/columns along the field direction (Fig. 11a). These chains/columns restrict the fluid flow perpendicular to the applied field direction, by which the ER fluid exhibits increased apparent viscosity and even solid-like behavior²⁹⁸. The rheological variation of ER fluid is reversible and displays fast response to electric field (response time down to milliseconds)²⁹⁵. Zhang *et al.*²⁹⁵ utilized a type of ER fluid with the giant ER effect (GER)²⁹⁸ as the continuous phase to actively control droplet generation. Upon the turn-on of a sufficiently-strong electric field, the continuous ER fluid stops flowing, and only the dispersed water advances into the junction. When the electric field is decreased to a critical value, the ER fluid rushes into the junction and breaks off the water thread. By programming the electrical control signal, tunable droplet size and droplet generation frequency are attainable. The GER fluid was also used as the dispersed phase by the same group²⁹⁹.

MR fluid exhibits tunable rheological characteristics due to the formation of chains/columns under a magnetic field. To display MR effect, the magnetic particles should be large enough (on the order of 100 nm to 10 μ m) to dominate over the effect of Brownian motion, different from those in ferrofluid with size smaller than 10 nm²⁹⁶. Despite the unique characteristics of MR fluid, few applications of MR fluids to droplet generation were reported in the literature. Although it is weak, MR effect exists in ferrofluid under certain circumstances. In droplet generation with the dispersed phase being water-based ferrofluid containing Fe_3O_4 nanoparticles (average size: 100 nm)²⁴¹, the nanoparticles align to form chains-like structure along the direction of magnetic field (Fig. 8c). Ferrofluid droplets with magnetically-increased viscosity are larger than water droplets with lower viscosity²⁴¹. However, the MR effect is limited by the small particle size and counteracted by the magnetic drag that tends to decrease the droplet size.

Thermal control exploits the dependence of fluid viscosity and interfacial tension on temperature. The microfluidic device is heated by microheaters²⁹⁷ or heat exchangers³⁰⁰. The resultant droplet size $D(T)$ is a function of the combined effect of temperature-dependent viscosity and interfacial tension^{297, 300}:

$$D(T) \propto Ca_c^{-1} = \frac{\gamma(T)}{\eta_c(T)u_c}, \quad (55)$$

where T is the temperature. For most liquids, viscosity and interfacial tension decrease as temperature increases, but the decrease of viscosity is faster than that of interfacial tension. Consequently, increasing temperature will enlarge the droplet size (Fig. 11b)^{297, 300}. By selectively choosing continuous fluids that have a much larger variation of viscosity than that of interfacial tension, Stan *et al.*³⁰⁰ demonstrated the increase in water droplet volume by almost two orders of magnitude when the temperature is increased from 0 to 80 °C. As droplet size increases with temperature, droplet generation mode could transition from dripping to squeezing³⁰¹. In squeezing mode, the droplet size is mainly determined by the flow rate ratio (Eq. (12)), thus depending weakly on temperature. Channel height also influences the correlation between droplet size and temperature. A decreased channel height shows an enhanced temperature dependence of droplet size due to the larger temperature gradient³⁰².

Besides water, water-based nanofluid droplet generation by thermal control has been investigated in both flow-focusing³⁰¹ and T-junction^{302, 303} configurations. In flow-focusing geometry, water and nanofluid droplet formation exhibit similar characteristics at different temperatures³⁰¹. In T-junctions, however, the size of water droplet can be either larger or smaller than that of nanofluid droplet, depending on the channel dimension and flow rate ratio^{302, 303}.

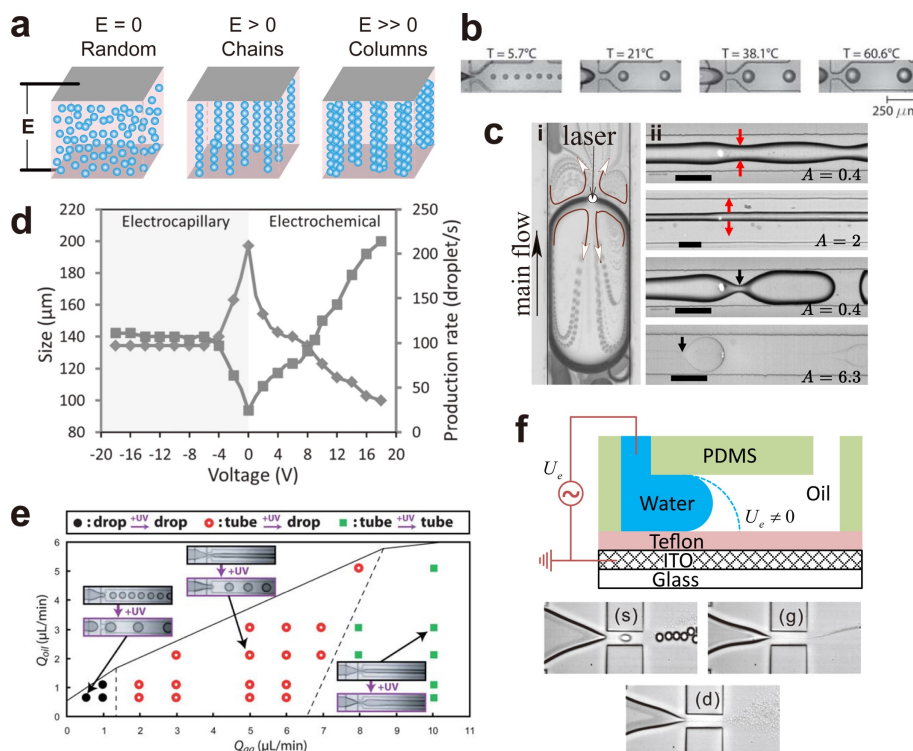


Fig. 11 Active droplet generation via modifying material properties. (a) Schematic of chains and columns formation with increasing electric field in ER fluid. (b) Images of water drops generated in mineral oil at different temperatures. (c) Laser control of droplet generation. (i) Overlaid images showing the motion of seeding particles near the hot spot. Note that the motion along the interface is directed towards the hot spot. (ii) Optocapillary deformation of the liquid interface by focused laser beam. Upper two images: a thick thread ($A < 1$) pinching while a thinner one ($A > 1$) bulges (indicated by red arrows). Lower two images: for large enough power, the thread eventually breaks in either pinching ($A = 0.4$) or bulging ($A = 6.3$) case (indicated by black arrows). (d) Plot of size (diamonds) and production rate (squares) versus DC voltage for both electrochemical and electrocapillary cases. (e) Effect of UV illumination for different flow rates of continuous fluid (Q_{oil}) and AzoTAB solution (Q_{Azo}) in a flow-focusing device. (f) EW actuation of droplet generation. Upper: schematic of EW implementation. Upon actuation, the contact angle of water on Teflon substrate decreases (dashed curve) and induces a smaller mean curvature of the water meniscus. Lower: images of dripping, tip-streaming and conical spray regimes (from left to right, top to down) under EW actuation. (b) is reproduced with permission from ref. 300. Copyright 2009, American Chemical Society. (c,i) is reproduced with permission from ref. 307. Copyright 2007, American Physical Society. (c,ii) is adapted with permission from ref. 310. Copyright 2015, American Physical Society. (d) is reproduced with permission from ref. 313. Copyright 2015, Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim. (e) is reproduced with permission from ref. 315. Copyright 2011, Royal Society of Chemistry. Images in (f) are adapted with permission from ref. 319. Copyright 2008, American Institute of Physics.

Moreover, the shape of nanoparticle impacts the dependence of droplet size on temperature. The droplet size increases with increasing temperature for nanofluids with spherical-shaped nanoparticles, but decreases for nanofluids with cylindrical-shaped nanoparticles³⁰². The underlying mechanism is unclear yet.

Different from above studies with both continuous and dispersed fluids heated, heating the dispersed phase exclusively can also be able to tune droplet size. Yeh *et al.*³⁰⁴ reported the generation of uniform gelatin emulsions (CV < 5%) by heating the gelatin phase. When the temperature is over 25 °C by heating, the gelatin transforms from a gel to an aqueous solution with decreased viscosity. The viscosity decrease is responsible for the hastened release of the gelatin droplet, which reduces the continuous phase supply during the formation of each droplet and renders the droplet larger³⁰⁴.

5.5.2 Interfacial tension. Active control of the interfacial tension can be achieved by thermal control^{297, 300, 305, 306}, optical control³⁰⁷⁻³¹² and electrical control³¹³. Thermal control involves either temperature-dependent interfacial tension^{297, 300} or Marangoni effect^{305, 306}. Similarly, optical control exploits the Marangoni effect by local laser heating³⁰⁷⁻³¹¹. In parallel, electrical control modulates the interfacial tension by electrochemical and electrocapillary effects³¹³. Note that laser heating is occasionally referred to as thermal control²⁵. We here group laser heating into optical control due to its use of a laser beam. In this subsection, we discuss exclusively the

manipulation of interfacial tension at liquid-liquid interface, distinct from that at liquid-solid interface which is classified into tunable channel wettability in next subsection.

Thermal control can be categorized into uniform and non-uniform heating. In uniform heating, the heating region covers the entire liquid-liquid interface, by which the interfacial tension is tuned homogeneously by temperature in the absence of interfacial tension gradient. In this situation, droplet size is controlled by Eq. (55), as we have discussed. In contrast, non-uniform heating involves locally-heating part of the liquid-liquid interface and producing a thermal gradient. The thermal gradient gives rise to a gradient of interfacial tension that generates Marangoni stress (tangential stress) along the liquid-liquid interface. The Marangoni stress, in turn, induces an interfacial flow that affects the droplet formation. In the simulation by Suryo and Basaran³⁰⁵, heating the liquid-nozzle contact line gave rise to droplet generation owing to the competition between Marangoni stress and capillary pressure. Moreover, in a T-junction configuration, Miralles *et al.*³⁰⁶ demonstrated the control over droplet size and generation frequency by heating the dispersed entrance of the junction.

In optical control involving laser heating, the laser beam is focused through a microscope objective, locally heating the centerline of the flow downstream of the junction. A dye (a fluorescein for example) is added to the dispersed aqueous phase to mediate laser absorption³⁰⁷⁻³⁰⁹. In the presence of Marangoni stress,

dispersed bubbles and/or droplets would be forced to migrate with velocity u_m . Recent experiments suggested that the migration velocity u_m is proportional to the product of channel width w and the mean temperature gradient $\langle \nabla T \rangle$ for aqueous droplets in confined rectangular microchannels³⁰⁹:

$$u_m \propto w \langle \nabla T \rangle, \quad (56)$$

with $\langle \nabla T \rangle$ proportional to the laser power. In the presence of surfactants (where interfacial tension was found to rise with increasing temperature^{307, 314}), locally heating of the advancing interface of a growing droplet leads to a migration velocity against the basic flow in the microchannel, thus slowing down or even blocking the droplet (Fig. 11c,i)³⁰⁷⁻³⁰⁹. As such, active tuning of the droplet generation is enabled by local laser heating.

Laser-heating-mediated droplet generation has been demonstrated in both dripping³⁰⁷⁻³⁰⁹ and jetting^{310, 311} flows, and in both flow-focusing³⁰⁷ and T-junction³⁰⁸ microchannels. In dripping regime, the advancing water-oil interface is blocked at the laser heating spot once the heating power exceeds the threshold, in which case the migration velocity u_m is larger than the velocity of the droplet imposed by the basic flow³⁰⁹. The net force exerted on the droplet due to the Marangoni effect is theoretically³⁰⁷ and experimentally³¹² estimated to be of the order $\sim 0.1 \mu\text{N}$, a sufficiently large force accounting for the blocking. For constant-pressure-driven basic flow, the length of the droplet with laser blocking (L_{block}) is shorter than that of the unblocked droplet (L_0), while for the constant-flow-rate-driven case, L_{block} is larger than L_0 and proportional to blocking time τ_b ³⁰⁹:

$$L_{\text{block}} = L_0 + \frac{Q_d}{hw} \tau_b. \quad (57)$$

The difference in droplet size variation arises from the different dynamics of neck width evolution. During blocking, the neck width thins exponentially but increases linearly with time at the constant-pressure and constant-flow-rate case, respectively³⁰⁹.

In jetting regime, the interface of the jet is deformed by the Marangoni stress³¹⁰. The orientation of the deformation is predicted by the following dimensionless ratio A ³¹⁰:

$$A = \frac{H_c}{H_d} \sqrt{\frac{\eta_d}{2\eta_c}}, \quad (58)$$

with H_d and H_c representing the height of the dispersed jet and that of the continuous fluid, respectively (H_c and H_d are related by $H_c + H_d = 0.5h$, with h being the channel height). Provided that $\rho_d > \rho_c$ and the laser heating spot has the largest interfacial tension, $A > 1$ (thin jet) indicates the deformation pointing from the dispersed to the continuous phase, and the jet bulges. Conversely, $A < 1$ (thick jet) stands for the pinching of the jet (Fig. 11c,ii)³¹⁰. When laser power is above a critical value, the Marangoni stress is sufficiently large to break the jet into droplets. In the bulging case, the breakup occurs upstream of the heating spot. On the contrary, fragmentation of the jet occurs downstream of the hot spot where the pinched neck becomes locally unconfined (Fig. 11c,ii)³¹⁰. With continuous heating, the droplet production exhibits a transition from period-1 with high monodispersity, to period-2 with two distinct droplets per sequence, and back to period-1 regime as laser power increases³¹⁰. In addition, Cordero *et al.* observed the breakup of an aqueous jet under sinusoidal laser heating³¹¹. The jet's oscillation can synchronize to laser frequency, and produce droplets with decreased polydispersity.

In electrical control employing electrochemical and electrocapillary effects³¹³, the liquid metal EGaIn (75% gallium, 25% indium) and glycerol-NaOH solution are used as the dispersed and continuous phase, respectively. The voltage is applied by connecting the anode and cathode to the liquid metal and glycerol-NaOH solution inlet tubes, respectively, or vice versa. Either case

can lower the interfacial tension between the two fluids. In the application of a positive (oxidative) bias to the liquid metal phase, electrochemically-induced formation of a surface oxide acts as a surfactant and lowers the interfacial tension. In the opposite case where the liquid metal phase is negatively charged, the interfacial tension is decreased due to electrocapillary effect. As such, the droplet size decreases with applied voltage in both cases (Fig. 11d) owing to the increase in capillary number Ca_c . The size decrease from electrochemical effect is in a larger range than that from electrocapillary effect (Fig. 11d). Different from DC voltage that produces uniform droplets, low-frequency AC control demonstrates its ability to generate sequences of droplets with various sizes. The size variation originates from the varying magnitude of the voltage.

5.5.3 Channel wettability. Both optical and electrical controls have been used to vary the wettability of channel wall/substrate, where the former utilizes a photosensitive surfactant, and the latter makes use of EW effect. In optical control, the surfactant azobenzene trimethylammonium bromide (AzoTAB), with a photosensitive apolar tail, is introduced into the dispersed aqueous phase³¹⁵. AzoTAB is available in two configurations: *trans* (less polar) and *cis* (more polar) forms. The initial *trans* (*cis*) form isomerizes to *cis* (*trans*) form upon UV (blue light) illumination, and the transition is reversible when the light is off. Compared to *trans*-AzoTAB, *cis*-AzoTAB has a lower substrate wettability. Consequently, the UV-induced dewetting transition, associated with imposed flow rates, results in three behaviors: (i) preserving the initially-dripping regime but producing larger droplets; (ii) breaking the initially-stable jet into successive droplets; (iii) stabilizing the jet but expanding in its radius (Fig. 11e). In contrast, a blue-light-induced *cis*-to-*trans* transition converts the dripping into the stable jet regime.

In electrical control using EW effect, high-frequency AC is applied³¹⁶ to change the wetting state of the microchannel substrate (Fig. 11f). Under applied voltage U_e , no droplet is generated until the dispersed pressure P_d exceeds an onset value $P_d^*(U_e, P_c)$ ^{316, 317}:

$$P_d^*(U_e, P_c) = P_L(U_e) + CP_c, \quad (59)$$

where P_c is the pressure of the continuous phase, $P_L(U_e)$ is the Laplace pressure, and C is a fitting parameter depending on the fluidic resistance of the channel. Laplace pressure $P_L(U_e)$ is expressed as $P_L(U_e) = \gamma (R_w^{-1} + R_h^{-1})$, with R_w and R_h being the principal radii of curvature in the horizontal and vertical planes, respectively. Upon applying the voltage, the contact angle of the dispersed liquid on the channel substrate decreases (Fig. 11f) and R_h , therefore, increases (R_h is initially positive). As γ and R_w are constant, $P_L(U_e)$ and $P_d^*(U_e, P_c)$ thus decreases with the increase of voltage U_e ^{316, 317}. With appropriate design of EW control, it is even possible to have negative values of R_h and $P_L(U_e)$ ³¹⁸. Consequently, spontaneous imbibition ($P_d^*(U_e, P_c) = 0$) of the dispersed fluid into the microchannel can be achieved³¹⁸.

Three regimes of droplet generation in EW have been identified: dripping, tip-streaming and conical spray (Fig. 11f)³¹⁹. Dripping occurs when the dispersed pressure P_d is relatively high, while tip-streaming and conical spray occur at intermediate P_d (at low P_d , no droplet generation). Compared with the tip-streaming, the conical spray occurs at a higher voltage where droplets are charged and repulse each other. While the size of generated droplets is comparable to the dimension of the orifice in dripping regime, they can be smaller than 10 μm in diameter in tip-streaming and conical spray regimes. Generally, the droplet size increases monotonically with increasing P_d ³¹⁹. Nevertheless, the influence of applied voltage is complicated. When the continuous-phase viscous force ($\sim \eta_c u_c / L$) is weak (e.g., low P_c) compared with the electrostatic force ($\sim \epsilon_c E^2$), the droplet size increases with increasing voltage U_e ³¹⁷; when the viscous force is comparable to electrostatic force, the droplet size is nearly constant as U_e increases^{317, 319}, but the frequency of droplet

generation can be varied by one order of magnitude³¹⁷. The latter offers possible on-demand generation of droplets³²⁰.

5.5.4 Fluid density. Modulating droplet size via varying fluid density is enabled by phase change process. Wang *et al.*³²¹ proposed a temperature-controlled vapor bubble condensation process by which the dispersed liquid (*n*-pentane) is first vaporized in the feeding pipe and then breaks up into vapor bubbles in the co-flow sub-millimeter fluidic channels, followed by a cooling process where vapor bubbles condense into microdroplets. The droplet diameter D is found to be proportional to the vapor bubble diameter D_{bubble} : $D = 0.24D_{\text{bubble}}$. This method demonstrates the capability of producing uniform (CV of 2.2 – 4.0%) microdroplets (43 – 200 μm in diameter) at a moderate generation frequency (1.3 – 2000 Hz) in sub-millimeter fluidic devices.

6. Concluding remarks

This review is on fundamentals of passive and active generation of microfluidic droplets. Five fundamental modes of droplet generation, squeezing, dripping, jetting, tip-streaming, tip-multi-breaking are identified in passive generation. Each of them has its unique characteristics owing to distinct flow instabilities. Suppression of capillary instability by confinement primarily leads to the superior control and monodispersity of droplet size in squeezing regime. Dripping, from an absolute flow instability that exhibits intrinsic oscillations, prevents external noises, thus producing monodisperse droplets, as well. Convectively-unstable flows that amplify perturbations of various frequencies render, on the other hand, droplets non-uniform in jetting regime. Tip-streaming mode enables droplet microfluidics to produce droplets of size (down to submicrometer-scale) unrestricted by the characteristic length scale of devices. The periodic oscillation of the globally-unstable tip offers precise control over droplet size distribution in tip-multi-breaking mode. A deep understanding regarding effects of viscous, inertial and capillary forces and governing dimensionless parameters (*e.g.*, capillary and Weber numbers) is critical for effective production of droplets with the desired outcome.

Active droplet generation is achieved by incorporating additional components into microfluidic systems. The state-of-the-art techniques mainly apply electrical, magnetic, centrifugal, optical, thermal and mechanical methods, by which electrical, magnetic and centrifugal forces are introduced, and the flow, viscosity, interfacial tension, channel wettability and fluid density are varied. In principle, any approach that could affect interfacial force balance by applying external forces and/or modifying intrinsic viscous, inertial and capillary forces is capable of actively controlling droplet generation. Based on this speculation, new techniques for active control can be developed, for example using magnetorheological effect²⁹⁶ to tune viscosity, thermo-sensitive surfactant (such as diethylhexyl sodium sulphosuccinate³²²) to tune interfacial tension, and magnetowetting effect²⁹⁶ to tune channel wettability. Active control, therefore, offers additional flexibility and tool in manipulating droplet formation.

The several outstanding issues remain to be resolved in order to produce droplets precisely and effectively. In passive generation, it is essential to design a system that enables a stable tip-streaming over a long period of time. As occurring at quite low flow rate ratio ($\phi \ll 1$), the tip-streaming is stable typically in less than several minutes before being destabilized by the variation in flow rates from syringe pumps. Using constant-pressure-driven flows in 3D devices is beneficial for the stabilization¹². Further studies of tip-multi-breaking dynamics are required to produce digital droplets with precise inter-correlation, and thus developing novel applications of droplet microfluidics. For active droplet generation, one of the current challenges is to miniaturize and parallelize the system. In addition, most microfluidic chips with active components suffer

from complicity in fabrication. Compared with the on-chip integration (*e.g.*, electrodes, microheaters, piezoelectric actuators), the off-chip-actuated devices are easier in fabrication. Moreover, in the on-demand droplet generation, limited systems are able to produce high-throughput ($> \text{kHz}$) monodisperse (CV $< 5\%$) droplets in a wide range of tunable volumes (*e.g.*, from femtolitre-scale to nanolitre-scale). This requires the smart design of microfluidic junctions and development of fast-responding actuation. The pulse laser-driven droplet generation²⁵⁴ and piezoelectric dispenser²⁶⁷ appear promising. The delivery of promising potential of droplet microfluidics relies on our fundamental understanding of mechanisms responsible for and developing of effective techniques for droplet generation.

7. Acknowledgements

The financial support from the Research Grants Council of Hong Kong (GRF 17237316, 17211115, 17207914 and 717613E) and the University of Hong Kong (URC 201511159108, 201411159074 and 201311159187) is gratefully acknowledged. The work is also supported in part by the Zhejiang Provincial, Hangzhou Municipal, and Lin'an County Governments.

8. References

1. S. Y. Teh, R. Lin, L. H. Hung and A. P. Lee, *Lab Chip*, 2008, **8**, 198-220.
2. C. N. Baroud, F. Gallaire and R. Danga, *Lab Chip*, 2010, **10**, 2032-2045.
3. S. S. Datta, A. Abbaspourrad, E. Amstad, J. Fan, S.-H. Kim, M. Romanowsky, H. C. Shum, B. Sun, A. S. Utada, M. Windbergs, S. Zhou and D. A. Weitz, *Adv. Mater.*, 2014, **26**, 2205-2218.
4. T. Y. Lee, T. M. Choi, T. S. Shim, R. A. M. Frijns and S.-H. Kim, *Lab Chip*, 2016, **16**, 3415-3440.
5. T. T. Kong, J. Wu, K. W. K. Yeung, M. K. T. To, H. C. Shum and L. Q. Wang, *Biomicrofluidics*, 2013, **7**, 044128.
6. T. T. Kong, J. Wu, M. To, K. W. K. Yeung, H. C. Shum and L. Q. Wang, *Biomicrofluidics*, 2012, **6**, 034104.
7. T. T. Kong, Z. Liu, Y. Song, L. Q. Wang and H. C. Shum, *Soft Matter*, 2013, **9**, 9780-9784.
8. J. Wu, T. T. Kong, K. W. K. Yeung, H. C. Shum, K. M. C. Cheung, L. Q. Wang and M. K. T. To, *Acta Biomater.*, 2013, **9**, 7410-7419.
9. Z. X. Kang, T. T. Kong, L. Y. Lei, P. A. Zhu, X. W. Tian and L. Q. Wang, *J. Micromech. Microeng.*, 2016, **26**, 075011.
10. J. Nunes, S. Tsai, J. Wan and H. Stone, *J. Phys. D: Appl. Phys.*, 2013, **46**, 114002.
11. T. D. Martz, D. Bardin, P. S. Sheeran, A. P. Lee and P. A. Dayton, *Small*, 2012, **8**, 1876-1879.
12. W.-C. Jeong, J.-M. Lim, J.-H. Choi, J.-H. Kim, Y.-J. Lee, S.-H. Kim, G. Lee, J.-D. Kim, G.-R. Yi and S.-M. Yang, *Lab Chip*, 2012, **12**, 1446-1453.
13. Y. Zeng, M. Shin and T. Wang, *Lab Chip*, 2013, **13**, 267-273.
14. P. Garstecki, H. A. Stone and G. M. Whitesides, *Phys. Rev. Lett.*, 2005, **94**, 164501.
15. A. S. Utada, A. Fernandez-Nieves, H. A. Stone and D. A. Weitz, *Phys. Rev. Lett.*, 2007, **99**, 094502.
16. R. Suryo and O. A. Basaran, *Phys. Fluids*, 2006, **18**, 082102.
17. P. A. Zhu, T. T. Kong, Z. X. Kang, X. W. Tian and L. Q. Wang, *Sci. Rep.*, 2015, **5**, 11102.

18. B.-U. Moon, S. G. Jones, D. K. Hwang and S. S. H. Tsai, *Lab Chip*, 2015, **15**, 2437-2444.
19. Y. Li, M. Jain, Y. Ma and K. Nandakumar, *Soft Matter*, 2015, **11**, 3884-3899.
20. D. J. Collins, A. Neild, A. deMello, A.-Q. Liu and Y. Ai, *Lab Chip*, 2015, **15**, 3439-3459.
21. G. Christopher and S. Anna, *J. Phys. D: Appl. Phys.*, 2007, **40**, R319-R336.
22. C.-X. Zhao and A. P. J. Middelberg, *Chem. Eng. Sci.*, 2011, **66**, 1394-1411.
23. R. Seemann, M. Brinkmann, T. Pfohl and S. Herminghaus, *Rep. Prog. Phys.*, 2012, **75**, 016601.
24. S. L. Anna, *Annu. Rev. Fluid Mech.*, 2016, **48**, 285-309.
25. Z. Z. Chong, S. H. Tan, A. M. Ganan-Calvo, S. B. Tor, N. H. Loh and N.-T. Nguyen, *Lab Chip*, 2016, **16**, 35-58.
26. A. M. Gañán-Calvo, *Phys. Rev. Lett.*, 1998, **80**, 285.
27. H. A. Stone, A. D. Stroock and A. Ajdari, *Annu. Rev. Fluid Mech.*, 2004, **36**, 381-411.
28. T. M. Squires and S. R. Quake, *Rev. Mod. Phys.*, 2005, **77**, 977.
29. G. Batchelor, *An Introduction to Fluid Dynamics*, Cambridge University Press, Cambridge, 1967.
30. P. Garstecki, M. J. Fuerstman and G. M. Whitesides, *Phys. Rev. Lett.*, 2005, **94**, 234502.
31. J. R. Lister and H. A. Stone, *Phys. Fluids*, 1998, **10**, 2758-2764.
32. B. Dollet, W. van Hoeve, J.-P. Raven, P. Marmottant and M. Versluis, *Phys. Rev. Lett.*, 2008, **100**, 034504.
33. T. Thorsen, R. W. Roberts, F. H. Arnold and S. R. Quake, *Phys. Rev. Lett.*, 2001, **86**, 4163.
34. J. Xu, S. Li, J. Tan, Y. Wang and G. Luo, *AIChE J.*, 2006, **52**, 3005-3010.
35. J. H. Xu, G. S. Luo, S. W. Li and G. G. Chen, *Lab Chip*, 2006, **6**, 131-136.
36. L. Q. Wang, Y. X. Zhang and L. Cheng, *Chaos Soliton. Fract.*, 2009, **39**, 1530-1537.
37. K. Khoshmanesh, A. Almansouri, H. Albloushi, P. Yi, R. Soffe and K. Kalantar-zadeh, *Sci. Rep.*, 2015, **5**, 9942.
38. A. R. Abate and D. A. Weitz, *Lab Chip*, 2011, **11**, 1713-1716.
39. L. Ménétrier-Deremble and P. Tabeling, *Phys. Rev. E*, 2006, **74**, 035303.
40. A. Abate, A. Poitzsch, Y. Hwang, J. Lee, J. Czerwinska and D. Weitz, *Phys. Rev. E*, 2009, **80**, 026310.
41. L. Shui, F. Mugele, A. van den Berg and J. C. T. Eijkel, *Appl. Phys. Lett.*, 2008, **93**, 153113.
42. L. Shui, A. van den Berg and J. C. T. Eijkel, *J. Appl. Phys.*, 2009, **106**, 124305.
43. M. L. Steegmans, K. G. Schroën and R. M. Boom, *Langmuir*, 2009, **25**, 3396-3401.
44. H. Song, D. L. Chen and R. F. Ismagilov, *Angew. Chem. Int. Ed.*, 2006, **45**, 7336-7356.
45. B. Zheng, J. D. Tice and R. F. Ismagilov, *Anal. Chem.*, 2004, **76**, 4977-4982.
46. Y. X. Zhang and L. Q. Wang, in *Advances in Transport Phenomena 2010*, Springer-Verlag, Berlin Heidelberg, 2011, pp. 171-294.
47. L. Frenz, J. Blouwolff, A. D. Griffiths and J.-C. Baret, *Langmuir*, 2008, **24**, 12073-12076.
48. R. Lin, J. S. Fisher, M. G. Simon and A. P. Lee, *Biomicrofluidics*, 2012, **6**, 024103.
49. M. Rhee, P. Liu, R. J. Meagher, Y. K. Light and A. K. Singh, *Biomicrofluidics*, 2014, **8**, 034112.
50. Y. Ding, X. C. i Solvas and A. deMello, *Analyst*, 2015, **140**, 414-421.
51. U. Tangen, A. Sharma, P. Wagler and J. S. McCaskill, *Biomicrofluidics*, 2015, **9**, 014119.
52. T. Nisisako and T. Torii, *Lab Chip*, 2008, **8**, 287-293.
53. D. Link, S. L. Anna, D. Weitz and H. Stone, *Phys. Rev. Lett.*, 2004, **92**, 054503.
54. P. B. Umbanhowar, V. Prasad and D. A. Weitz, *Langmuir*, 2000, **16**, 347-351.
55. A. Utada, L.-Y. Chu, A. Fernandez-Nieves, D. Link, C. Holtze and D. Weitz, *MRS Bull.*, 2007, **32**, 702-708.
56. J. R. Anderson, D. T. Chiu, H. Wu, O. Schueller and G. M. Whitesides, *Electrophoresis*, 2000, **21**, 27-40.
57. C. Cramer, P. Fischer and E. J. Windhab, *Chem. Eng. Sci.*, 2004, **59**, 3045-3058.
58. J. M. Gordillo, A. Sevilla and F. Campo-Cortés, *J. Fluid Mech.*, 2014, **738**, 335-357.
59. E. Castro-Hernández, V. Gundabala, A. Fernández-Nieves and J. M. Gordillo, *New J. Phys.*, 2009, **11**, 075021.
60. A. M. Gañán-Calvo and J. M. Gordillo, *Phys. Rev. Lett.*, 2001, **87**, 274501.
61. S. Takeuchi, P. Garstecki, D. B. Weibel and G. M. Whitesides, *Adv. Mater.*, 2005, **17**, 1067-1072.
62. L. Yobas, S. Martens, W.-L. Ong and N. Ranganathan, *Lab Chip*, 2006, **6**, 1073-1079.
63. S. L. Anna, N. Bontoux and H. A. Stone, *Appl. Phys. Lett.*, 2003, **82**, 364.
64. A. Utada, E. Lorenceau, D. Link, P. Kaplan, H. Stone and D. Weitz, *Science*, 2005, **308**, 537-541.
65. M. Hein, J.-B. Fleury and R. Seemann, *Soft Matter*, 2015, **11**, 5246-5252.
66. E. M. Chan, A. P. Alivisatos and R. A. Mathies, *J. Am. Chem. Soc.*, 2005, **127**, 13854-13861.
67. C. Priest, S. Herminghaus and R. Seemann, *Appl. Phys. Lett.*, 2006, **88**, 024106.
68. M. Leman, F. Abouakil, A. D. Griffiths and P. Tabeling, *Lab Chip*, 2015, **15**, 753-765.
69. B. Shen, J. Ricouvier, F. Malloggi and P. Tabeling, *Adv. Sci.*, 2016, **3**, 1600012.
70. F. Malloggi, N. Pannacci, R. Attia, F. Monti, P. Mary, H. Willaime, P. Tabeling, B. Cabane and P. Poncet, *Langmuir*, 2010, **26**, 2369-2373.
71. L. Shui, A. van den Berg and J. C. Eijkel, *Microfluid. Nanofluid.*, 2011, **11**, 87-92.
72. L. Shui, E. Stefan Kooij, D. Wijnperle, A. van den Berg and J. C. T. Eijkel, *Soft Matter*, 2009, **5**, 2708-2712.
73. J. Guzowski and P. Garstecki, *Phys. Rev. Lett.*, 2015, **114**, 188302.
74. K. C. van Dijke, G. Veldhuis, K. Schroën and R. M. Boom, *AIChE J.*, 2010, **56**, 833-836.
75. V. Chokkalingam, S. Herminghaus and R. Seemann, *Appl. Phys. Lett.*, 2008, **93**, 254101.
76. C. Cohen, R. Giles, V. Sergeyeve, N. Mittal, P. Tabeling, D. Zerrouki, J. Baudry, J. Bibette and N. Bremond, *Microfluid. Nanofluid.*, 2014, **17**, 959-966.
77. N. Mittal, C. Cohen, J. Bibette and N. Bremond, *Phys. Fluids*, 2014, **26**, 082109.
78. R. Dangla, S. C. Kayi and C. N. Baroud, *Proc. Natl. Acad. Sci. U.S.A.*, 2013, **110**, 853-858.
79. K. van Dijke, G. Veldhuis, K. Schroen and R. Boom, *Lab Chip*, 2009, **9**, 2824-2830.
80. K. van Dijke, R. de Ruiter, K. Schroen and R. Boom, *Soft Matter*, 2010, **6**, 321-330.
81. F. Dutka, A. S. Opalski and P. Garstecki, *Lab Chip*, 2016, **16**, 2044-2049.
82. F. Schuler, F. Schwemmer, M. Trotter, S. Wadle, R. Zengerle, F. von Stetten and N. Paust, *Lab Chip*, 2015, **15**, 2759-2766.
83. F. Schuler, N. Paust, R. Zengerle and F. von Stetten,

- Micromachines*, 2015, **6**, 1180-1188.
84. F. Schuler, M. Trotter, M. Geltman, F. Schwemmer, S. Wadle, E. Dominguez-Garrido, M. Lopez, C. Cervera-Acedo, P. Santibanez, F. von Stetten, R. Zengerle and N. Paust, *Lab Chip*, 2016, **16**, 208-216.
 85. S. Kahkeshani and D. Di Carlo, *Lab Chip*, 2016, **16**, 2474-2480.
 86. G. T. Vladislavjević, I. Kobayashi and M. Nakajima, *Microfluid. Nanofluid.*, 2012, **13**, 151-178.
 87. G. T. Vladislavjević, N. Khalid, M. A. Neves, T. Kuroiwa, M. Nakajima, K. Uemura, S. Ichikawa and I. Kobayashi, *Adv. Drug Deliv. Rev.*, 2013, **65**, 1626-1663.
 88. I. Kobayashi, Y. Wada, K. Uemura and M. Nakajima, *Microfluid. Nanofluid.*, 2010, **8**, 255-262.
 89. Y. Kikuchi, H. Ohki, T. Kaneko and K. Sato, *Biorheology*, 1989, **26**, 1055.
 90. T. Kawakatsu, Y. Kikuchi and M. Nakajima, *J. Am. Oil Chem. Soc.*, 1997, **74**, 317-321.
 91. T. Kawakatsu, H. Komori, M. Nakajima, Y. Kikuchi and T. Yonemoto, *J. Chem. Eng. Jpn.*, 1999, **32**, 241-244.
 92. T. Kawakatsu, G. Trägårdh, Y. Kikuchi, M. Nakajima, H. Komori and T. Yonemoto, *J. Surfactants Deterg.*, 2000, **3**, 295-302.
 93. I. Kobayashi, M. Nakajima, K. Chun, Y. Kikuchi and H. Fujita, *AIChE J.*, 2002, **48**, 1639-1644.
 94. I. Kobayashi, S. Mukataka and M. Nakajima, *Langmuir*, 2005, **21**, 7629-7632.
 95. S. Sugiura, T. Oda, Y. Izumida, Y. Aoyagi, M. Satake, A. Ochiai, N. Ohkohchi and M. Nakajima, *Biomaterials*, 2005, **26**, 3327-3331.
 96. I. Kobayashi, Y. Murayama, T. Kuroiwa, K. Uemura and M. Nakajima, *Microfluid. Nanofluid.*, 2009, **7**, 107-119.
 97. I. Kobayashi, S. Mukataka and M. Nakajima, *J. Colloid Interface Sci.*, 2004, **279**, 277-280.
 98. T. Nakashima, M. Shimizu and M. Kukizaki, *Key Eng. Mater.*, 1991, **61-62**, 513-516.
 99. T. Nakashima, M. Shimizu and M. Kukizaki, *Adv. Drug Deliv. Rev.*, 2000, **45**, 47-56.
 100. K. Suzuki, I. Shuto and Y. Hagura, *Food Sci. Technol. Int. Tokyo*, 1996, **2**, 43-47.
 101. M. Yasuda, T. Goda, H. Ogino, W. R. Glomm and H. Takayanagi, *J. Colloid Interface Sci.*, 2010, **349**, 392-401.
 102. S. R. Kosvintsev, G. Gasparini, R. G. Holdich, I. W. Cumming and M. T. Stillwell, *Ind. Eng. Chem. Res.*, 2005, **44**, 9323-9330.
 103. M. M. Dragosavac, M. N. Sovilj, S. R. Kosvintsev, R. G. Holdich and G. T. Vladislavjević, *J. Membr. Sci.*, 2008, **322**, 178-188.
 104. G. T. Vladislavjević and R. A. Williams, *J. Colloid Interface Sci.*, 2006, **299**, 396-402.
 105. M. S. Manga, O. J. Cayre, R. A. Williams, S. Biggs and D. W. York, *Soft Matter*, 2012, **8**, 1532-1538.
 106. J. Zhu and D. Barrow, *J. Membr. Sci.*, 2005, **261**, 136-144.
 107. R. G. Holdich, M. M. Dragosavac, G. T. Vladislavjević and S. R. Kosvintsev, *Ind. Eng. Chem. Res.*, 2010, **49**, 3810-3817.
 108. N. A. Wagdare, A. T. Marcelis, O. B. Ho, R. M. Boom and C. J. van Rijn, *J. Membr. Sci.*, 2010, **347**, 1-7.
 109. G. T. Vladislavjević and H. Schubert, *J. Dispersion Sci. Technol.*, 2003, **24**, 811-819.
 110. P. J. Dowding, J. W. Goodwin and B. Vincent, *Colloid Surf. A Physicochem. Eng. Asp.*, 2001, **180**, 301-309.
 111. X. Liu, D. Bao, W. Xue, Y. Xiong, W. Yu, X. Yu, X. Ma and Q. Yuan, *J. Appl. Polym. Sci.*, 2003, **87**, 848-852.
 112. I. Kobayashi, M. Yasuno, S. Iwamoto, A. Shono, K. Satoh and M. Nakajima, *Colloid Surf. A Physicochem. Eng. Asp.*, 2002, **207**, 185-196.
 113. L. Giorno, N. Li and E. Drioli, *J. Membr. Sci.*, 2003, **217**, 173-180.
 114. M. M. Dragosavac, G. T. Vladislavjević, R. G. Holdich and M. T. Stillwell, *Langmuir*, 2011, **28**, 134-143.
 115. G. T. Vladislavjević, *Adv. Colloid Interface Sci.*, 2015, **225**, 53-87.
 116. G. De Luca, F. P. Di Maio, A. Di Renzo and E. Drioli, *Chem. Eng. Process.*, 2008, **47**, 1150-1158.
 117. N. C. Christov, K. D. Danov, D. K. Danova and P. A. Kralchevsky, *Langmuir*, 2008, **24**, 1397-1410.
 118. A. Timgren, G. Trägårdh and C. Trägårdh, *Chem. Eng. Res. Des.*, 2010, **88**, 229-238.
 119. A. Abrahamse, A. Van der Padt, R. Boom and W. De Heij, *AIChE J.*, 2001, **47**, 1285-1291.
 120. M. Rayner, G. Trägårdh, C. Trägårdh and P. Dejmeck, *J. Colloid Interface Sci.*, 2004, **279**, 175-185.
 121. E. Amstad, S. Datta and D. Weitz, *Lab Chip*, 2014, **14**, 705-709.
 122. L. Wu, G.-P. Li, W. Xu and M. Bachman, *Appl. Phys. Lett.*, 2006, **89**, 144106.
 123. X. Huang, W. Hui, C. Hao, W. Yue, M. Yang, Y. Cui and Z. Wang, *Small*, 2014, **10**, 758-765.
 124. Y. X. Zhang and L. Q. Wang, *Nanoscale Microscale Thermophys. Eng.*, 2009, **13**, 228-242.
 125. T. Ward, M. Faivre, M. Abkarian and H. A. Stone, *Electrophoresis*, 2005, **26**, 3716-3724.
 126. P. Garstecki, I. Gitlin, W. DiLuzio, G. M. Whitesides, E. Kumacheva and H. A. Stone, *Appl. Phys. Lett.*, 2004, **85**, 2649-2651.
 127. S. L. Anna and H. C. Mayer, *Phys. Fluids*, 2006, **18**, 121512.
 128. J. D. Tice, A. D. Lyon and R. F. Ismagilov, *Anal. Chim. Acta*, 2004, **507**, 73-77.
 129. P. Garstecki, M. J. Fuerstman, H. A. Stone and G. M. Whitesides, *Lab Chip*, 2006, **6**, 437-446.
 130. M. De Menech, P. Garstecki, F. Jousse and H. Stone, *J. Fluid Mech.*, 2008, **595**, 141-161.
 131. M. Zagnoni, J. Anderson and J. M. Cooper, *Langmuir*, 2010, **26**, 9416-9422.
 132. J. Fan, Y. X. Zhang and L. Q. Wang, *Nano*, 2010, **5**, 175-184.
 133. P. Guillot, A. Colin, A. S. Utada and A. Ajdari, *Phys. Rev. Lett.*, 2007, **99**, 104502.
 134. B. R. Benson, H. A. Stone and R. K. Prud'homme, *Lab Chip*, 2013, **13**, 4507-4511.
 135. J. Sivasamy, T.-N. Wong, N.-T. Nguyen and L. T.-H. Kao, *Microfluid. Nanofluid.*, 2011, **11**, 1-10.
 136. A. R. Abate, P. Mary, V. van Steijn and D. A. Weitz, *Lab Chip*, 2012, **12**, 1516-1521.
 137. P. A. Romero and A. R. Abate, *Lab Chip*, 2012, **12**, 5130-5132.
 138. J. Xu, S. Li, J. Tan and G. Luo, *Microfluid. Nanofluid.*, 2008, **5**, 711-717.
 139. S. Van der Graaf, T. Nisisako, C. Schroen, R. Van Der Sman and R. Boom, *Langmuir*, 2006, **22**, 4144-4152.
 140. J. D. Tice, H. Song, A. D. Lyon and R. F. Ismagilov, *Langmuir*, 2003, **19**, 9127-9133.
 141. A. Gupta and R. Kumar, *Microfluid. Nanofluid.*, 2010, **8**, 799-812.
 142. Y. Ba, H. Liu, J. Sun and R. Zheng, *International Journal of Heat and Mass Transfer*, 2015, **90**, 931-947.
 143. H. Liu and Y. Zhang, *J. Appl. Phys.*, 2009, **106**, 034906.
 144. V. van Steijn, C. R. Kleijn and M. T. Kreutzer, *Phys. Rev. Lett.*, 2009, **103**, 214501.
 145. V. van Steijn, M. T. Kreutzer and C. R. Kleijn, *Chem. Eng.*

- Sci.*, 2007, **62**, 7505-7514.
146. V. van Steijn, C. R. Kleijn and M. T. Kreutzer, *Lab Chip*, 2010, **10**, 2513-2518.
 147. X. Chen, T. Glawdel, N. Cui and C. L. Ren, *Microfluid. Nanofluid.*, 2015, **18**, 1341-1353.
 148. D. Funfschilling, H. Debas, H.-Z. Li and T. Mason, *Phys. Rev. E*, 2009, **80**, 015301.
 149. K. Xu, C. P. Tostado, J.-H. Xu, Y.-C. Lu and G.-S. Luo, *Lab Chip*, 2014, **14**, 1357-1366.
 150. W. Lan, S. Li and G. Luo, *Chem. Eng. Sci.*, 2015, **134**, 76-85.
 151. H. Liu and Y. Zhang, *Phys. Fluids*, 2011, **23**, 082101.
 152. G. F. Christopher, N. N. Noharuddin, J. A. Taylor and S. L. Anna, *Phys. Rev. E*, 2008, **78**, 036317.
 153. R. M. Erb, D. Obrist, P. W. Chen, J. Studer and A. R. Studart, *Soft Matter*, 2011, **7**, 8757-8761.
 154. Y. Liu and L. Yobas, *Appl. Phys. Lett.*, 2015, **106**, 174101.
 155. J. Husny and J. J. Cooper-White, *J. Non-Newton. Fluid Mech.*, 2006, **137**, 121-136.
 156. Y. X. Zhang, J. Fan and L. Q. Wang, *Curr. Nanosci.*, 2009, **5**, 519-526.
 157. A. Gupta, S. M. S. Murshed and R. Kumar, *Appl. Phys. Lett.*, 2009, **94**, 164107.
 158. T. Fu, Y. Ma, D. Funfschilling, C. Zhu and H. Z. Li, *Chem. Eng. Sci.*, 2010, **65**, 3739-3748.
 159. T. Glawdel, C. Elbuken and C. L. Ren, *Phys. Rev. E*, 2012, **85**, 016323.
 160. T. Glawdel, C. Elbuken and C. L. Ren, *Phys. Rev. E*, 2012, **85**, 016322.
 161. A. Sauret and H. C. Shum, *Int. J. Nonlinear Sci. Numer. Simul.*, 2012, **13**, 351-362.
 162. T. Cubaud and T. G. Mason, *Phys. Fluids*, 2008, **20**, 053302.
 163. S. Tomotika, *Proc. R. Soc. Lond. A*, 1935, **150**, 322-337.
 164. G. Taylor, *Proc. R. Soc. Lond. A*, 1934, **146**, 501-523.
 165. H. A. Stone, *Annu. Rev. Fluid Mech.*, 1994, **26**, 65-102.
 166. A. M. Gañán-Calvo, R. González-Prieto, P. Riesco-Chueca, M. A. Herrada and M. Flores-Mosquera, *Nat. Phys.*, 2007, **3**, 737-742.
 167. O. A. Basaran and R. Suryo, *Nat. Phys.*, 2007, **3**, 679-680.
 168. E. Castro-Hernández, F. Campo-Cortés and J. M. Gordillo, *J. Fluid Mech.*, 2012, **698**, 423-445.
 169. A. G. Marín, F. Campo-Cortés and J. M. Gordillo, *Colloid Surf. A Physicochem. Eng. Asp.*, 2009, **344**, 2-7.
 170. J.-u. Shim, R. T. Ranasinghe, C. A. Smith, S. M. Ibrahim, F. Hollfelder, W. T. Huck, D. Klennerman and C. Abell, *ACS Nano*, 2013, **7**, 5955-5964.
 171. R. A. De Bruijn, *Chem. Eng. Sci.*, 1993, **48**, 277-284.
 172. J. Janssen, A. Boon and W. Agterof, *AIChE J.*, 1994, **40**, 1929-1939.
 173. J. Janssen, A. Boon and W. Agterof, *AIChE J.*, 1997, **43**, 1436-1447.
 174. C. D. Eggleton, T.-M. Tsai and K. J. Stebe, *Phys. Rev. Lett.*, 2001, **87**, 048302.
 175. M. Booty and M. Siegel, *J. Fluid Mech.*, 2005, **544**, 243-275.
 176. W. Lee, L. M. Walker and S. L. Anna, *Phys. Fluids*, 2009, **21**, 032103.
 177. T. M. Moyle, L. M. Walker and S. L. Anna, *Phys. Fluids*, 2012, **24**, 082110.
 178. T. Ward, M. Faivre and H. A. Stone, *Langmuir*, 2010, **26**, 9233-9239.
 179. W. W. Zhang, *Phys. Rev. Lett.*, 2004, **93**, 184502.
 180. J. F. d. l. Mora, *Annu. Rev. Fluid Mech.*, 2007, **39**, 217-243.
 181. A. M. Gañán-Calvo and J. M. Montanero, *Phys. Rev. E*, 2009, **79**, 066305.
 182. A. M. Gañán-Calvo, *Phys. Rev. E*, 2008, **78**, 026304.
 183. Y.-H. Tseng and A. Prosperetti, *J. Fluid Mech.*, 2015, **776**, 5-36.
 184. P. A. Zhu, T. T. Kong, L. Y. Lei, X. W. Tian, Z. X. Kang and L. Q. Wang, *Sci. Rep.*, 2016, **6**, 21527.
 185. P. Garstecki, M. J. Fuerstman and G. M. Whitesides, *Nat. Phys.*, 2005, **1**, 168-171.
 186. M. Hashimoto and G. M. Whitesides, *Small*, 2010, **6**, 1051-1059.
 187. A. Evangelio, F. Campo-Cortes and J. M. Gordillo, *J. Fluid Mech.*, 2015, **778**, 653-668.
 188. M. A. Herrada, A. M. Gañán-Calvo, A. Ojeda-Monge, B. Bluth and P. Riesco-Chueca, *Phys. Rev. E*, 2008, **78**, 036323.
 189. J. M. Montanero, N. Rebollo-Muñoz, M. A. Herrada and A. M. Gañán-Calvo, *Phys. Rev. E*, 2011, **83**, 036309.
 190. E. J. Vega, J. M. Montanero, M. A. Herrada and A. M. Gañán-Calvo, *Phys. Fluids*, 2010, **22**, 064105.
 191. T. Si, F. Li, X.-Y. Yin and X.-Z. Yin, *J. Fluid Mech.*, 2009, **629**, 1-23.
 192. J. E. Kreutz, T. Munson, T. Huynh, F. Shen, W. Du and R. F. Ismagilov, *Anal. Chem.*, 2011, **83**, 8158-8168.
 193. F. Shen, B. Sun, J. E. Kreutz, E. K. Davydova, W. Du, P. L. Reddy, L. J. Joseph and R. F. Ismagilov, *J. Am. Chem. Soc.*, 2011, **133**, 17705-17712.
 194. J. Guzowski, K. Gizynski, J. Gorecki and P. Garstecki, *Lab Chip*, 2016, **16**, 764-772.
 195. K. J. Humphry, A. Ajdari, A. Fernández-Nieves, H. A. Stone and D. A. Weitz, *Phys. Rev. E*, 2009, **79**, 056310.
 196. D. Saeki, S. Sugiura, T. Kanamori, S. Sato, S. Mukataka and S. Ichikawa, *Langmuir*, 2008, **24**, 13809-13813.
 197. Z. Li, A. M. Leshansky, L. M. Pismen and P. Tabeling, *Lab Chip*, 2015, **15**, 1023-1031.
 198. R. Dangla, E. Fradet, Y. Lopez and C. N. Baroud, *J. Phys. D: Appl. Phys.*, 2013, **46**, 114003.
 199. M. Hein, S. Afkhami, R. Seemann and L. Kondic, *Microfluid. Nanofluid.*, 2014, **18**, 911-917.
 200. J.-C. Baret, *Lab Chip*, 2012, **12**, 422-433.
 201. M. L. Steegmans, A. Warmerdam, K. G. Schroen and R. M. Boom, *Langmuir*, 2009, **25**, 9751-9758.
 202. J. H. Xu, P. F. Dong, H. Zhao, C. P. Tostado and G. S. Luo, *Langmuir*, 2012, **28**, 9250-9258.
 203. Y. Chen, G.-T. Liu, J.-H. Xu and G.-S. Luo, *Chem. Eng. Sci.*, 2015, **132**, 1-8.
 204. Y. Chen, J.-H. Xu and G.-S. Luo, *Chem. Eng. Sci.*, 2015, **138**, 655-662.
 205. K. Wang, Y. Lu, J. Xu and G. Luo, *Langmuir*, 2009, **25**, 2153-2158.
 206. K. Wang, L. Zhang, W. Zhang and G. Luo, *Langmuir*, 2016, **32**, 3174-3185.
 207. J.-C. Baret, F. Kleinschmidt, A. El Harrak and A. D. Griffiths, *Langmuir*, 2009, **25**, 6088-6093.
 208. Y.-C. Liao, E. I. Franses and O. A. Basaran, *Phys. Fluids*, 2006, **18**, 022101.
 209. M. R. de Saint Vincent, J. Petit, M. Aytouna, J.-P. Delville, D. Bonn and H. Kellay, *J. Fluid Mech.*, 2012, **692**, 499-510.
 210. M. Roché, M. Aytouna, D. Bonn and H. Kellay, *Phys. Rev. Lett.*, 2009, **103**, 264501.
 211. Q. Xu, Y.-C. Liao and O. A. Basaran, *Phys. Rev. Lett.*, 2007, **98**, 054503.
 212. F. Jin, N. R. Gupta and K. J. Stebe, *Phys. Fluids*, 2006, **18**, 022103.
 213. N. M. Kovalchuk, E. Nowak and M. J. H. Simmons, *Langmuir*, 2016, **32**, 5069-5077.
 214. N.-T. Nguyen, S. Lassemono, F. A. Chollet and C. Yang,

- IEEE Sens. J.*, 2007, **7**, 692-697.
215. K. Muijltwijk, E. Hinderink, D. Ershov, C. Berton-Carabin and K. Schroën, *J. Colloid Interface Sci.*, 2016, **470**, 71-79.
 216. T. Glawdel and C. L. Ren, *Phys. Rev. E*, 2012, **86**, 026308.
 217. X. Wang, A. Riaud, K. Wang and G. Luo, *Microfluid. Nanofluid.*, 2015, **18**, 503-512.
 218. W. Lan, C. Wang, X. Guo, S. Li and G. Luo, *AIChE J.*, 2016, **62**, 2542-2549.
 219. J. M. Park, M. Hulsen and P. Anderson, *The European Physical Journal Special Topics*, 2013, **222**, 199-210.
 220. P. Huerre and P. A. Monkewitz, *Annu. Rev. Fluid Mech.*, 1990, **22**, 473-537.
 221. J. Eggers and E. Villermaux, *Rep. Prog. Phys.*, 2008, **71**, 036601.
 222. P. Guillot, A. Colin and A. Ajdari, *Phys. Rev. E*, 2008, **78**, 016307.
 223. M. A. Herrada, A. M. Gañán-Calvo and P. Guillot, *Phys. Rev. E*, 2008, **78**, 046312.
 224. A. M. Gañán-Calvo and P. Riesco-Chueca, *J. Fluid Mech.*, 2006, **553**, 75-84.
 225. A. S. Utada, A. Fernandez-Nieves, J. M. Gordillo and D. A. Weitz, *Phys. Rev. Lett.*, 2008, **100**, 014502.
 226. J. Chomaz, P. Huerre and L. Redekopp, *Phys. Rev. Lett.*, 1988, **60**, 25.
 227. D. R. Link, E. Grasland-Mongrain, A. Duri, F. Sarrazin, Z. Cheng, G. Cristobal, M. Marquez and D. A. Weitz, *Angew. Chem. Int. Ed.*, 2006, **45**, 2556-2560.
 228. S. H. Tan, B. Semin and J.-C. Baret, *Lab Chip*, 2014, **14**, 1099-1106.
 229. M. He, J. S. Kuo and D. T. Chiu, *Appl. Phys. Lett.*, 2005, **87**, 031916.
 230. P. He, H. Kim, D. Luo, M. Marquez and Z. Cheng, *Appl. Phys. Lett.*, 2010, **96**, 174103.
 231. H. Kim, D. Luo, D. Link, D. A. Weitz, M. Marquez and Z. Cheng, *Appl. Phys. Lett.*, 2007, **91**, 133106.
 232. C.-H. Yeh, M.-H. Lee and Y.-C. Lin, *Microfluid. Nanofluid.*, 2012, **12**, 475-484.
 233. M. He, J. S. Kuo and D. T. Chiu, *Langmuir*, 2006, **22**, 6408-6413.
 234. S. H. Tan, F. Maes, B. Semin, J. Vignon and J.-C. Baret, *Sci. Rep.*, 2014, **4**, 4787.
 235. E. Castro-Hernández, P. García-Sánchez, J. Alzaga-Gimeno, S. H. Tan, J.-C. Baret and A. Ramos, *Biomechanics*, 2016, **10**, 043504.
 236. E. Castro-Hernández, P. García-Sánchez, S. H. Tan, A. M. Gañán-Calvo, J.-C. Baret and A. Ramos, *Microfluid. Nanofluid.*, 2015, **19**, 787-794.
 237. S.-H. Tan, N.-T. Nguyen, L. Yobas and T. G. Kang, *J. Micromech. Microeng.*, 2010, **20**, 045004.
 238. Y. Wu, T. Fu, Y. Ma and H. Z. Li, *Soft Matter*, 2013, **9**, 9792-9798.
 239. J. Liu, S.-H. Tan, Y. F. Yap, M. Y. Ng and N.-T. Nguyen, *Microfluid. Nanofluid.*, 2011, **11**, 177-187.
 240. S. H. Tan and N.-T. Nguyen, *Phys. Rev. E*, 2011, **84**, 036317.
 241. Q. Yan, S. Xuan, X. Ruan, J. Wu and X. Gong, *Microfluid. Nanofluid.*, 2015, **19**, 1377-1384.
 242. C.-P. Lee, T.-S. Lan and M.-F. Lai, *J. Appl. Phys.*, 2014, **115**, 17B527.
 243. J. Liu, Y. F. Yap and N.-T. Nguyen, *Phys. Fluids*, 2011, **23**, 072008.
 244. O. Strohmeier, M. Keller, F. Schwemmer, S. Zehnle, D. Mark, F. von Stetten, R. Zengerle and N. Paust, *Chem. Soc. Rev.*, 2015, **44**, 6187-6229.
 245. M. Madou, J. Zoval, G. Jia, H. Kido, J. Kim and N. Kim, *Annu. Rev. Biomed. Eng.*, 2006, **8**, 601-628.
 246. J. Ducrée, S. Haeberle, S. Lutz, S. Pausch, F. Von Stetten and R. Zengerle, *J. Micromech. Microeng.*, 2007, **17**, S103-S115.
 247. D. Mark, S. Haeberle, R. Zengerle, J. Ducrece and G. T. Vladislavljević, *J. Colloid Interface Sci.*, 2009, **336**, 634-641.
 248. K. Maeda, H. Onoe, M. Takinoue and S. Takeuchi, *Adv. Mater.*, 2012, **24**, 1340-1346.
 249. S. Haeberle, L. Naegel, R. Burger, F. v. Stetten, R. Zengerle and J. Ducrée, *J. Microencapsul.*, 2008, **25**, 267-274.
 250. S. Haeberle, R. Zengerle and J. Ducrée, *Microfluid. Nanofluid.*, 2007, **3**, 65-75.
 251. D. Chakraborty and S. Chakraborty, *Appl. Phys. Lett.*, 2010, **97**, 234103.
 252. S. Kar, S. Joshi, K. Chaudhary, T. K. Maiti and S. Chakraborty, *Appl. Phys. Lett.*, 2015, **107**, 244101.
 253. Y. Ren and W. W. F. Leung, *Micromachines*, 2016, **7**, 17.
 254. S.-Y. Park, T.-H. Wu, Y. Chen, M. A. Teitell and P.-Y. Chiou, *Lab Chip*, 2011, **11**, 1010-1012.
 255. A. Sauret, C. Spandagos and H. C. Shum, *Lab Chip*, 2012, **12**, 3380-3386.
 256. H. C. Shum, J. Varnell and D. A. Weitz, *Biomechanics*, 2012, **6**, 012808.
 257. A. Sauret and H. C. Shum, *Appl. Phys. Lett.*, 2012, **100**, 154106.
 258. J. Li, N. Mittal, S. Y. Mak, Y. Song and H. C. Shum, *J. Micromech. Microeng.*, 2015, **25**, 084009.
 259. P. A. Zhu, X. Tang and L. Q. Wang, *Microfluid. Nanofluid.*, 2016, **20**, 47.
 260. P. A. Zhu, X. Tang, Y. Tian and L. Q. Wang, *Sci. Rep.*, 2016, **6**, 31436.
 261. H. Zhou and S. Yao, *Microfluid. Nanofluid.*, 2014, **16**, 667-675.
 262. K. Churski, P. Korczyk and P. Garstecki, *Lab Chip*, 2010, **10**, 816-818.
 263. M. E. Dolega, S. Jakiela, M. Razew, A. Rakszewska, O. Cybulski and P. Garstecki, *Lab Chip*, 2012, **12**, 4022-4025.
 264. M. Yu, Y. Hou, H. Zhou and S. Yao, *Lab Chip*, 2015, **15**, 1524-1532.
 265. S. Jakiela, P. R. Debski, B. Dabrowski and P. Garstecki, *Micromachines*, 2014, **5**, 1002-1011.
 266. S.-Y. Jung, S. T. Retterer and C. P. Collier, *Lab Chip*, 2010, **10**, 2688-2694.
 267. J. Xu and D. Attinger, *J. Micromech. Microeng.*, 2008, **18**, 065020.
 268. A. Bransky, N. Korin, M. Khoury and S. Levenberg, *Lab Chip*, 2009, **9**, 516-520.
 269. Y. N. Cheung and H. Qiu, *J. Micromech. Microeng.*, 2012, **22**, 125003.
 270. Y. N. Cheung and H. Qiu, *Phys. Rev. E*, 2011, **84**, 066310.
 271. I. Ziemecka, V. van Steijn, G. J. M. Koper, M. Rosso, A. M. Brizard, J. H. van Esch and M. T. Kreutzer, *Lab Chip*, 2011, **11**, 620-624.
 272. J. Shemesh, A. Nir, A. Bransky and S. Levenberg, *Lab Chip*, 2011, **11**, 3225-3230.
 273. J. C. Brenker, D. J. Collins, H. Van Phan, T. Alan and A. Neild, *Lab Chip*, 2016, **16**, 1675-1683.
 274. D. J. Collins, T. Alan, K. Helmersson and A. Neild, *Lab Chip*, 2013, **13**, 3225-3231.
 275. L. Schmid and T. Franke, *Lab Chip*, 2013, **13**, 1691-1694.
 276. L. Schmid and T. Franke, *Appl. Phys. Lett.*, 2014, **104**, 133501.
 277. A. R. Abate, M. B. Romanowsky, J. J. Agresti and D. A. Weitz, *Appl. Phys. Lett.*, 2009, **94**, 023503.
 278. C.-Y. Lee, Y.-H. Lin and G.-B. Lee, *Microfluid. Nanofluid.*,

- 2009, **6**, 599-610.
279. H.-W. Wu, Y.-C. Huang, C.-L. Wu and G.-B. Lee, *Microfluid. Nanofluid.*, 2009, **7**, 45-56.
280. A. Raj, R. Halder, P. Sajeesh and A. K. Sen, *Microfluid. Nanofluid.*, 2016, **20**, 102.
281. B. Cai, R. He, X. Yu, L. Rao, Z. He, Q. Huang, W. Liu, S. Guo and X.-Z. Zhao, *Microfluid. Nanofluid.*, 2016, **20**, 56.
282. J.-W. Choi, S. Lee, D.-H. Lee, J. Kim and S.-I. Chang, *RSC Adv.*, 2014, **4**, 20341-20345.
283. S. Zeng, B. Li, X. o. Su, J. Qin and B. Lin, *Lab Chip*, 2009, **9**, 1340-1343.
284. K. Churski, J. Michalski and P. Garstecki, *Lab Chip*, 2010, **10**, 512-518.
285. F. Guo, K. Liu, X.-H. Ji, H.-J. Ding, M. Zhang, Q. Zeng, W. Liu, S.-S. Guo and X.-Z. Zhao, *Appl. Phys. Lett.*, 2010, **97**, 233701.
286. J.-C. Galas, D. Bartolo and V. Studer, *New J. Phys.*, 2009, **11**, 075027.
287. H. Willaime, V. Barbier, L. Kloul, S. Maine and P. Tabeling, *Phys. Rev. Lett.*, 2006, **96**, 054501.
288. J.-H. Wang and G.-B. Lee, *Micromachines*, 2013, **4**, 306-320.
289. W. S. Lee, S. Jambovane, D. Kim and J. W. Hong, *Microfluid. Nanofluid.*, 2009, **7**, 431-438.
290. B.-C. Lin and Y.-C. Su, *J. Micromech. Microeng.*, 2008, **18**, 115005.
291. J.-H. Choi, S.-K. Lee, J.-M. Lim, S.-M. Yang and G.-R. Yi, *Lab Chip*, 2010, **10**, 456-461.
292. C.-T. Chen and G.-B. Lee, *J. Microelectromech. Syst.*, 2006, **15**, 1492-1498.
293. S.-K. Hsiung, C.-T. Chen and G.-B. Lee, *J. Micromech. Microeng.*, 2006, **16**, 2403-2410.
294. C.-H. Lee, S.-K. Hsiung and G.-B. Lee, *J. Micromech. Microeng.*, 2007, **17**, 1121-1129.
295. M. Zhang, J. Wu, X. Niu, W. Wen and P. Sheng, *Phys. Rev. E*, 2008, **78**, 066305.
296. N.-T. Nguyen, *Microfluid. Nanofluid.*, 2012, **12**, 1-16.
297. N.-T. Nguyen, T.-H. Ting, Y.-F. Yap, T.-N. Wong, J. C.-K. Chai, W.-L. Ong, J. Zhou, S.-H. Tan and L. Yobas, *Appl. Phys. Lett.*, 2007, **91**, 084102.
298. P. Sheng and W. Wen, *Annu. Rev. Fluid Mech.*, 2012, **44**, 143-174.
299. X. Niu, M. Zhang, J. Wu, W. Wen and P. Sheng, *Soft Matter*, 2009, **5**, 576-581.
300. C. A. Stan, S. K. Tang and G. M. Whitesides, *Anal. Chem.*, 2009, **81**, 2399-2402.
301. S.-H. Tan, S. S. Murshed, N.-T. Nguyen, T. N. Wong and L. Yobas, *J. Phys. D: Appl. Phys.*, 2008, **41**, 165501.
302. S. S. Murshed, S. H. Tan, N. T. Nguyen, T. N. Wong and L. Yobas, *Microfluid. Nanofluid.*, 2009, **6**, 253-259.
303. S. S. Murshed, S.-H. Tan and N.-T. Nguyen, *J. Phys. D: Appl. Phys.*, 2008, **41**, 085502.
304. C.-H. Yeh, K.-R. Chen and Y.-C. Lin, *Microfluid. Nanofluid.*, 2013, **15**, 775-784.
305. R. Suryo and O. A. Basaran, *Phys. Rev. Lett.*, 2006, **96**, 034504.
306. V. Miralles, A. Huerre, H. Williams, B. Fournie and M.-C. Jullien, *Lab Chip*, 2015, **15**, 2133-2139.
307. C. N. Baroud, J.-P. Delville, F. Gallaire and R. Wunenburger, *Phys. Rev. E*, 2007, **75**, 046302.
308. C. N. Baroud, M. R. de Saint Vincent and J.-P. Delville, *Lab Chip*, 2007, **7**, 1029-1033.
309. M. Robert de Saint Vincent and J.-P. Delville, *Phys. Rev. E*, 2012, **85**, 026310.
310. M. R. de Saint Vincent, H. Chraïbi and J.-P. Delville, *Phys. Rev. Appl.*, 2015, **4**, 044005.
311. M. L. Cordero, F. Gallaire and C. N. Baroud, *Phys. Fluids*, 2011, **23**, 094111.
312. E. Verneuil, M. a. Cordero, F. Gallaire and C. N. Baroud, *Langmuir*, 2009, **25**, 5127-5134.
313. S.-Y. Tang, I. D. Joshipura, Y. Lin, K. Kalantar-Zadeh, A. Mitchell, K. Khoshmanesh and M. D. Dickey, *Adv. Mater.*, 2016, **28**, 604-609.
314. B. Berge, O. Kononov, J. Lajzerowicz, A. Renault, J. Rieu, M. Vallade, J. Als-Nielsen, G. Grübel and J. Legrand, *Phys. Rev. Lett.*, 1994, **73**, 1652.
315. A. Diguët, H. Li, N. Queyriaux, Y. Chen and D. Baigl, *Lab Chip*, 2011, **11**, 2666-2669.
316. F. Malloggi, S. A. Vanapalli, H. Gu, D. van den Ende and F. Mugele, *J. Phys.: Condens. Matter*, 2007, **19**, 462101.
317. F. Malloggi, H. Gu, A. Banpurkar, S. Vanapalli and F. Mugele, *Eur. Phys. J. E*, 2008, **26**, 91-96.
318. H. Gu, M. H. Duits and F. Mugele, *Lab Chip*, 2010, **10**, 1550-1556.
319. H. Gu, F. Malloggi, S. A. Vanapalli and F. Mugele, *Appl. Phys. Lett.*, 2008, **93**, 183507.
320. H. Gu, C. U. Murade, M. H. Duits and F. Mugele, *Biomechanics*, 2011, **5**, 011101.
321. K. Wang, L. Xie, Y. Lu and G. Luo, *Lab Chip*, 2013, **13**, 73-76.
322. G. Bolognesi, A. Hargreaves, A. D. Ward, A. K. Kirby, C. D. Bain and O. Ces, *RSC Adv.*, 2015, **5**, 8114-8121.